

GE Healthcare

Signa HD, HDx & HDxt 1.5T Block Diagrams &
Supplemental Schematics
(Release 12.x, 14.x & 15.x)
Technical Publication



5133310
Rev 12

DAMAGE IN TRANSPORTATION

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- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
 - IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
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-
- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
 - SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
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- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
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- NÃO TENDE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1. . .	Sep 29, 2004	Initial Release
2 . . .	Apr 14, 2005	Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added XW8200 host computer, removed CAN Link for ACGD, added HFD–S CAN Link, and other minor corrections. Operator Workspace Tab – Added XW8200 host computer and other minor corrections. Patient Handling Tab – Made minor corrections. Gradient Tab – Removed ACGD cabinet pages and inserted HFD–S Cabinet page. MGD/RRF/RF Tab – Corrected minor errors and added informational callouts. Added MRC II Tool & Cable and MRC II Tool – 8 Channel Connector pages.
3 . . .	Oct 15, 2005	System Tab – Added J116 and J119 to TAC Cabinet and relocated Terminator. Patient Handling Tab – Added test points and LED's to Magnet Enclosure and PHPS. Added MNS connection information to SSM. MGD/RRF/RF Tab – Added LED info to Exciter and Receiver boards. Corrected RRF Receive Chain connection labels. Added MNS connection info to Driver Module and SSM. Added MC Preamp Bias Filter pinouts. Added LED and Pinout info for 16 Ca'hannel Switch. Corrected MNS connection info on ASC Chassis. Revised and added interconnects for forward production and upgrade MNS options. Major update to manual with the addition of HDx (Release 14.x) product. Changed title page and banners on all pages.
4 . . .	Mar 31, 2006	Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added HP9300 computer info, 32 Channel option info, and the new RFS cabinet power distribution using two power strips. Revised all sheet references. OW Tab – Added HP9300 computer info. MGD/RRF/RF Tab – Added HDx (14.x) info as needed which includes VRE modules: ICN's, 16/24 Port Ethernet Switch, Infiniband board and module. Added HDx A, B, and C port connection information for the HDx LPCA. Added MCR III Tool
5 . . .	Jun 30, 2006	Overview Tab – Added definitions to Section 3. Added TOC's to System, Operator Workspace and Patient Handling Sections. System Tab – Added ASC Module to System/RFS cabinet depictions. Patient Handling Tab – Updated J20 Pin Signal table. Added SSM block diagram. MGD/RRF/RF Tab – Added 32 Channel information and other minor corrections and updates. Added Environmental Sensor to System/RFS cabinet in System. Operator Workspace, and MGD/RRF/RF Sections (MRlhc19337). MGD/RRF/RF Tab –Added Port 2 to Port 2 connection for 4 ICN configuration. Added 4 ICN configuration without Infiniband switch. Updated Multi–Coil Adapter by adding MC Bias 17–24 for 14.x systems (MRlhc18216). Updated several pages with additional MC signal information (MRlhc18217).
7 . . .	Dec 20, 2006	SYSTEM TAB – Added 32 Channel configuration without Infiniband Switch. GRADIENT Tab – Added Pang block configuration info (MRlhc20731). OW TAB – Added Audio IN/OUT signals for PC Host for 14.x system. MGD/RRF/RF Tab – Revised ASC Chassis to reflect functionality using new Detector Board. 5251054.
8	Jun 6, 2007	SYSTEM Tab – Added note for designating components required for 16 channel systems. MGD/RRF/RF Tab –Added notes to designate parts not included with newer 8 Channel systems and for parts that are required for 16 Channel systems.
9 . . .	Oct 26, 2007	SYSTEM Tab – Added CAM Chassis designation to MGD Chassis where applicable. Added CAN LINK block diagram for RFS cabinet with CAM Chassis. MGD/RF/RRF Tab – Added board slot locations for CAM Chassis.
10	Sep 11, 2008	Changed title of manual and all page headers to <i>Signa HD, HDx & HDxt 1.5T (12.x, 14.x & 15.x) Block Diagrams & Supplemental/Schematics</i> . HDi and Signa Vibrant are information is also included with this name change. Updates for HDXT (15.x) in all sections. Added information on Sun ICN's. Added depopulated 16–channel switch for 8–channel installations.
11	Dec 17, 2008	SYSTEM Tab – Added new environmental sensor diagram (ECO 2055310). PATIENT HANDLING Tab – Revised E–Stop circuit voltage from 24V to 12V in Teal PDU (PQR 13207104).
12	Jun 30, 2010	SYSTEM Tab – Added new Z400 computer page (ECO2056587) and added 5337894–3 and –4 ICN's (ECO 2101939). MGD/RRF/RF Tab – Added new ICN's (ECO 2101939).

LIST OF EFFECTIVE PAGES

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* This revision number/letter corresponds to the indicated document's revision control system.

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Purpose
Standard Conventions
Abbreviations



SYSTEM

System Block Diagrams



OPERATOR WORKSPACE

Operator Workspace Block Diagrams



PATIENT HANDLING

Patient Handling Block Diagrams



GRADIENT

Gradient Block Diagrams (HFD & ACGD)



PDU

PDU Block Diagrams

Note:

Phoenix PDU found in Vendor Manuals only. (SEE Service Methods CDROM)



MGD/RRF/RF (1.5T)

MGD/RRF/RF Block Diagrams (1.5T)
RF Supplemental Schematics (1.5T)



COOLING SYSTEM

Water Chillers Block Diagrams

Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)



OVERVIEW

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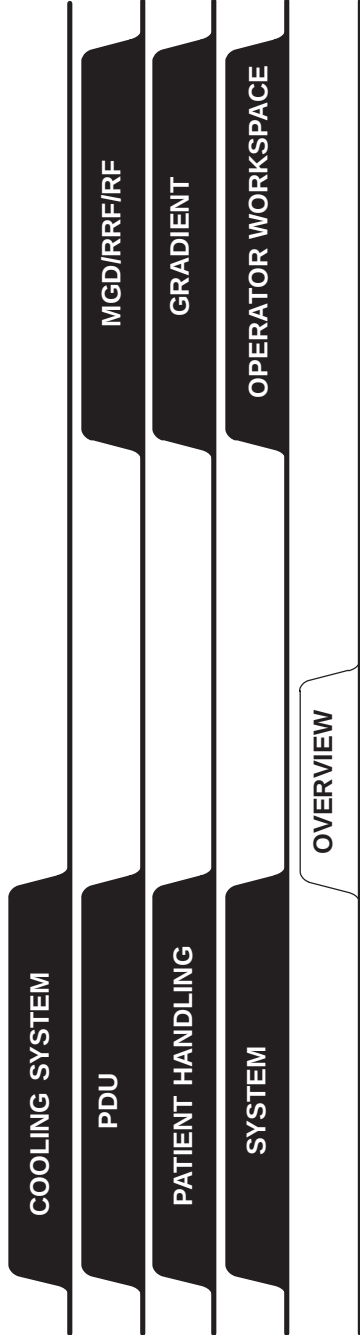
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SECTION 1 – PURPOSE

The purpose of this document is to provide an overview of the hardware which makes up the MR Signa System, and a quick means to troubleshoot major signals within the system.

It is recommended that you page through this manual to become familiar with its content and organization, beginning with System Tab, OVERALL SYSTEM. This section shows how the major subsystems are grouped by cabinet, and the main communication lines between them.

The remaining block diagrams are organized into sections which correspond to major subsystems. Some diagrams show hardware from another subsystem in order to complete an entire signal flow route. For such exceptions, the board/module title of the circuitry shown will generally help identify the applicable subsystem. See Illustration 1–1 for tab organization of this manual.



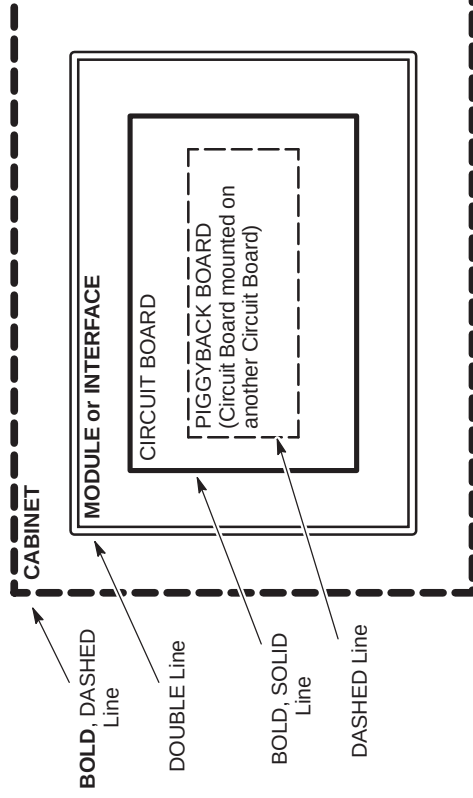
TAB ORGANIZATION FOR THIS DIRECTION
ILLUSTRATION 1

These block diagrams, along with provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level on down to the specific module or board level. Since some of the RF subsystem block diagrams are presently insufficient for troubleshooting to the FRU (Field Replaceable Unit) level, schematics have been provided to supplement those block diagrams. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

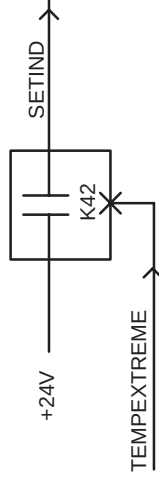
SECTION 2 – STANDARD CONVENTIONS

The following standard conventions are used within the block diagrams:

- The type of line used to enclose a major box on a block diagram indicates the following:



- The component designator for cabinets, modules, circuit boards, etc. is enclosed in “()” after or below their name on the block diagram.
- Logic levels: A “0” indicates a logic “low” level (typ. <0.8VDC), a “1” indicates a logic “high” level (typ. > 2.0VDC).
- An “*” is used to identify an “active low” signal (e.g. “INT*”) signal when at a logic “0” indicates that an interrupt is present; a logic “1” level indicates that no interrupt is present).
- A “#/#” indicates the number of wires in a cable (e.g. $\xrightarrow{9}$ indicates 9 wires).
- Signal flow is generally from left to right on a block diagram. Arrows are used to identify exceptions to this rule and to aid in identifying input/output signal lines on the diagram.
- The “x” symbol (e.g. —x) shown on boxes in a block diagram are used to indicate that the function within the box is enabled by the “active” signal on this input line.



For this example, when signal TEMPEXTREME goes active (“1”), relay K42 will close (become enabled) and the signal SETIND now becomes active to set a temperature indicator.

SECTION 3 – ABBREVIATIONS

Note

This list has been compiled from a wide variety of sources. Not all of these abbreviations will apply to your system. For an electronic copy of over 1000 acronyms, see *Direction 2124201–2*.

A

AC = Alternating Current
ACGD = Advanced Concept Gradient Driver
ACK = Acknowledge
ACQ = Acquisition
A/D = Analog-to-Digital
ADC = Analog to Digital Converter
ADD or ADDR = Address
ADJ = Adjust
ADV = Advance
AGC = Automatic Gain Control
AGP = Applications Gateway Processor (Board)
AI = Applications Interface, Artificial Intelligence
ALGN = Align

ALU = Arithmetic Logic Unit
AM = Amplitude Modulation
AMP = ampere, amplifier
A/N = Alpha-numeric
ANSI = American National Standards Institute
ANT = Antenna

AP = Array Processor
APM = Analog Power Monitor
APS = Auto Pre-Scan or Acquisition Processing
System (Board)

APSGT = Auto Pre-Scan Gain Control
ARCH = Archive/Remove Process
ASC = Automated Support Center or
Amplifier Support Controller
ASCII = American Standard Code for Information
Interchange

ASM or ASSY = Assembly
ASM = Analog Service Module
ATTEN = Attenuator
AUTO = Automatic
AUX = Auxiliary

B

B = magnetic field
BATT = Battery
BB = Buffered Data or Broadband
BC = Body Coil

BD = Board

BDS = Buffered Data Strobe

BLK = Black

BNC = BNC Connector

BRM = Body Resonance Module

BRT = Brightness

BUF = Buffer

B/W = BandWidth

C

C = centigrade, Celsius

CAB = Cabinet

CAL = Calibration

CALC = Calculate

CAN = Controller Area Network

CATTEN = Course Attenuation

CB = Circuit Breaker

CCC = CAN Core Controller card

CCW = Counter Clockwise

CD-ROM = Compact Disk Read–Only Memory

CE = Chip Enable

CF = Center Frequency

CFA = Center Frequency Adjust

CHAN or CHNL = Channel

CHK = Check

CKT = Circuit

CLR = Clear

CMD = Command

CMMR = Common Mode Rejection Ratio

CMOS = Complimentary Symmetry Metal Oxide
Semiconductor

CMP = Compare or Compressor

CNT = Contrast

COAX = Coaxial

COF = Cut Off Frequency

CONFIG = Configuration

CONT = Continuous

CONV = Converter

CPD = Communication Pin Driver

CPU = Central Processing Unit
CRT = Cathode-Ray Tube
CTL = Clock Time Control
CTRL = Control
CV = Configuration Variable
CW = Clockwise or Controller Wave
CYL = Cylinder

D

D/A = Digital-to-Analog
DAC = Digital to Analog Converter
dB = Decibel
dBm = Decibels above or below one Milliwatt

DC = Direct Current
DCB = Driver Control Board
DCKSTP = Dock Stop
DD = Dynamic Disable
DDS = Direct Digital Synthesizer
DDSB = Dynamic Disable Switch Box
DEC = Decimal
DEG = Degree(s)
DEMUX = Demultiplexer
DEV = device

DIA = Diagnostic Imaging Accessories
DIAG = Diagnostic
DIFF = Differential

DIG = Digit
Dig B0 = Digital B0 Adjustment
DIP = Dual In-line Package
DIR = Direction
DIS = Disable

DM = Driver Module
DMA = Direct Memory Access
DMAC = Direct Memory Access Controller

DMM = Digital Multi-Meter
DPR or DPRAM = Dual Port Random Access Memory
DQA = Daily Quality Assurance
DRAM = Dynamic Random Access Memory
DRC = Dynamic Ram Controller
DRF = Digital Receive Filter (Board)
DRVR = Driver

DSP = Digital Signal Processor
DVM = Digital Volt Meter
DWG = Drawing

E

ECC = Extended Cursor Control or Error Checking and Correction
ECG = Electro-Cardiogram
EEPROM = Electrically Erasable Programmable Read Only Memory

EFB = Envelop Feedback
EPI = Echo Planer Imaging
e.g. = for example
EM or EMERG = Emergency
EMI = Electro-Magnetic Interference
EN or ENA = Enable

ENV = Envelope

EOS = End Of Sequence

EOT = End Of Travel

EOW = End Of Waveform

EPO = Emergency Power Off

EPROM = Erasable Programmable Read Only Memory

ERU = Emergency Rundown Unit

EXER = Exerciser

EXCITE = Expanding Applications with Multi-channel Imaging Technology

F

f = farad

F = Fahrenheit, Fuse (designator)

FBD = Functional Block Diagram

FBK = Feedback

FC = footcandle

Fc = Center Frequency

FDBK = Feedback

FDO = Future Delivery Order

FE = Field Engineer

FET = Field-Effect Transistor

FF or FIFO = First In – First Out

FFD = Full Field Distortion

FFT = Fast Fourier Transform

FID = Free Induction Decay

FIDAT = FIFO In Data

FIFO or FF = First In – First Out

FILT = Filter

FM = Frequency Modulated

FMI = Field Modification Instruction

FOV = Field Of View

FREQ = Frequency

FRNT = Front

FRU = Field Replaceable Unit

ft = foot

FT = Feet

FTP = File Transfer Protocol

G

G = gauss
G/A = Gradient Amplifier
GEN = Generator
GND = Ground
GOC AA = Global Operator Console Audio Assembly
GRAD = Gradient
GRN = Green
GRY = Grey
GP = Gradient Processor
GSW = Gradient Switch

H

H = henry
H = Hexadecimal
HART = Hardware Tracker
HC = Head Coil
HD = Head or High Definition
He = Helium
He-ALARM = Helium Alarm
HEX = Hexadecimal
He-WARNING = Helium Warning
HFA = High Fidelity Amplifier
HFD = High Fidelity Driver
HFD-S = High Fidelity Driver – Special
HORIZ or HORZ = Horizontal
HPC = Hardware and Protocol Control
HPFS = Host Platform Software
hr = hour
HSN = High Speed Network
HSS (D) = High Speed Serial Data Link
HTR = Heater
HV = High Voltage
HV PSU = High Voltage Power Supply Unit
HVAC = Heating, Ventilation, & Air Conditioning
Hz = hertz

I

I = Current or Phase or Real
IAMP = Amplifier Current (Gradient)
ICD = Installation Certification Document
ICL = Coil Overcurrent (Gradient)
ICN = Image Compute Node
ID = inner diameter
ID = Identity, Identifier
i.e. = that is to say
IEC = International Electrical Code
IEEE = Institute of Electrical & Electronics Engineers
I/F = Interface
in. = inch
INFO = Information

INS, INSTR, or INSTR = Instruction
INT or INTR = Interrupt
I/O = Input/Output
IP = Image Processor or Plate Current (RF amp) or
Instruction Processor or Internet Protocol
IPA = Intermediate Power Amplifier
IPC = Interprocess Communication
IQ = Image Quality
IR = Image Reject
IRF = Interface and Remote Functions (Board)
IRF I/O = Interface and Remote Functions
Input/Output (Board)
IRQ = Interrupt Request
ISA = Interrupt Start Address
ISI = Inter-Sequence Interrupt
ISO = Isolator or Isolation
IT-MGD = Image Transfer-MDG (Board)
IT-Host = Image Transfer-Host (Board)
IVI = Interactive Vascular Imaging

J

J = Jack Connector (designator)
JP = Jumper

K

K = Kelvin (unit of temperature) or Kilo (Thousand)
KB = one thousand bytes of memory
KCal = kiloCalorie
kg = kilogram
kHz = kilohertz
KVA = kilo Volt Amperes
kW = Kilowatt

L

L or LFT = Left
LAN = Local Area Network
lb = pound
LBK = Loopback
LCA = Logic Cell Arrays
LED = Light Emitting Diode
LFC = Load From Cold
LHe =Liquid Helium
LIN = Linear
LM or LMK = Landmark
LMP = Lamp
LNK = Link
LPCA = Low Profile Carriage Assembly
LO = Local Oscillator
LOEP = List of Effective Pages
LONG = Longitudinal
LVLE = Low Voltage, Low Energy

M

m = meter
M = mega – one million
mA = millamps
MAG = Magnet
MB = megabyte
MC = Multi-Coil
MCD = Multi-Coil switch Driver
MCR = Multicoil Receive (Tool or Diagnostic)
MDS = Multidrop Serial interface
MEM = Memory
Mfg. = Manufacturing
MFU = Multi-Format Unit
MGD = Multi-Generational Data acquisition (Chassis)
MGMT = Management
MHz = megahertz
mm = millimeter
MOD = Modifier
MON = Monitor
MOSFET = Metal Oxide Semiconductor Field Effect Transistor
MOV = Metal Oxide Varistor
MPS = Manual Pre-Scan
MR = Magnetic Resonance
MSG = Message
MTR = Motor
MUX = Multiplexer
mV = millivolts
MW = megawatt
mW = milliwatt

OT = Overtemp
OV = Over voltage
OVR = Over
OVRD or OL = Overload
oz = ounce
OW = Operator Workspace

P

P = Positive
PA = Patient or Power Amplifier
PAC = Physiological Acquisition Controller
PAL = Programmable Array Logic or Patient Alignment Lights
PAN = Primary Archive Node
PAT = Patient
PCI = Programmable Communications Interface or Peripheral Component Interconnect
PCT = Pulse Controller Task
PDI = Product Delivery Instruction
PDU = Power Distribution Unit
PEN = Penetration
PHPS = Patient Handling Power Supply
PIN = Positive-Intrinsic-Negative
PIO = Programmed Input/Output
PLL = Phase Lock Loop
PIXEL = Picture Element
PM = Phase Modulation, or Planned Maintenance
PMC = PCI Mezzanine Connector
P/N = Positive/Negative
PNL = Panel
POS = Position
POT = Potentiometer
PP = Penetration Panel
PPC = Power PC
P-P = Peak to Peak
ppm = parts per million
PREAMP = Preamplifier
PROC = Process, Processor
PROG = Program, –ed, –able
PROM = Programmable Read Only Memory or Program
PS = Power Supply
PSD = Pulse Sequence Database
PSG = Pulse Sequence Generator
PSU = Power Supply Unit
PSS = Power Supply System Board
PT = Patient Transport or Patient Table
PUR = Purple
PVC = Polyvinylchloride

N

N = Negative
N2 = Nitrogen
n.c. = normally closed
NEC = National Electrical Code
Ni-Cad = Nickel Cadmium
n.o. = normally open
NO. = Number
NOPROC = No Processing
NOREC = No Reconstruction
nS = Nanosecond

O

O2 = oxygen
OC = Operator's Console
OD = outside diameter
OM = Oxygen Monitor
OPI = Oblique Plane Imaging
OPTO = Optical
ORG = Orange

PW = Pulse Width

PWA = Printed Wiring Assembly

PWB = printed wiring board

PWR = Power

Q

Q = Quadrature, Quality Factor or Imaginary

Qty = Quantity

QUAD = Quadrature or Quadrant

R

R or RT = Right

RAM = Random Access Memory

RCV = Receive

RD = Read

RDY = Ready

RECON = Reconstruction

REF = Reference or Reflected

REG = Register or Regulator

RET or RTN = Return

REV = Revision or Reverse

RF = Radio Frequency

RFAMP = Radio Frequency Amplifier

RFA = RF Amplifier

RFI = Radio Frequency Interference or RF Interface
(Module)

RFS = RF System Cabinet

RGB = Red Green Blue

RLA or RLY = Relay

rms = root mean square

ROI = Region of Interest

ROM = Read Only Memory

RPC = Remote Control Panel

RRF = Remote RF (Chassis)

RRF-DIF = Remote RF-Digital Interface board

RSR = Register

RST = Reset

R/W = Read/Write

RX = Receive

RXD = Receive Data

S

S = Switch (designator)

SACK = Slave Acknowledge

SAG = Sagittal

SAR = Specific Absorption Rate

S/C = Superconducting

SCA = Scan Control Assembly

SCP = Scan Control Processor (Board)

SCR = Silicon Controlled Rectifier

SCSI = Small Computer System Interface

SEC = second

SEG = Segment

SEL = Select

SEPI = Spin Echo Planer Imaging

SGA = Switchable Gradient Amplifier

SGI = Silicon Graphics Inc.

SIG = Signal

SII = "S-2" (type of magnet)

SIII = "S-3" (type of magnet)

SIMMS = Single In-line Memory Module

SIP = Single In-line Package

SL = Slice

SMC = Status Monitoring and Control

SMPT E = Society of Motion Picture and Television
Engineers

SN = Service Note

SNR = Signal-to-Noise Ratio

SPI = Serial Peripheral Interface

SPT = System Performance Test

SPU = Signal Processing Unit

SRAM = Static Random Access Memory

SRF = Solid State RF Amplifier SGA or Scalable RF
(Cabinet)

SRF/TRF = Sequencer Related Function / Trigger and
Rotation Function (Board)

SRI = Scan Room Interface

SRV = Service

SSM = System Support Module

SSP = Sequence Synchronous Protocol

SST = Small Sample Test

STAB = Stability

STAT = Status

STD = Standard

STD DEV = Standard Deviation

STIF Bd = SRF/TRF InterFace Board

SUB-D = Subminiature D (connector)

SW = Switch or Software

SW = Software

SYNC = Synchronous

SYS = System

SYSID = System Identification

T

T = tesla
TAC = Twin Accessory Cabinet
TBD = To Be Determined
TBL = Table
TEMP = Temperature
TFU = Thyristor Firing Unit
TG = Transmit Gain
TGWG = TwinSpeed Gradient Water Cooler
TNS = Transient Noise Suppressor
TP = Test Point (designator)
TR or T/R = Transmit–Receive
TRMAP = Transmit/Receive Map
TRM = Twin Coil Resonator Module
TS = Terminal Strip
TSCC = TwinSpeed Cooling Cabinet
TST = Test
TTL = Transistor to Transistor Logic
TX = Transmit
TXD = Transmit Data

U

UART = Universal Asynchronous Receiver–Transmitter
uf = microfarad
UFI = Ultra Fast Imaging
UNBLK = Unblank
uP = Microprocessor
UPLMT = Up Limit
UPM = Universal Power Monitor
USART = Universal Synchronous/Asynchronous
Receiver–Transmitter
uS or usec = microsecond
UTNS = Universal Transient Noise Suppressor (Board)
UV = Under voltage

V

VAC = Volts Alternating Current
VAR = Variable
VBUS = Bus Voltage
VCR = Video Cassette Recorder
VCNTRL = Control Voltage for 8645 Techtrons
VDC = Volts Direct Current
VERT = Vertical
Vp-p = Volts Peak-to–Peak
VRE = Volume Recon Engine
VRMS = Volts, Root Mean Square
VTR = Video Tape Recorder

W

WARP = Waveform And Rotation Processor
WC = Water Chiller
WR = Write
WHT = White
WIM = Workspace Interface Module
WND = Window
WORM = Write Once Read Many
WRT = Write

X

XCVR = Transceiver
XFMR = Transformer
XMT or XMIT = Transmit

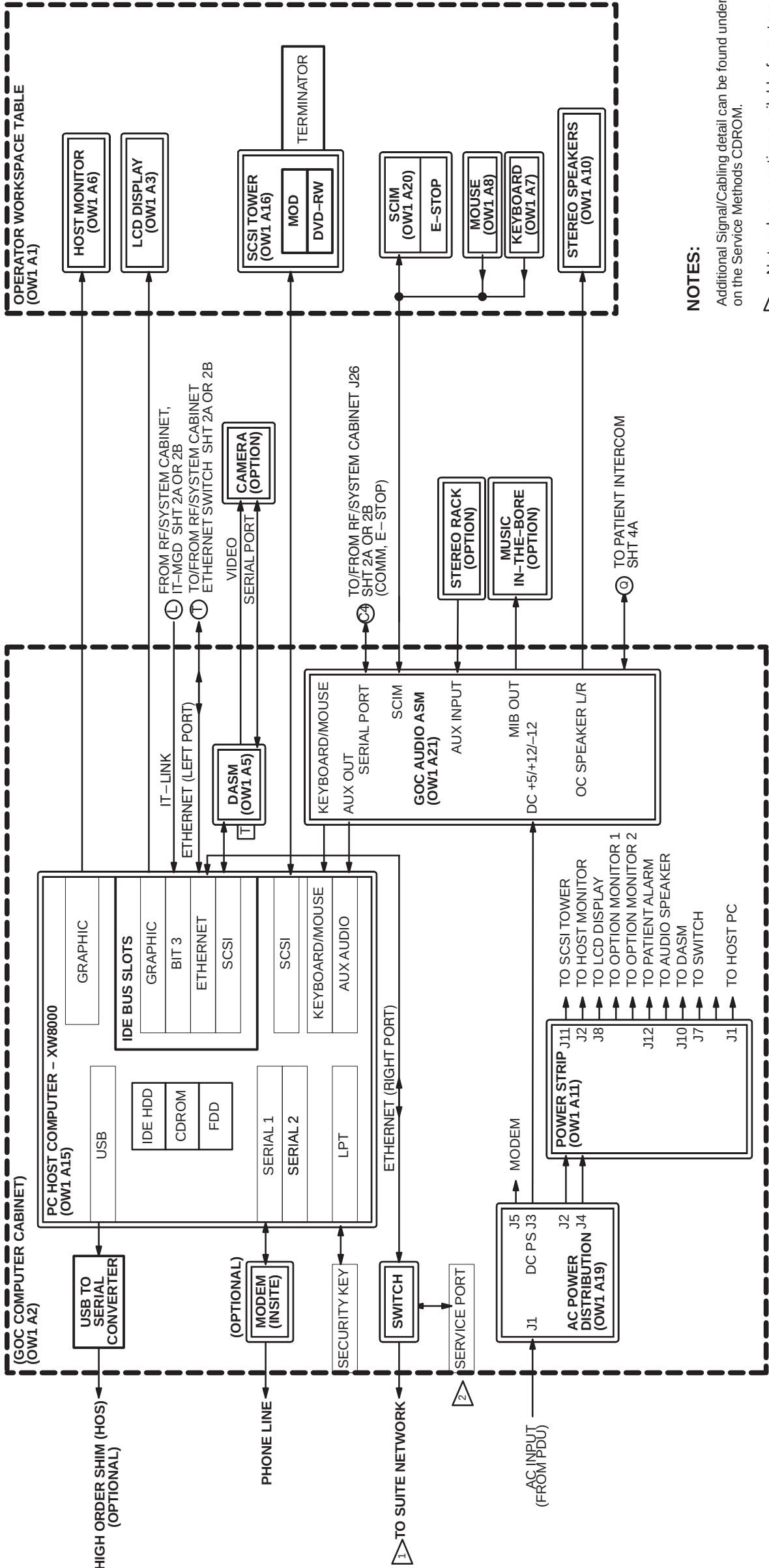
Y

YEL = Yellow

SYSTEM

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NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

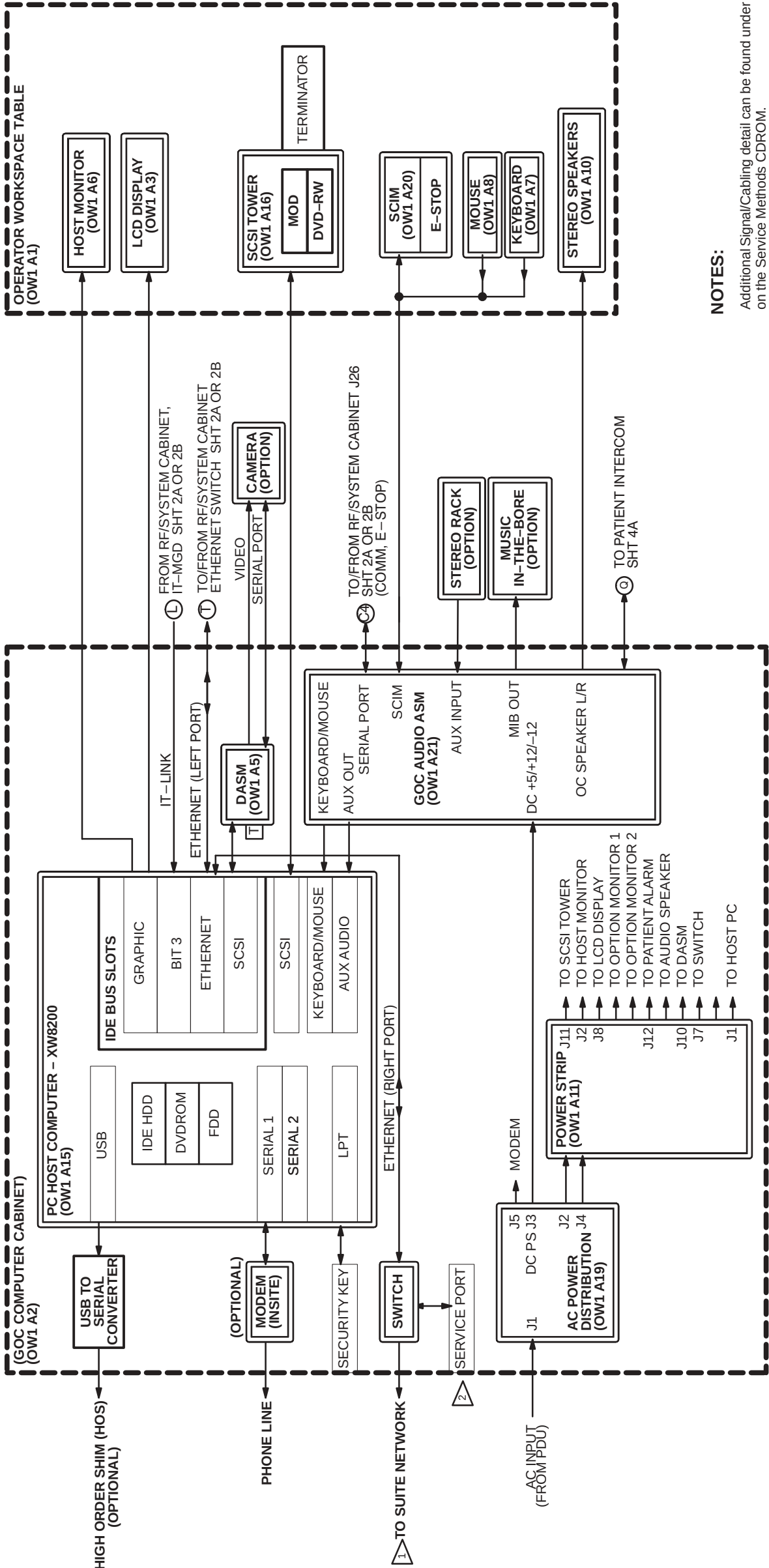
1) PATIENT HANDLING

2) MGD/RRF/RF

EXCITE HD (12.x) OVERALL SYSTEM

Forward Production or Pre-EXCITE to HD Upgrades
(OPERATOR WORKSPACE CABINET w/8000 & TABLE)

SHEET 1A OF 5



NOTES:

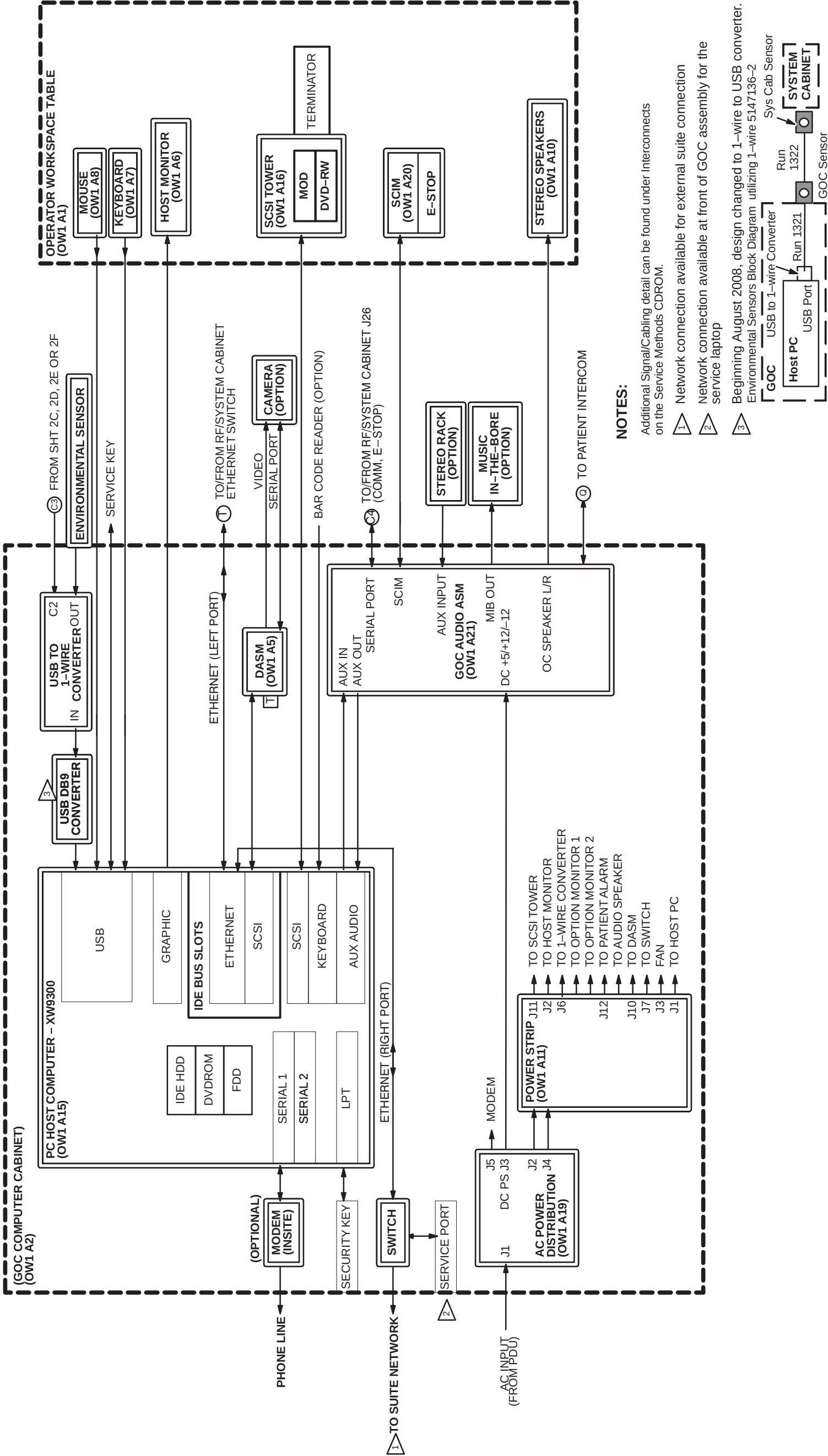
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- 1 Network connection available for external suite connection
- 2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/RRF/RF

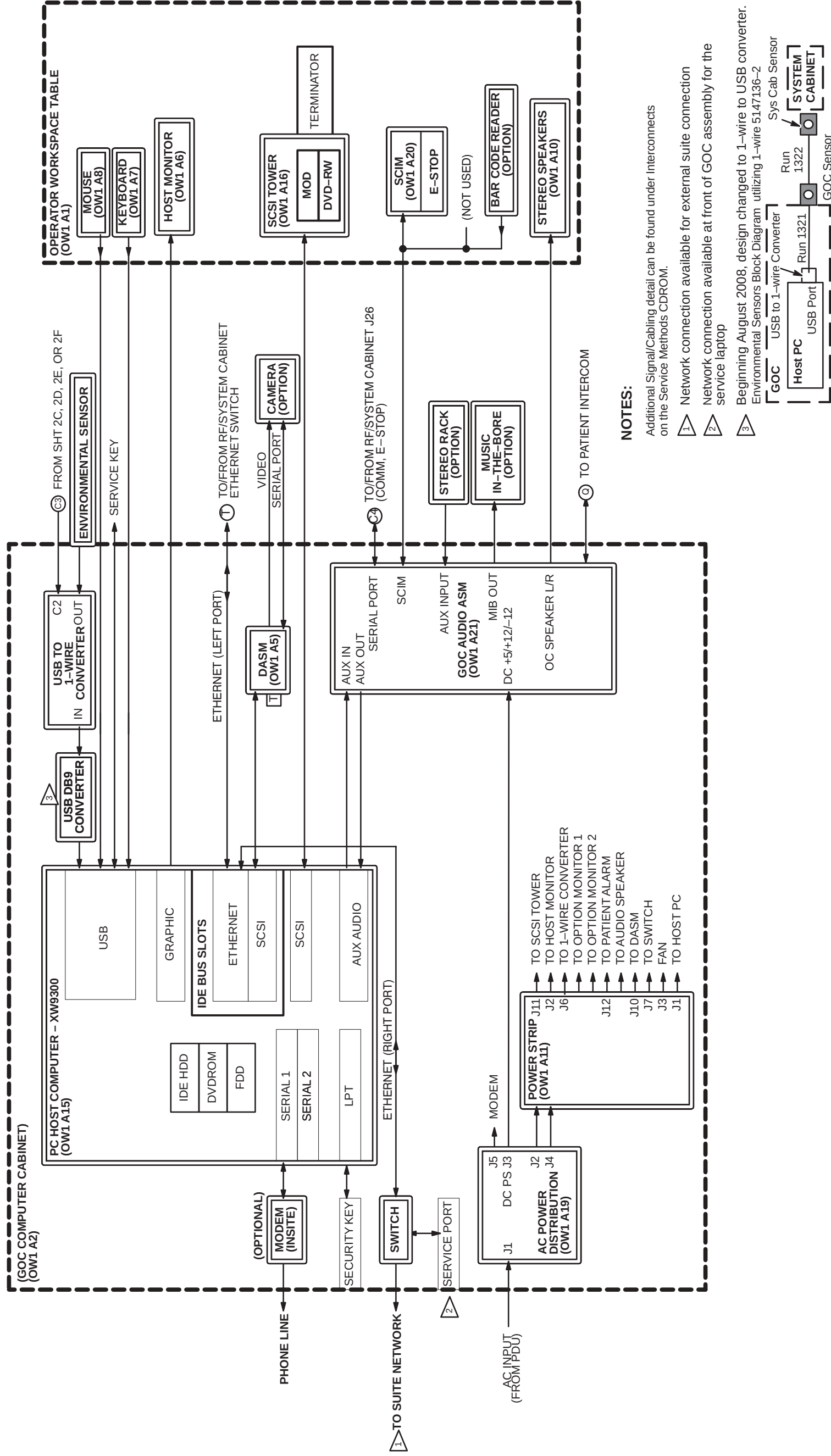


HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
 Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
 (OPERATOR WORKSPACE CABINET w/9300 & TABLE)

SHEET 1D OF 5

Refer to the following functional block diagram for additional information:

- NOTE:**
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF



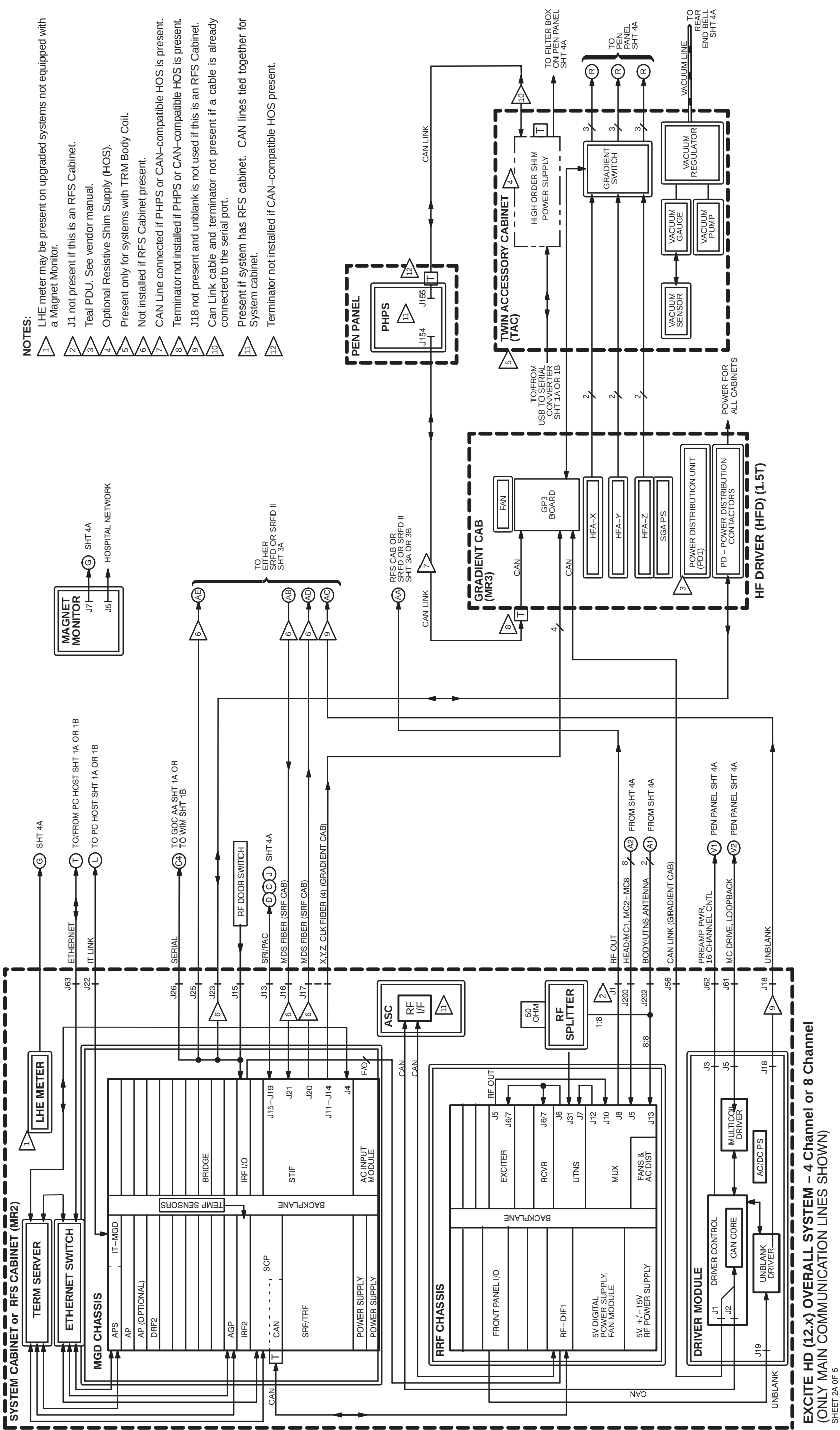
HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
EXCITE HD to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & TABLE)

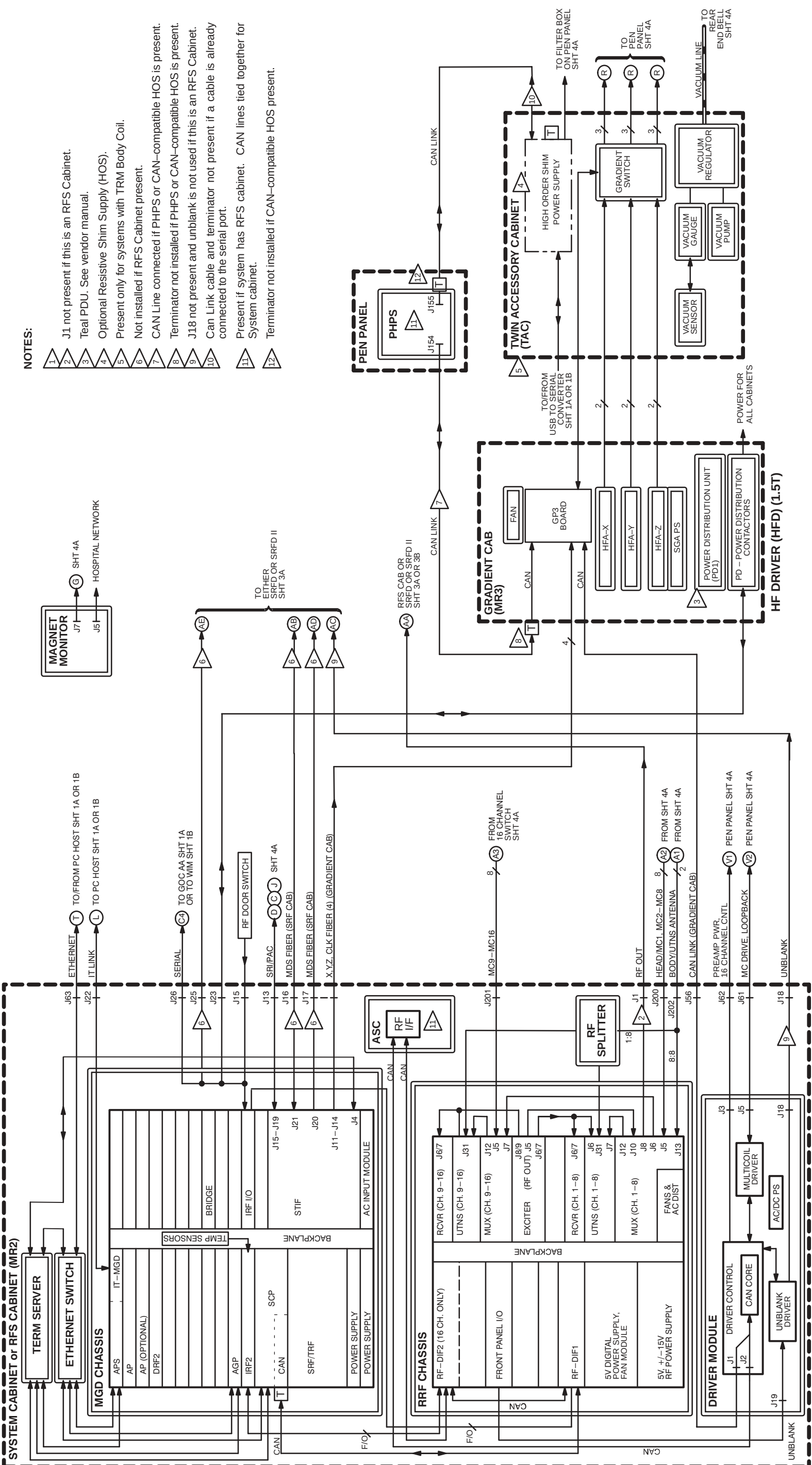
SHEET 1E OF 5

Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

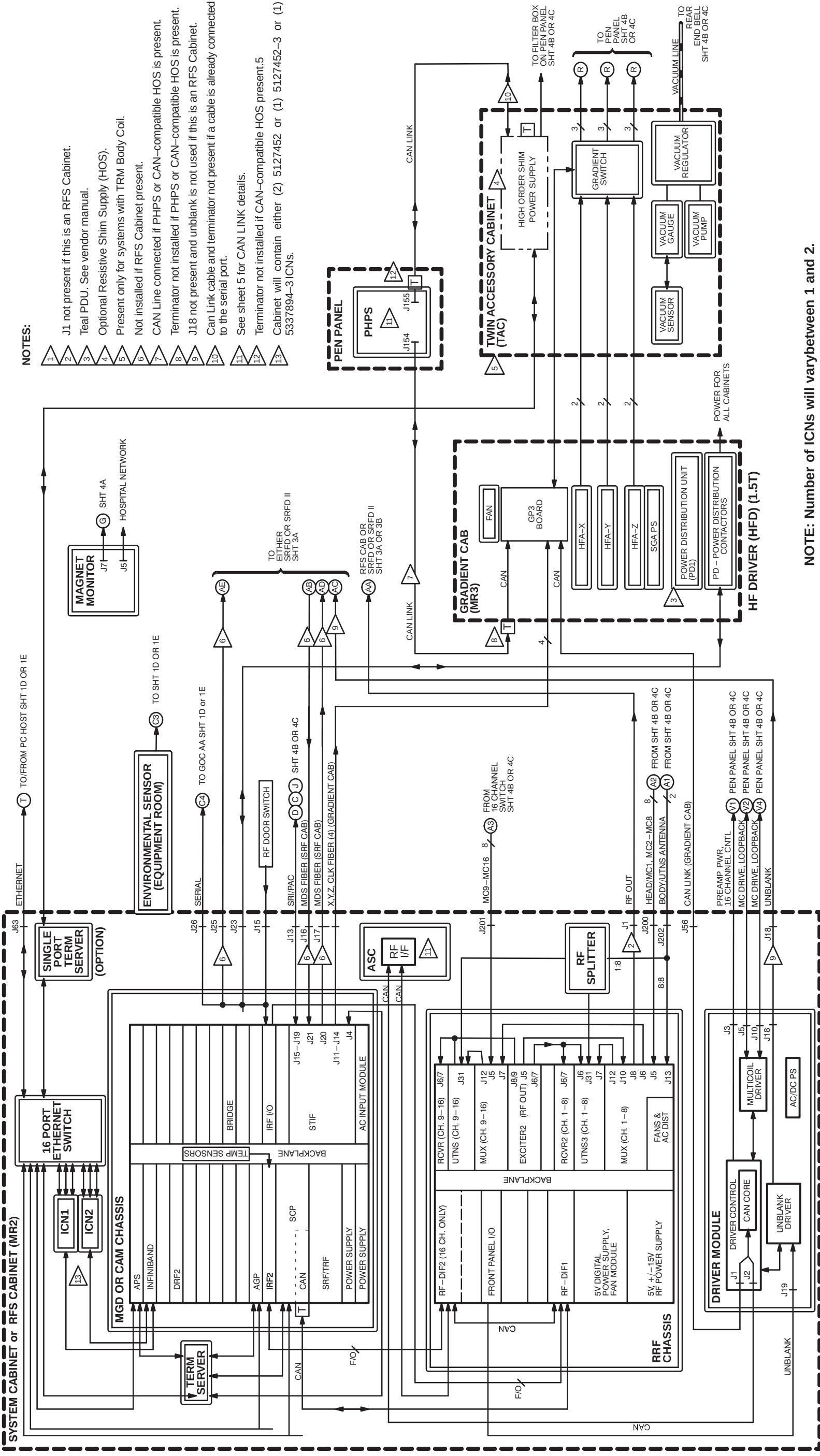
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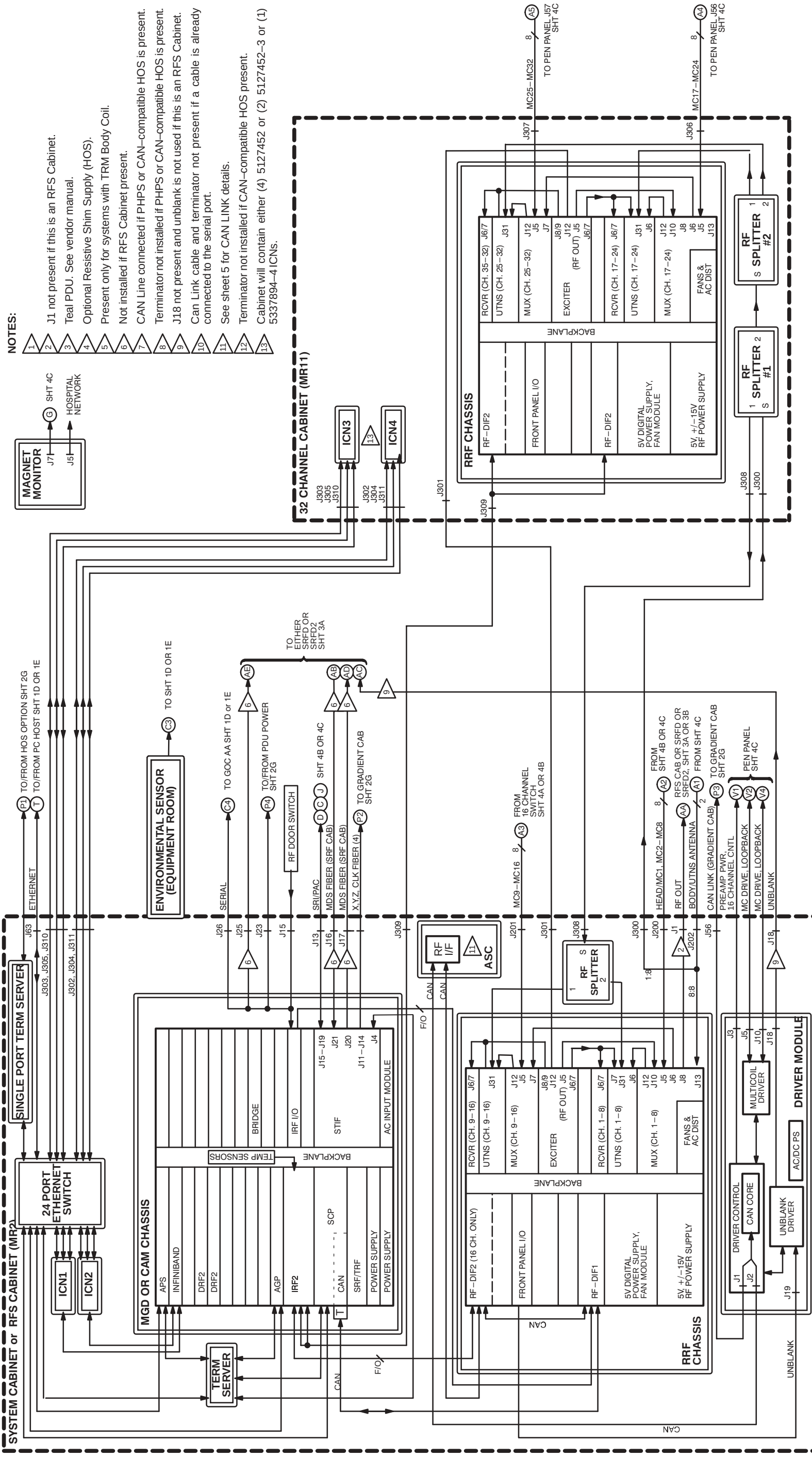
NOTES:

- 1 J1 not present if this is an RFS Cabinet.
- 2 Teal PDU. See vendor manual.
- 3 Optional Resistive Shim Supply (HOS).
- 4 Present only for systems with TRM Body Coil.
- 5 Not installed if RFS Cabinet present.
- 6 CAN Line connected if PHPS or CAN-compatible HOS is present.
- 7 Terminator not installed if PHPS or CAN-compatible HOS is present.
- 8 J18 not present and unblank is not used if this is an RFS Cabinet.
- 9 Can Link cable and terminator not present if a cable is already connected to the serial port.
- 10 Present if system has RFS cabinet. CAN lines tied together for System cabinet.
- 11 Terminator not installed if CAN-compatible HOS present.
- 12



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 16 Channel
 (ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2D OF 5

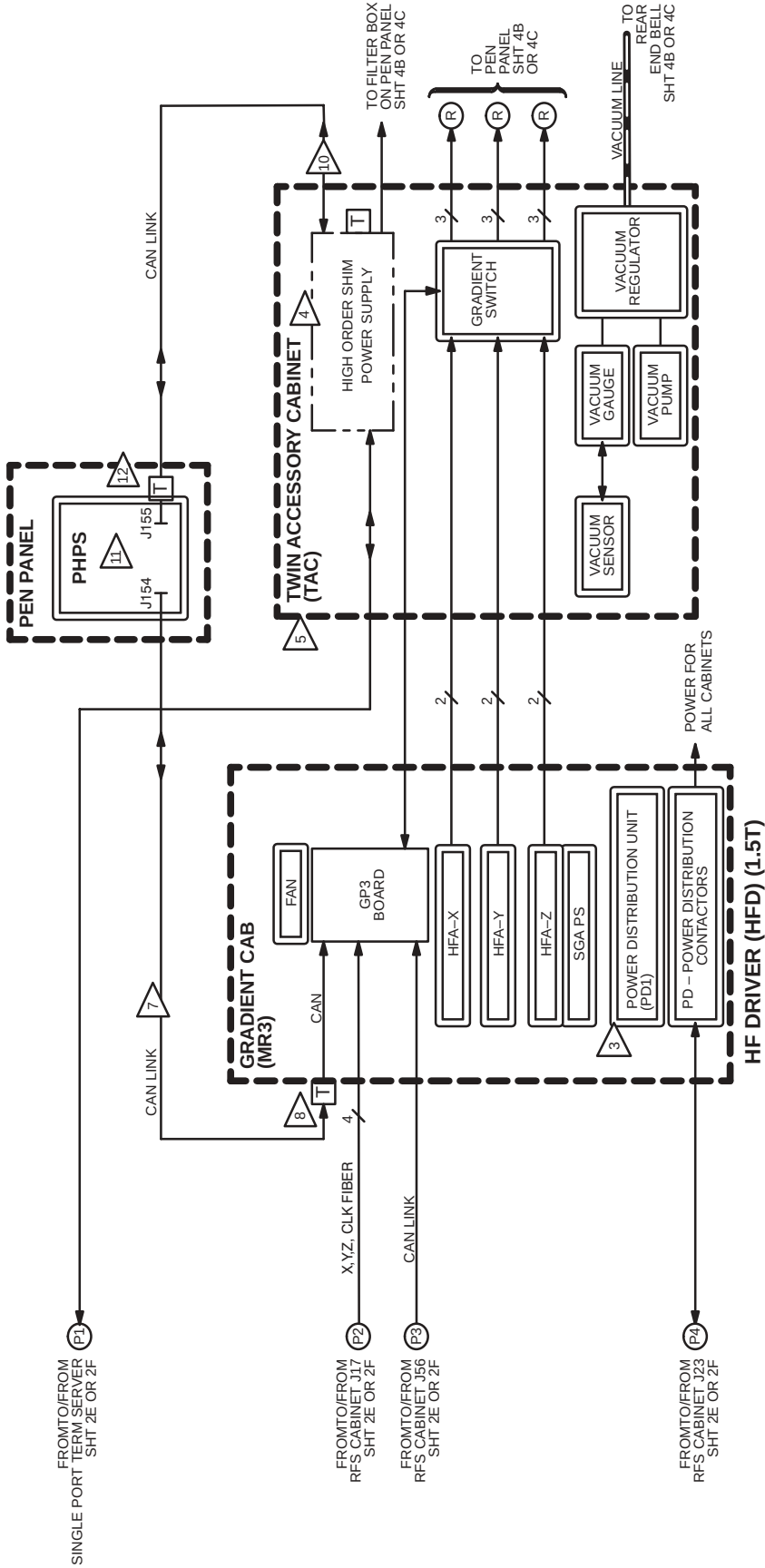


NOTES:

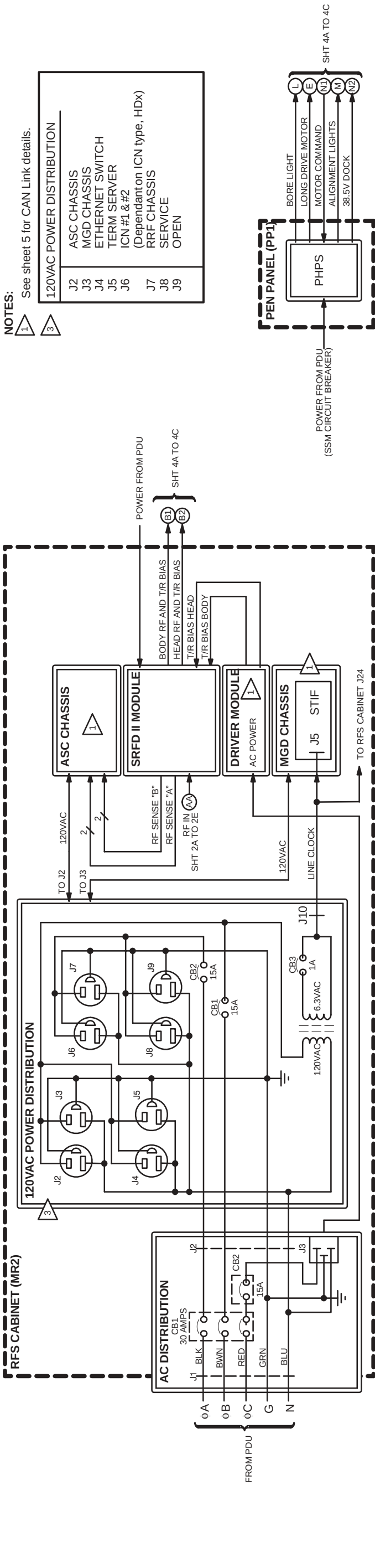
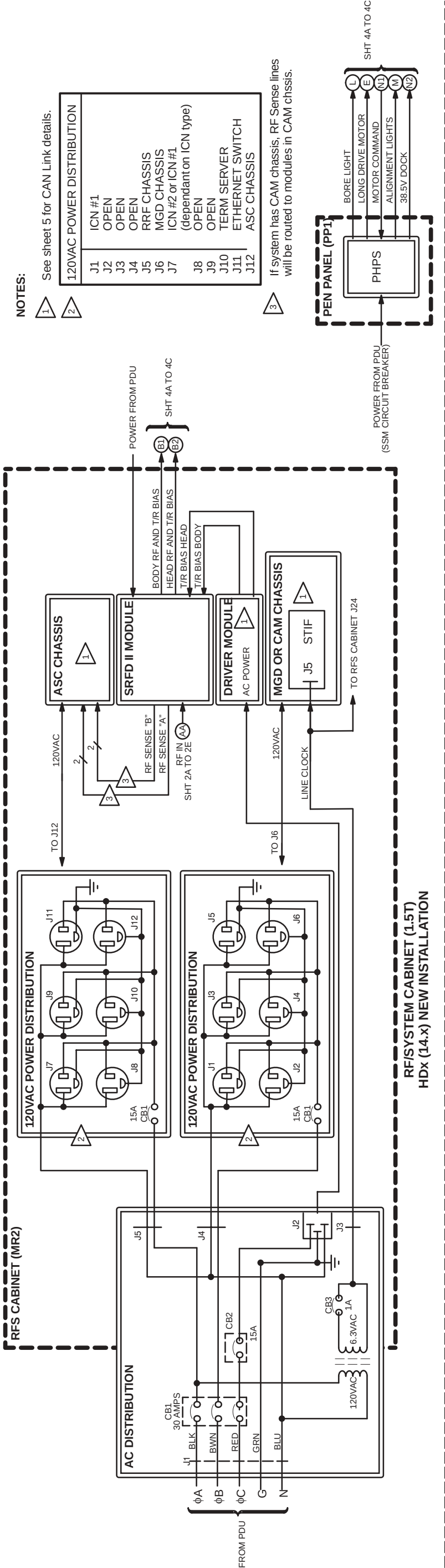
- 1 J1 not present if this is an RFS Cabinet.
- 2 Teal PDU. See vendor manual.
- 3 Optional Resistive Shim Supply (HOS).
- 4 Present only for systems with TRM Body Coil.
- 5 Not installed if RFS Cabinet present.
- 6 CAN Line connected if PHPS or CAN-compatible HOS is present.
- 7 Terminator not installed if PHPS or CAN-compatible HOS is present.
- 8 J18 not present and unblank is not used if this is an RFS Cabinet.
- 9 Can Link cable and terminator not present if a cable is already connected to the serial port.
- 10 See sheet 5 for CAN LINK details.
- 11 Terminator not installed if CAN-compatible HOS present.
- 12 Cabinet will contain either (4) 5127452 or (2) 5127452-3 or (1) 5337894-4 ICNs.

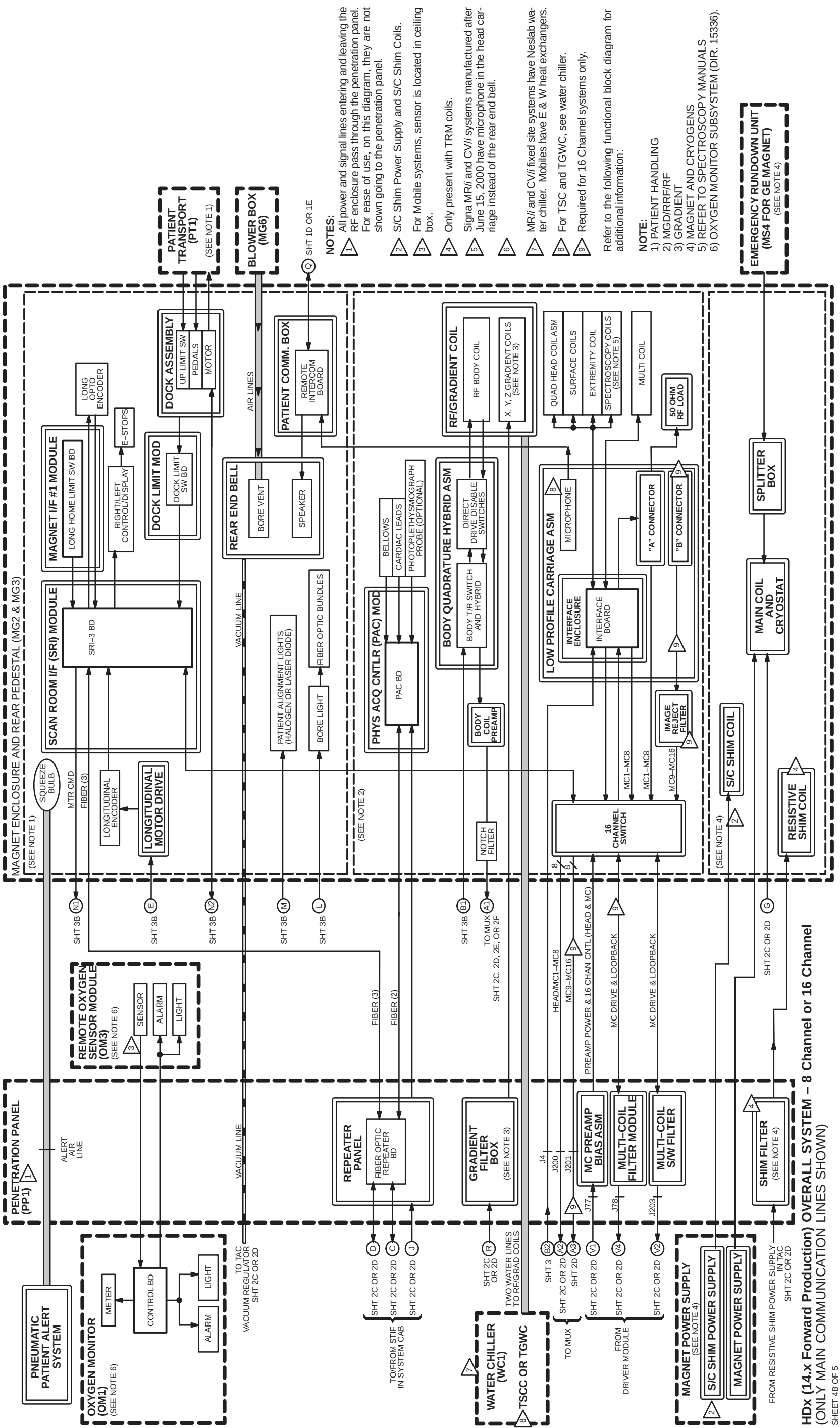
NOTE: Number of ICNs will vary between 1, 2 or 4.

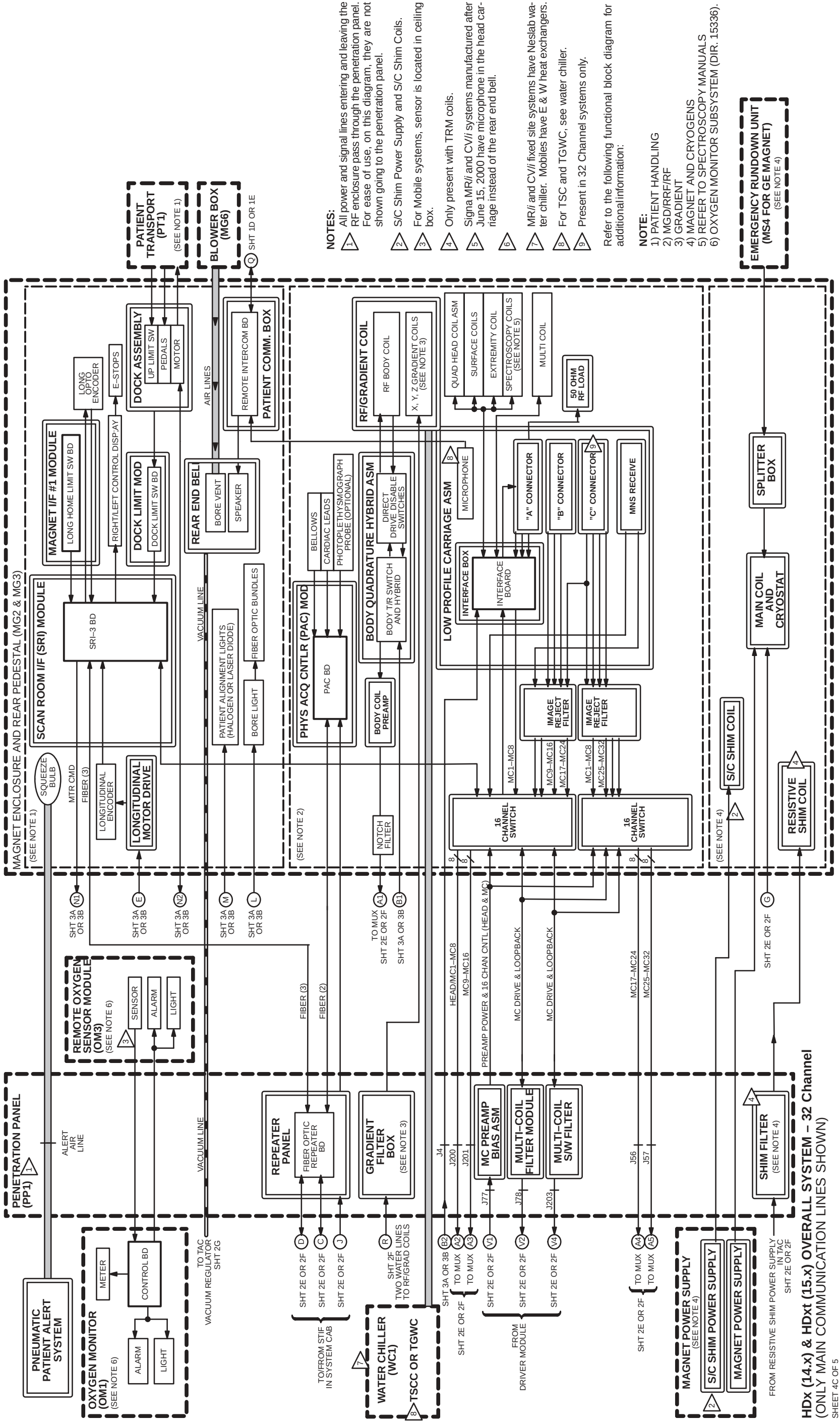
HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 32 Channel without Infinband Switch
(ONLY MAIN COMMUNICATION LINES SHOWN)



- NOTES:
- 4 Teal PDU. See vendor manual.
 - 5 Optional Resistive Shim Supply (HOS).
 - 10 Can Link cable and terminator not present if a cable is already connected to the serial port.
 - 11 PHPS present if system has RFS cabinet.
 - 12 Terminator not installed if CAN-compatible HOS present.







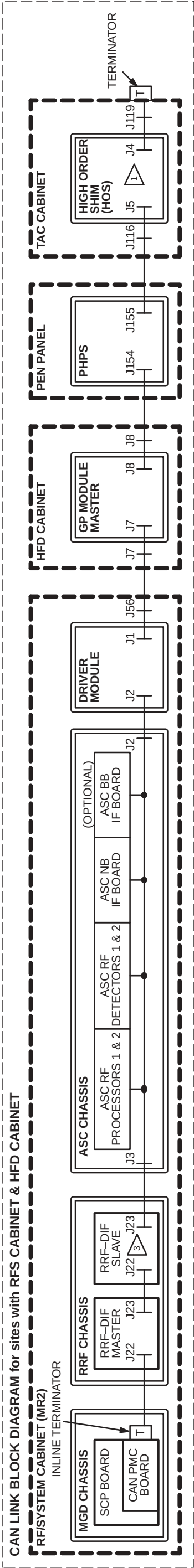
NOTES:

- 1 All power and signal lines entering and leaving the RF enclosure pass through the penetration panel. For ease of use, on this diagram, they are not shown going to the penetration panel.
- 2 S/C Shim Power Supply and S/C Shim Coils.
- 3 For Mobile systems, sensor is located in ceiling box.
- 4 Only present with TRM coils.
- 5 Signa MR/i and CV/i systems manufactured after June 15, 2000 have microphone in the head carriage instead of the rear end bell.
- 6 MR/i and CV/i fixed site systems have Neslab water chiller. Mobiles have E & W heat exchangers.
- 8 For TSC and TGWC, see water chiller.
- 9 Present in 32 Channel systems only.

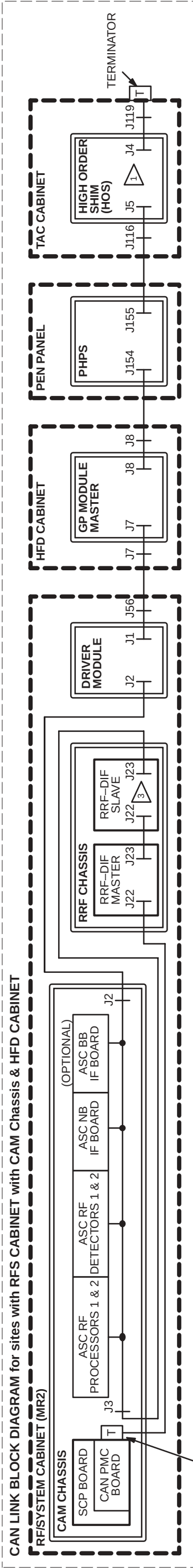
Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/IRRF/RF
- 3) GRADIENT
- 4) MAGNET AND CRYOGENS
- 5) REFER TO SPECTROSCOPY MANUALS
- 6) OXYGEN MONITOR SUBSYSTEM (DIR. 15336).

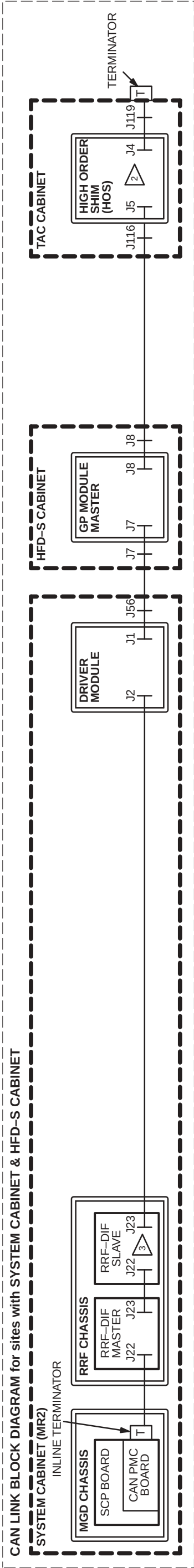


OR



INLINE TERMINATOR

OR



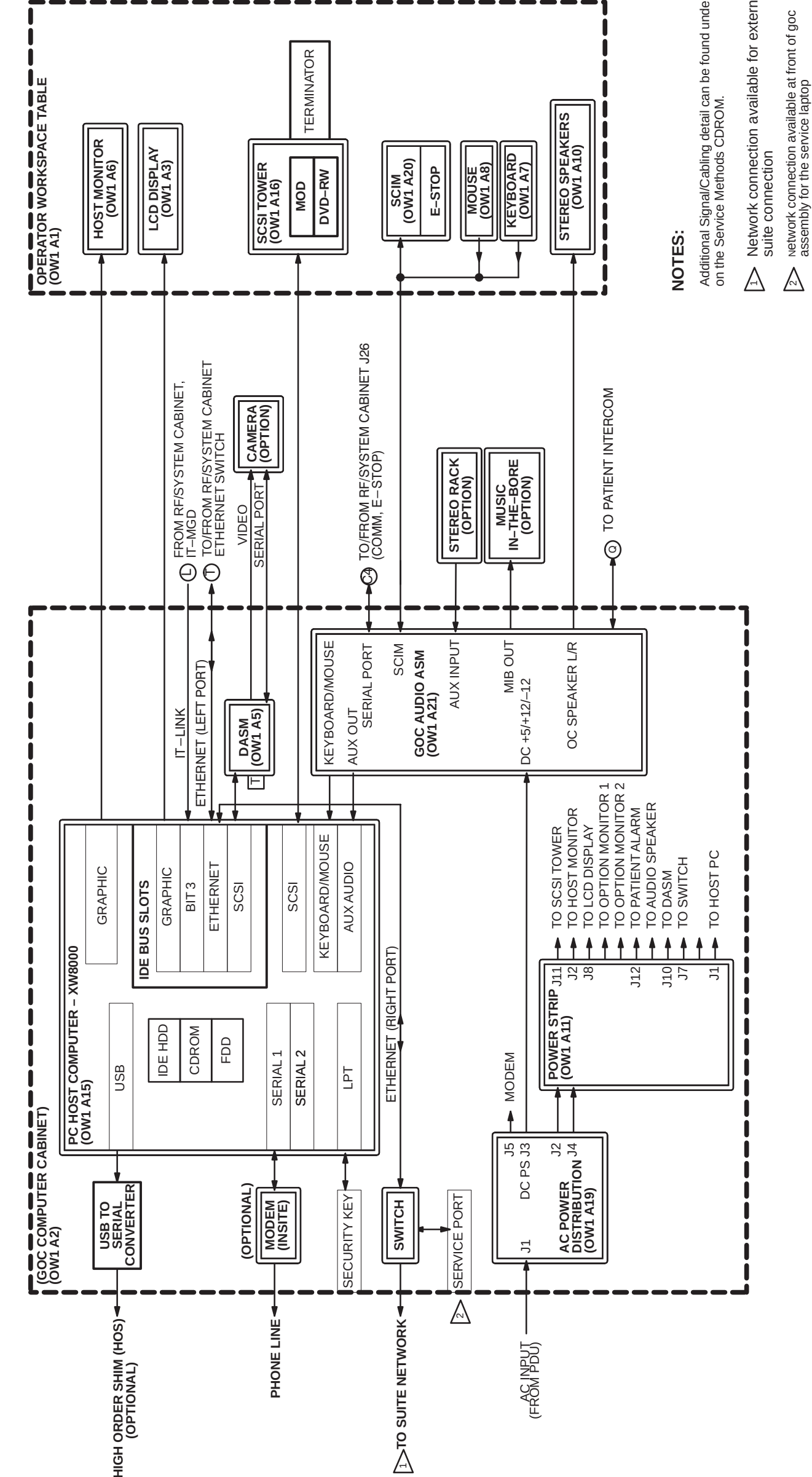
NOTES:

- 1 HOS is optional. Terminator must be connected to J155 on PHPS if HOS not present.
- 2 HOS is optional. Terminator must be connected to J8 on HFD Cabinet /IF if HOS not present.
- 3 Only present in systems equipped with 16 Channels. If system is 8 Channels or less, then RF-DIF Master J23 connects to next unit in link as shown.

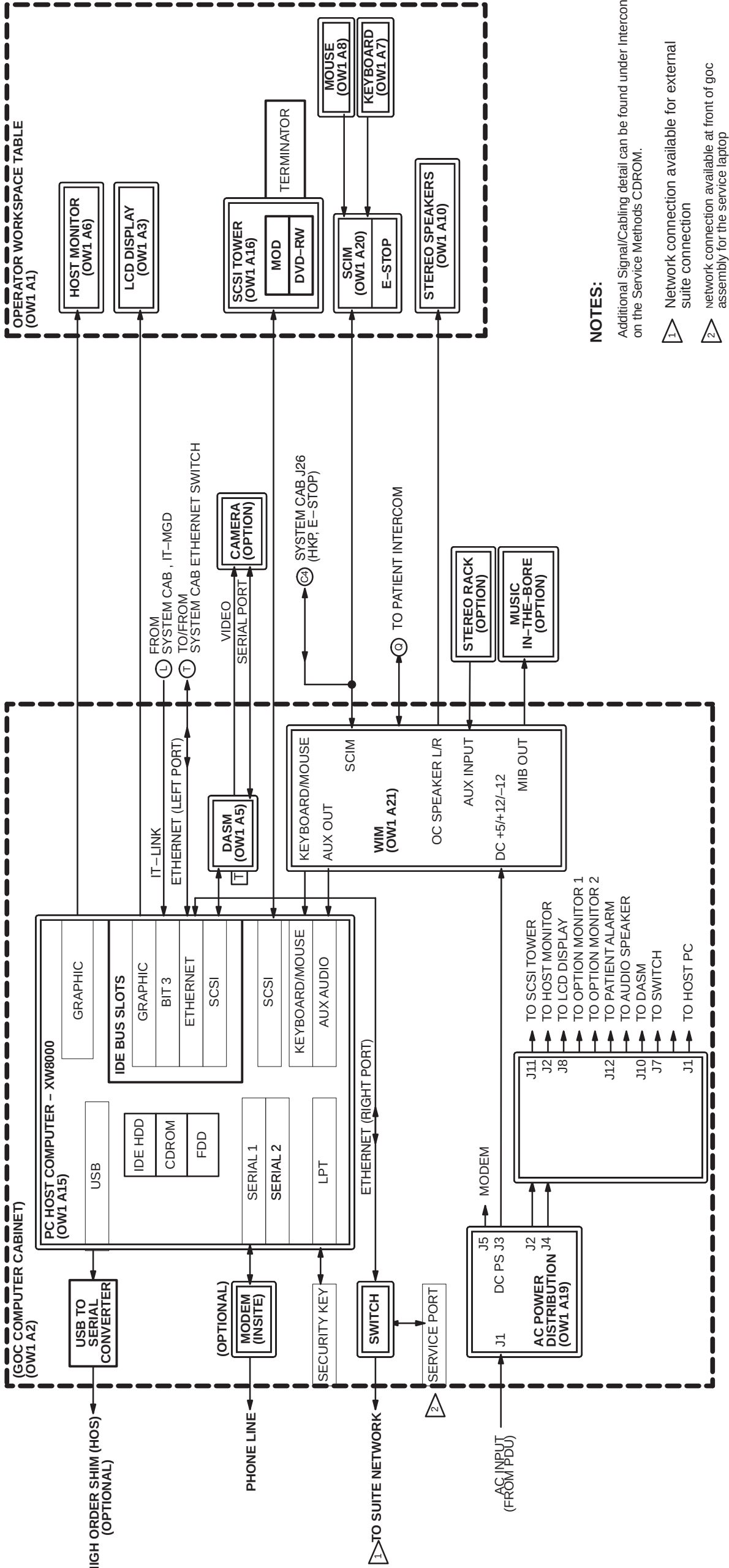
OPERATOR WORKSPACE

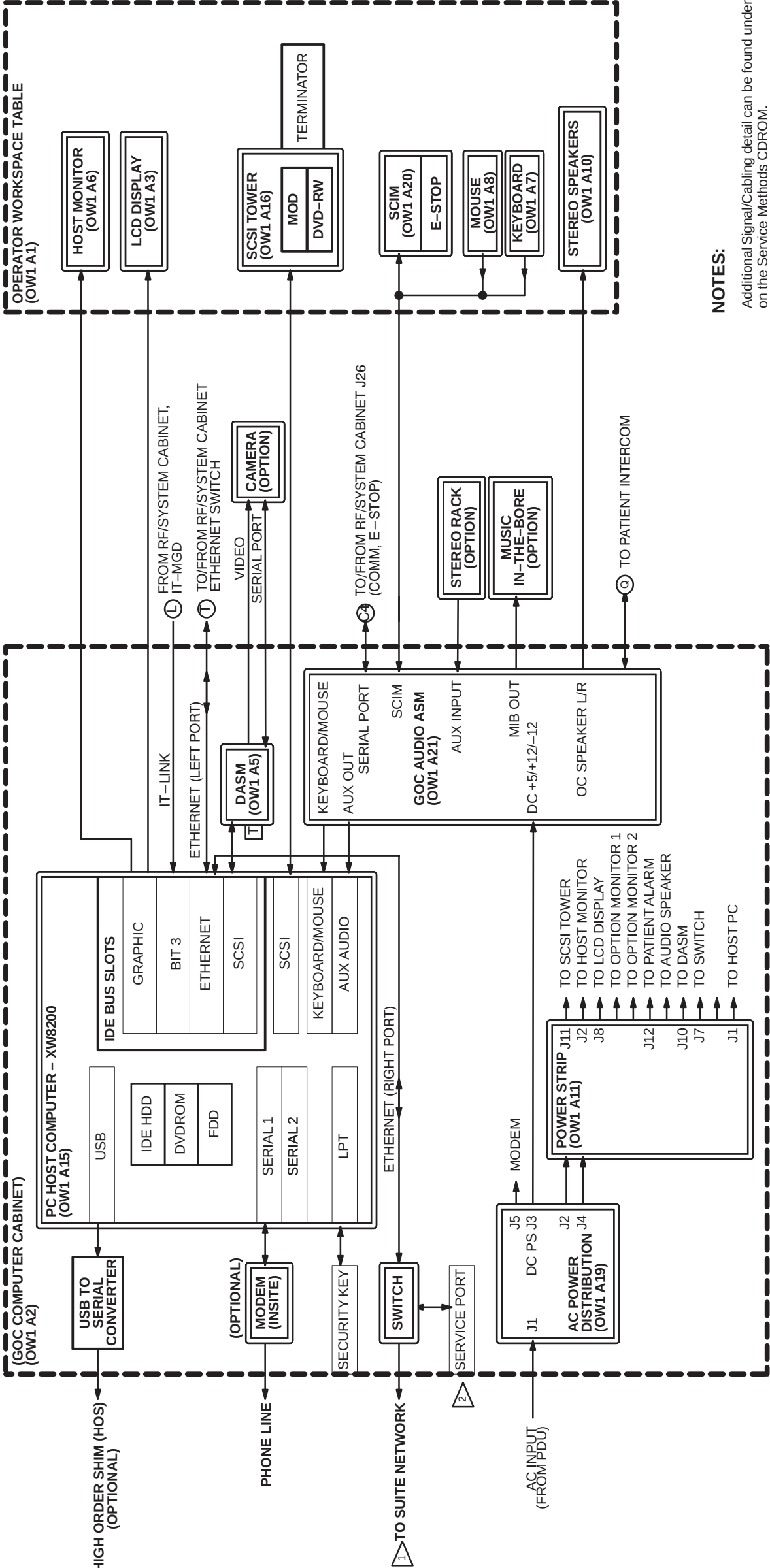
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EXCITE HD (12.x) OVERALL SYSTEM
 Forward Production or Pre-EXCITE to HD Upgrades
 (OPERATOR WORKSPACE CABINET w/8000 & TABLE)
 SHEET 1A OF 5





NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDFROM.

1 Network connection available for external suite connection

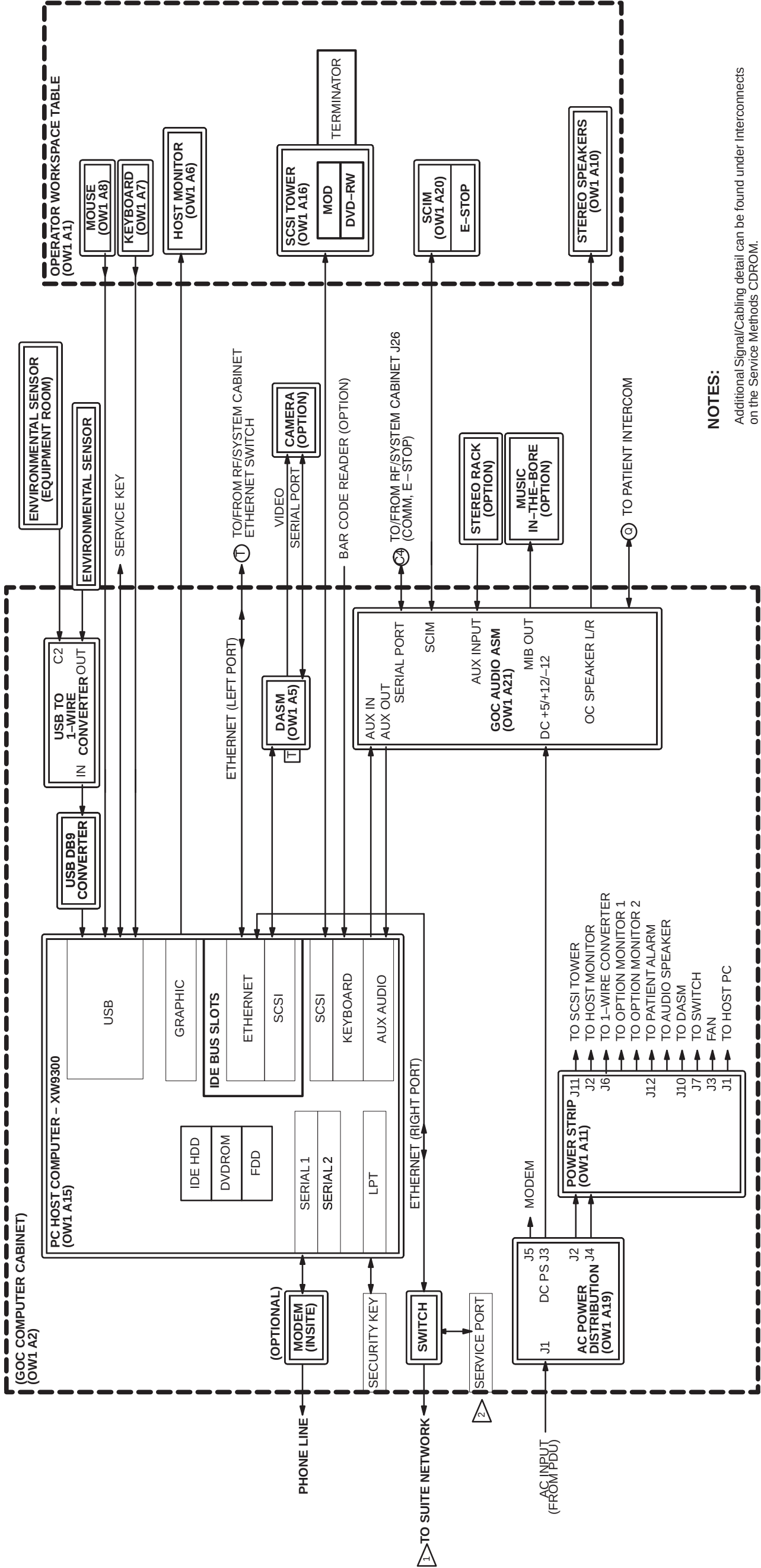
2 Network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

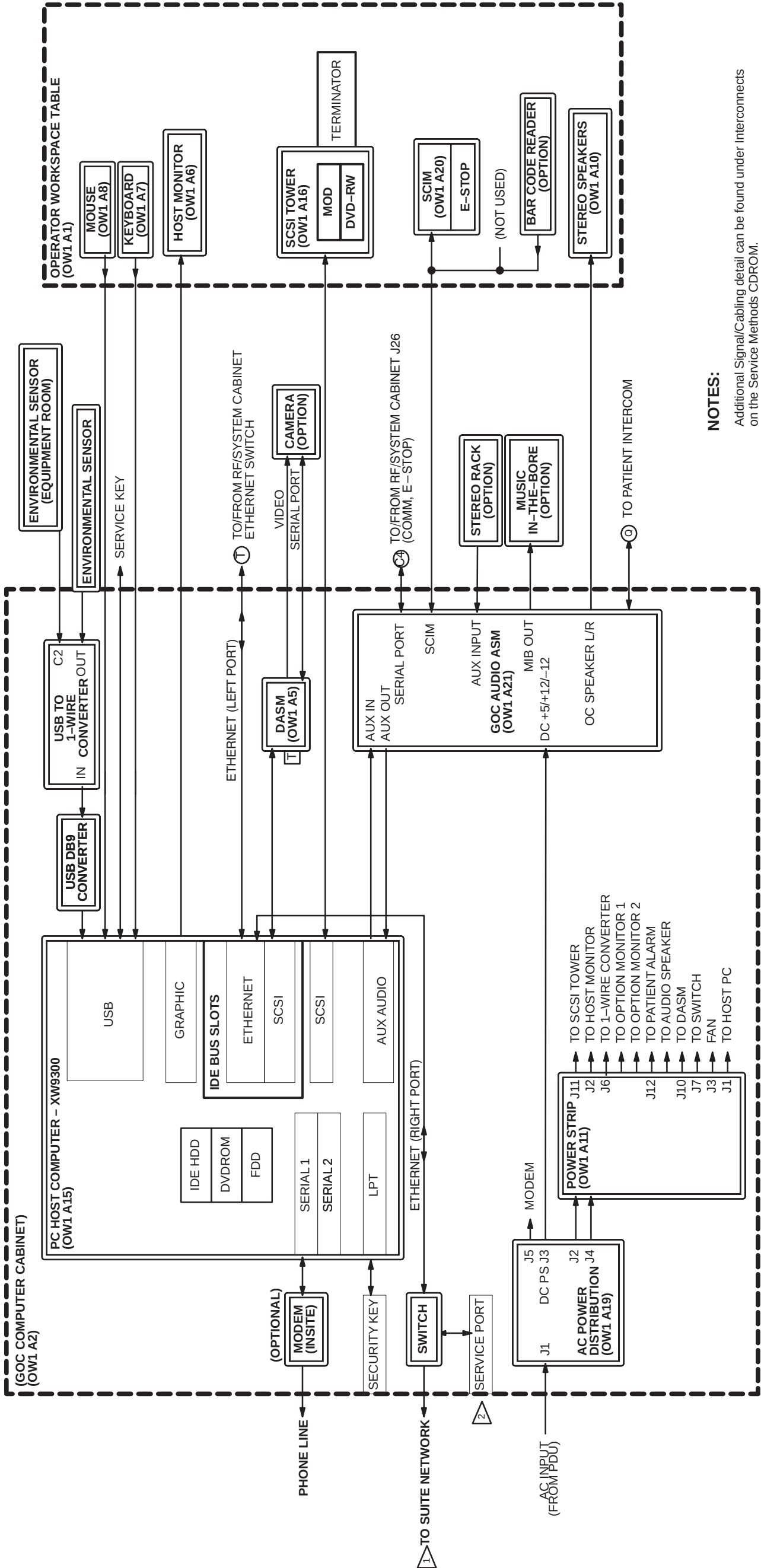
NOTE:

1) PATIENT HANDLING

2) MGD/RRF/RF



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
 Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
 (OPERATOR WORKSPACE CABINET w/9300 & TABLE)

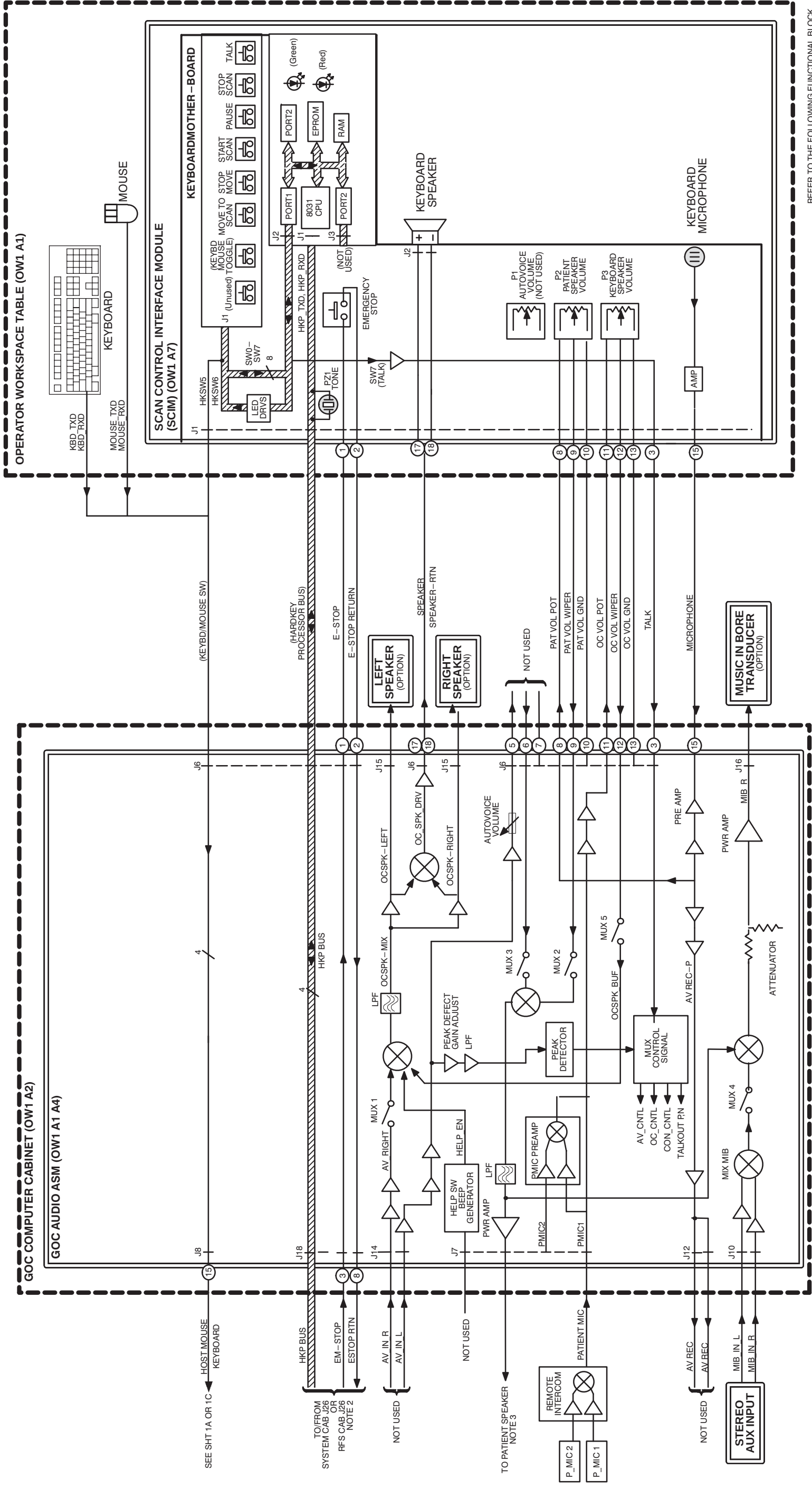


NOTES:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection
2 Network connection available at front of GOC assembly for the service laptop

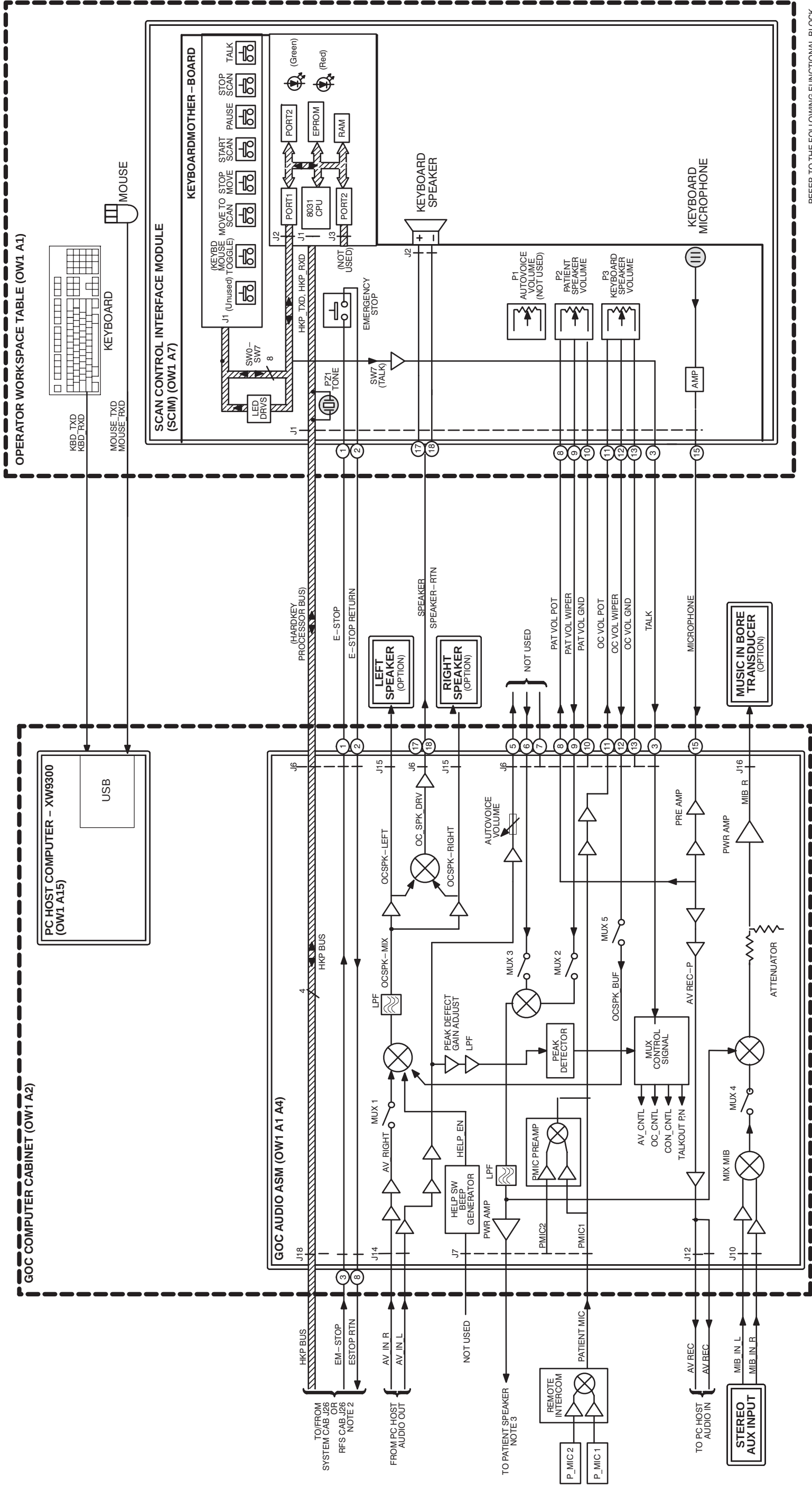
Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF



EXCITE HD (12.x) OVERALL SYSTEM
 (OPERATOR WORKSPACE CABINET/TABLE with GOC AUDIO ASM)

SHEET 2B OF 2



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
 (OPERATOR WORKSPACE CABINET/TABLE with GOC AUDIO ASM)

SHEET 2C OF 2

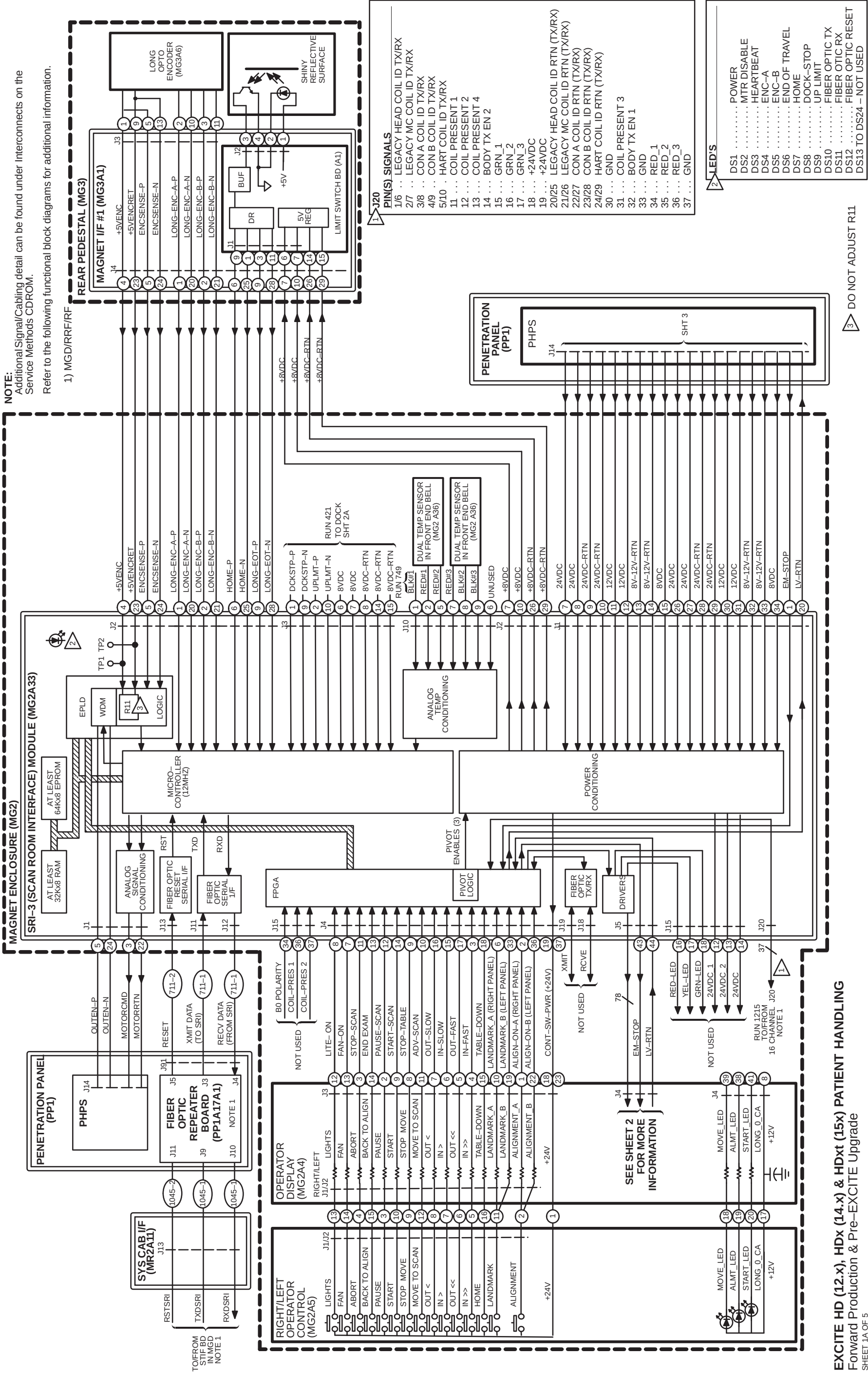
REFER TO THE FOLLOWING FUNCTIONAL BLOCK
 DIAGRAM FOR ADDITIONAL INFORMATION:

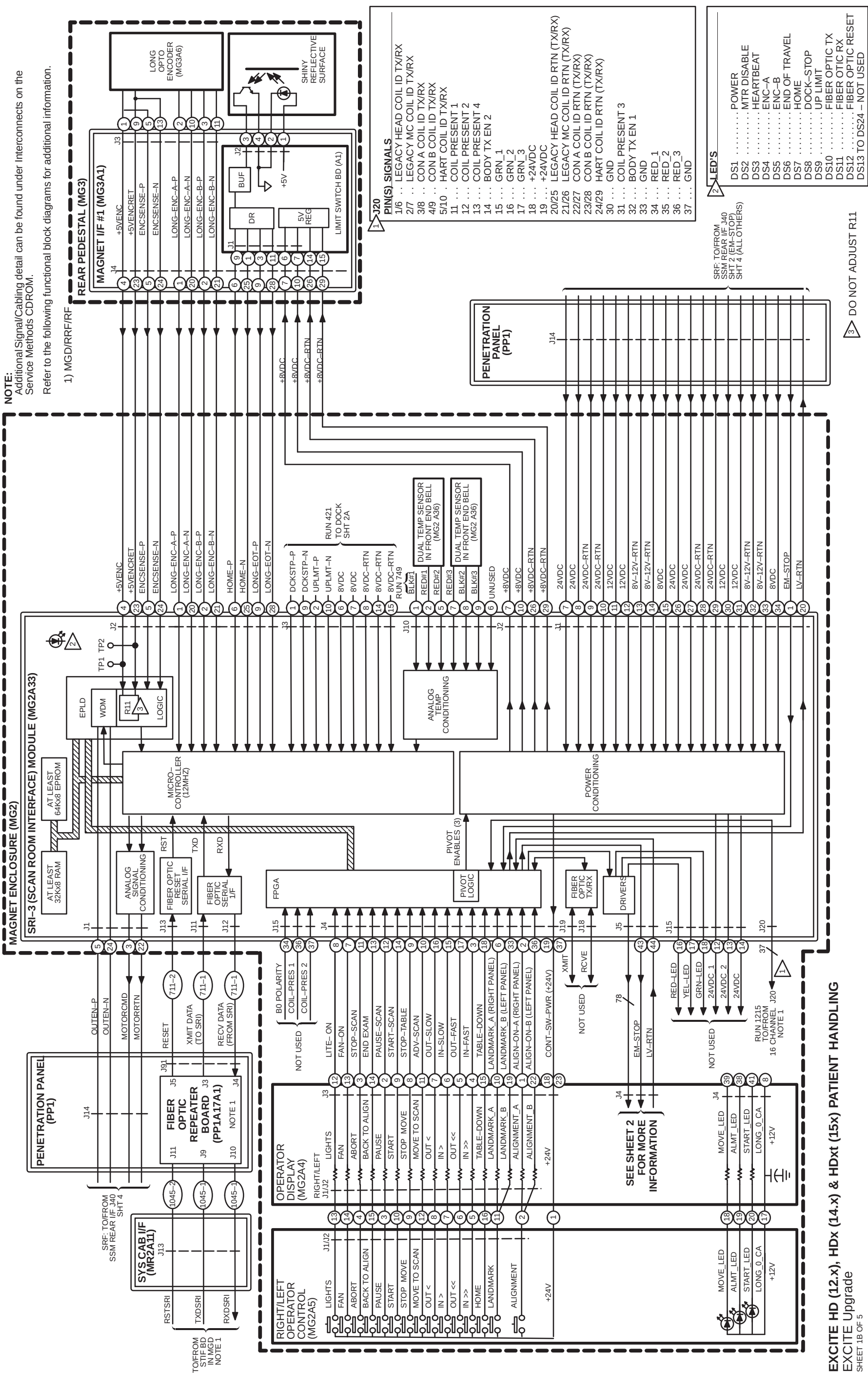
- NOTE:
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF
 - 3) OVERALL SYSTEM

PATIENT HANDLING

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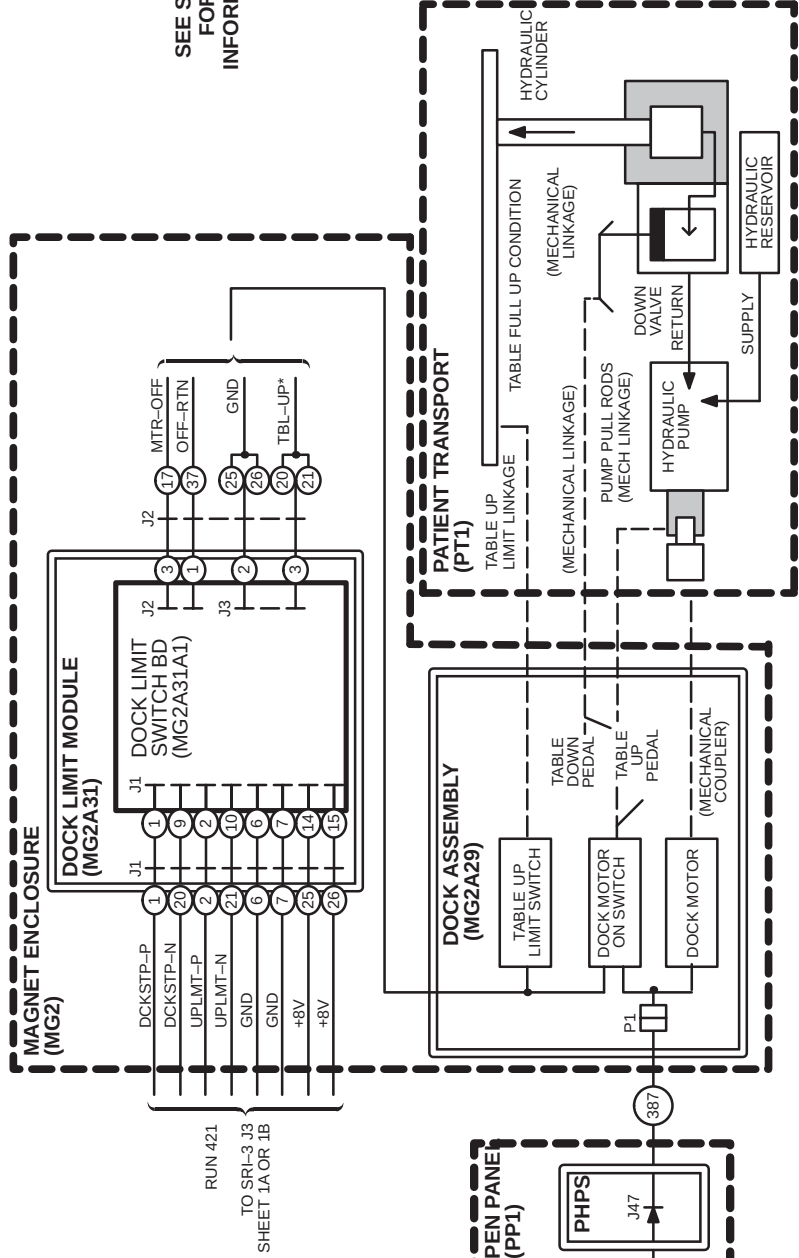
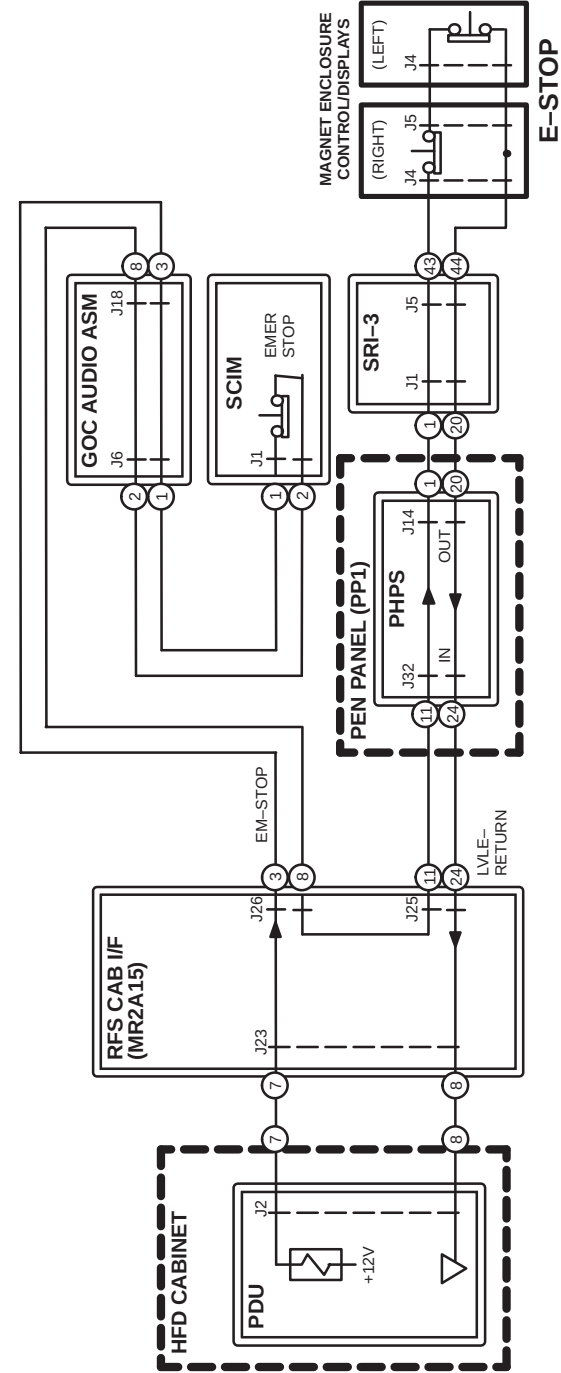




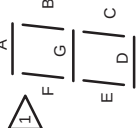
DO NOT ADJUST R11

PATIENT HANDLING

1-2

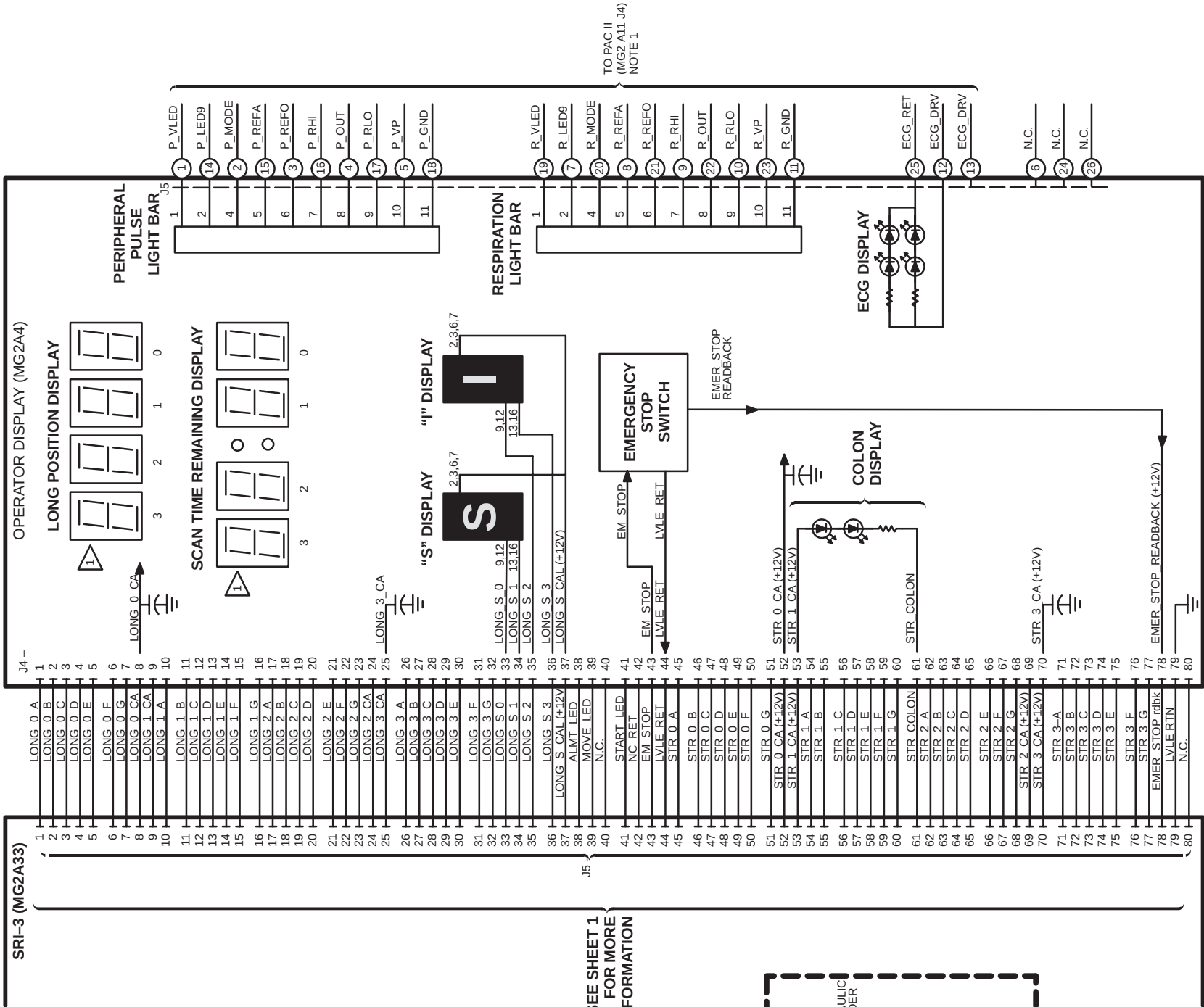


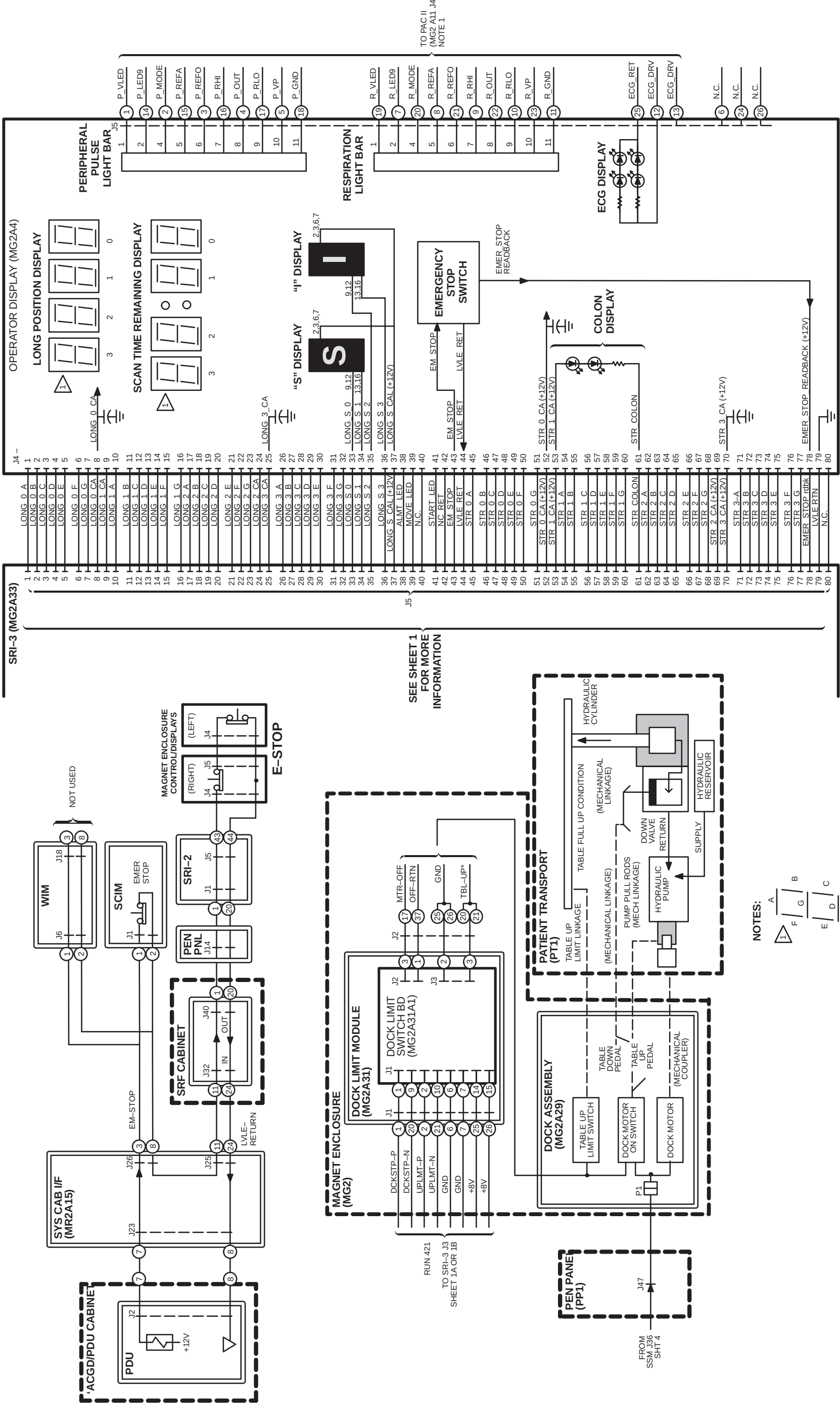
NOTES:



Refer to the following functional block diagrams for additional information. :

1) MGD/RRF/RF







Refer to the following Functional Block Diagrams for additional information:

- 1** **OPTIONAL:** Used only if Patient Handling Power Supply (PHPS) is not installed. J8 terminator removed when PHPS is installed.

- NOT
USED



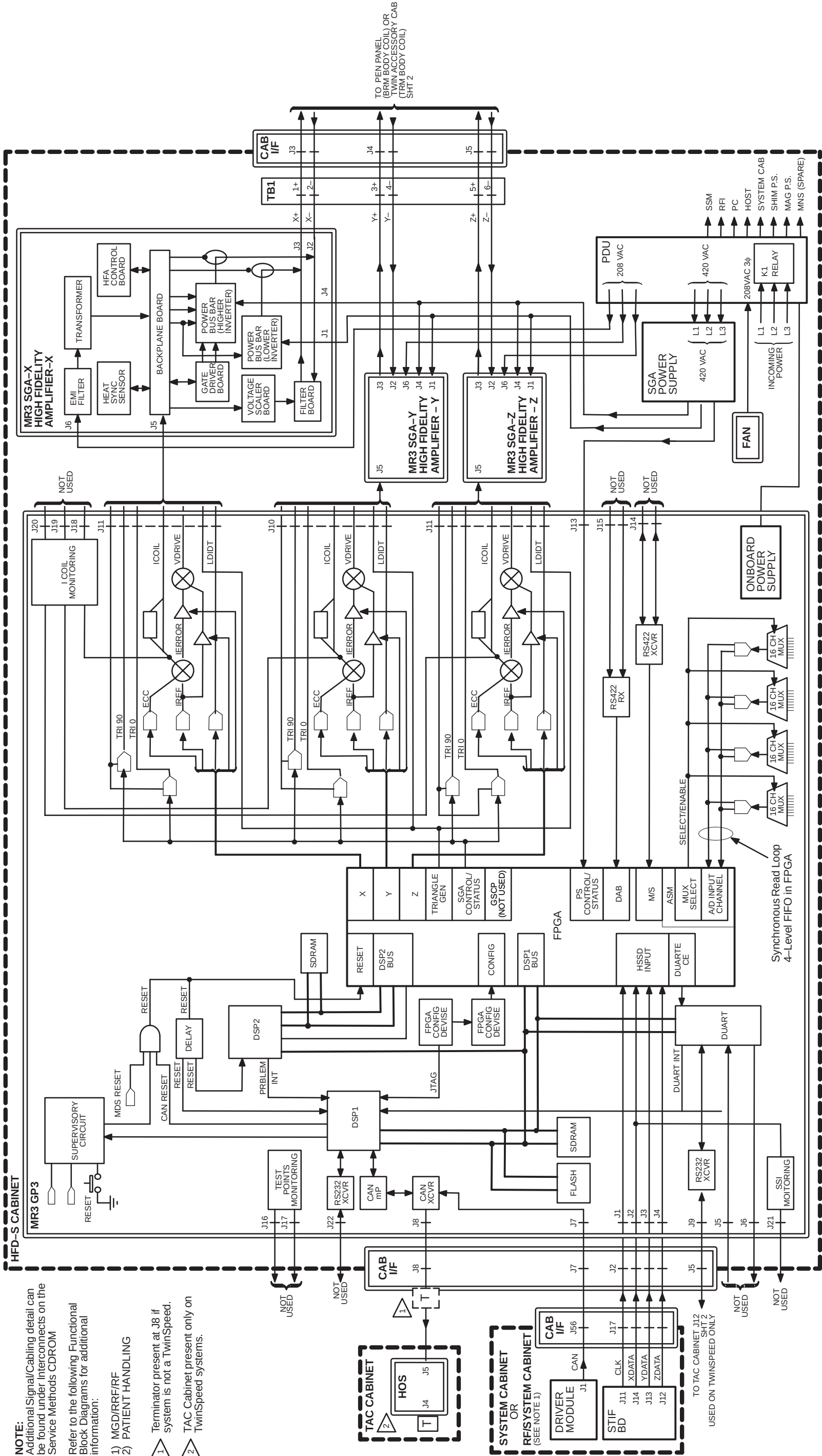
NOTE:
 Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM

Refer to the following Functional Block Diagrams for additional information:

- MGD/RRF/RF
- PATIENT HANDLING

Terminator present at J8 if system is not a TwinSpeed.

TAC Cabinet present only on TwinSpeed systems.



PDU

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Note: PDU found in Vendor Manuals only. (SEE SERVICE METHODS CDROM)

MGD/RRF/RF (1.5T)

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NOTE:

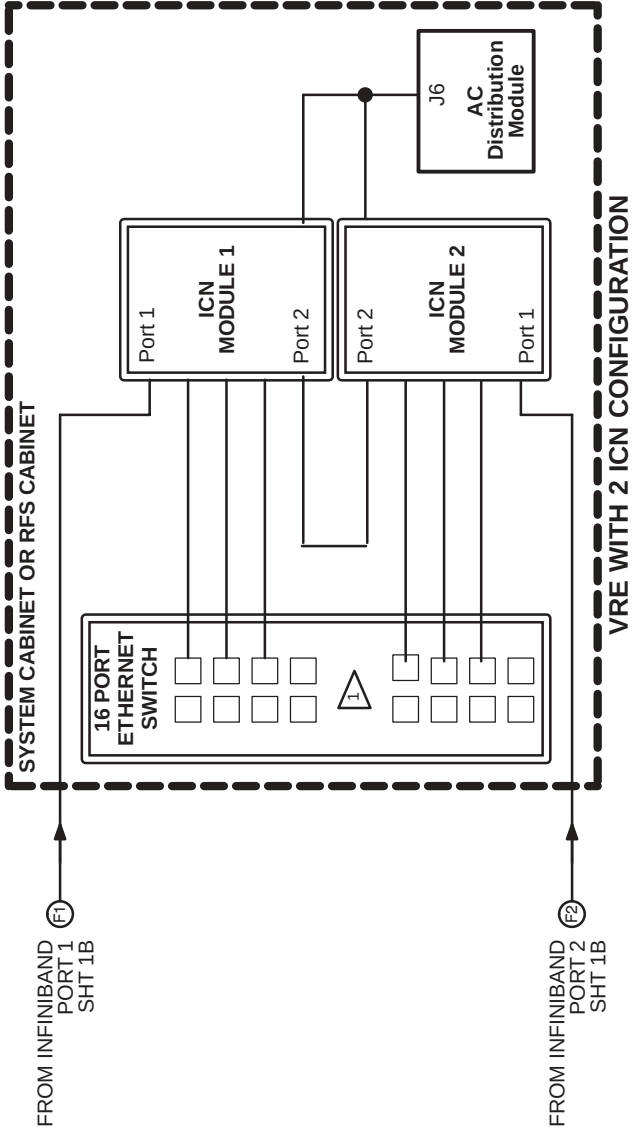
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

2 – MGD/RRF/RF SUBSYSTEM SUPPLEMENTAL SCHEMATICS (1.5T)

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T/R BOARD SCHEMATIC (2133404S)	2–6
MULTICOIL MODULE (2295846)	2–7
MOTHER BOARD (2295860, 5 SHEETS)	2–8

NOTE:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

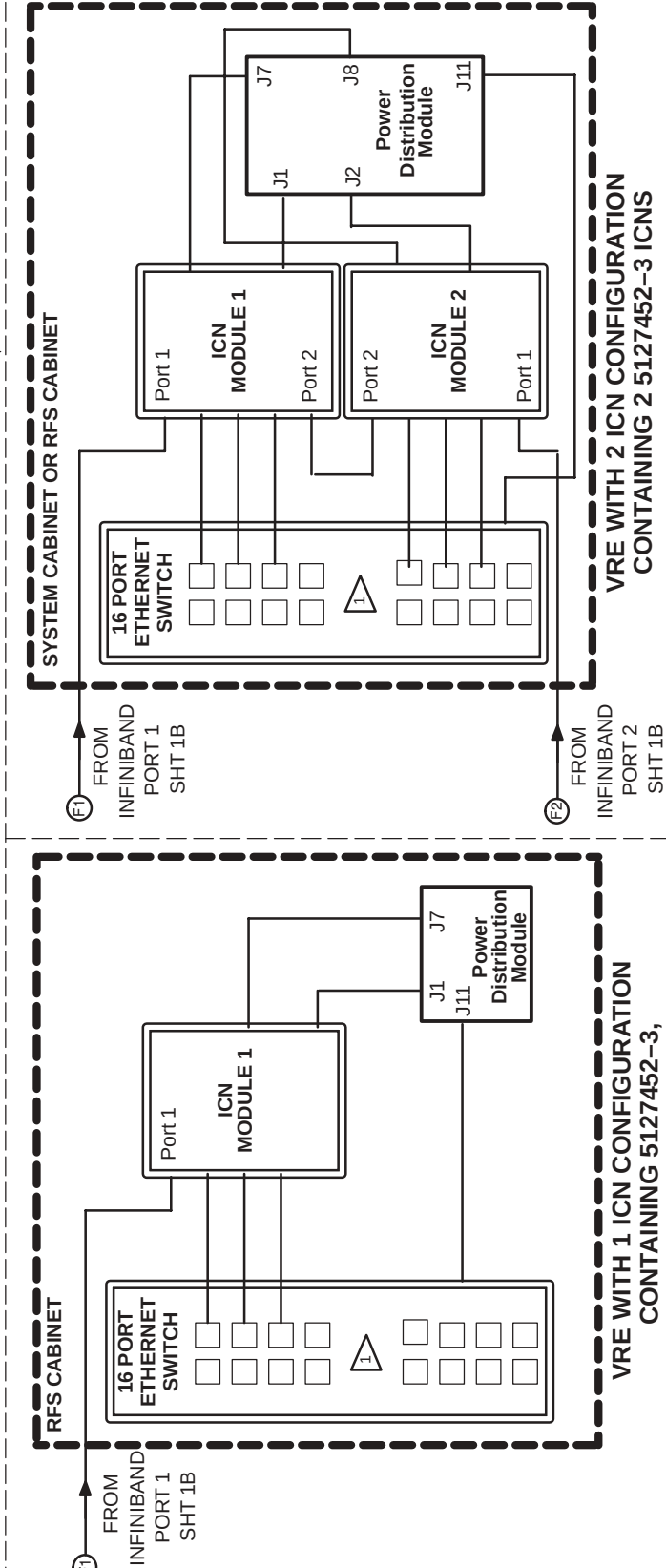


NOTES:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.
Refer to following Block Diagrams for additional information:
1)OPERATOR WORKSPACE

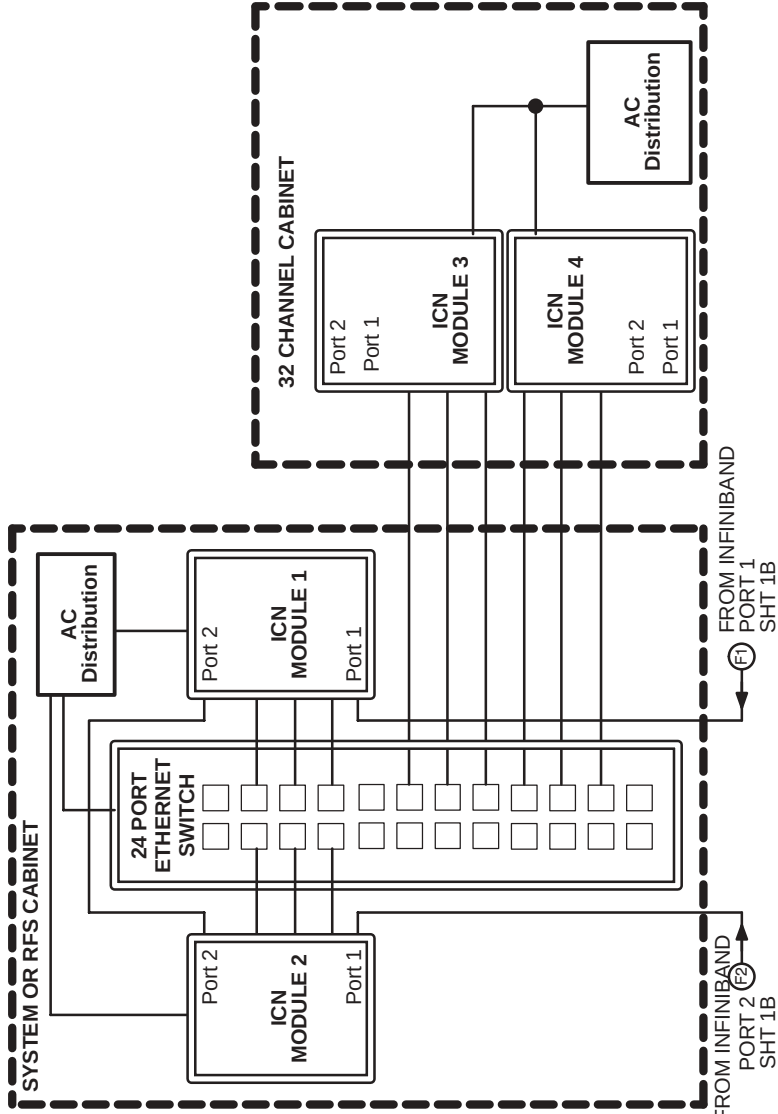
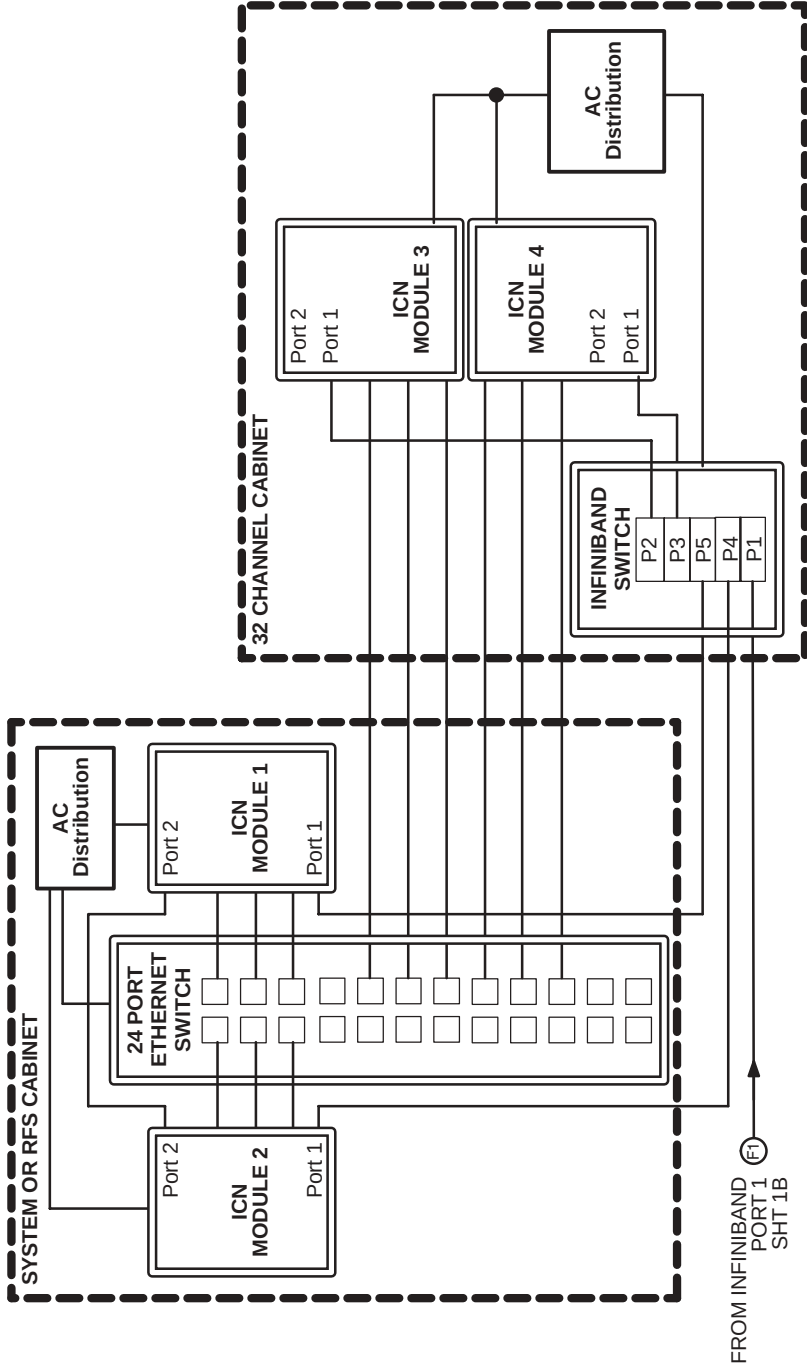


VRE WITH 2 ICN CONFIGURATION

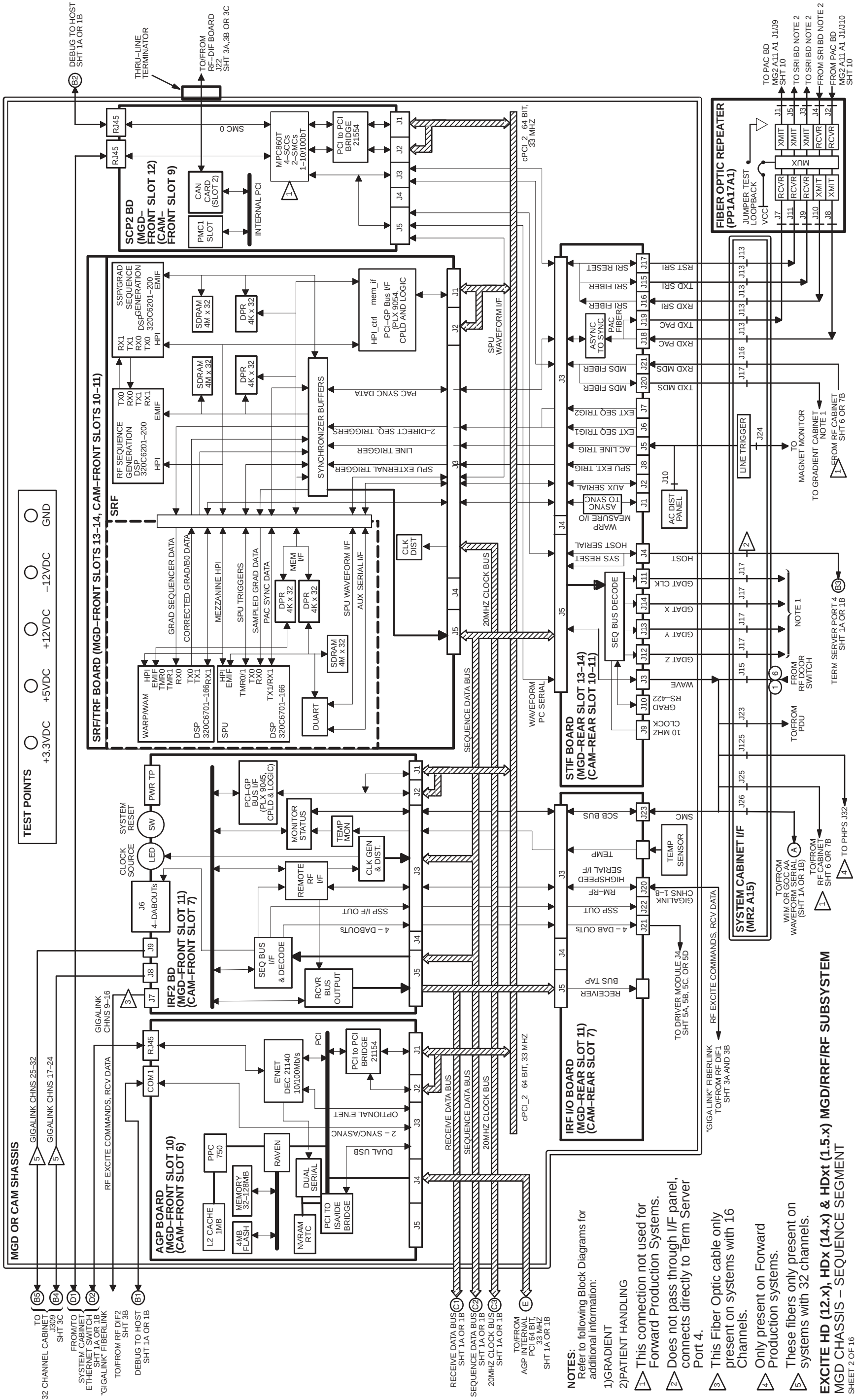
VRE WITH 4 ICN CONFIGURATION WITH INFINIBAND SWITCH



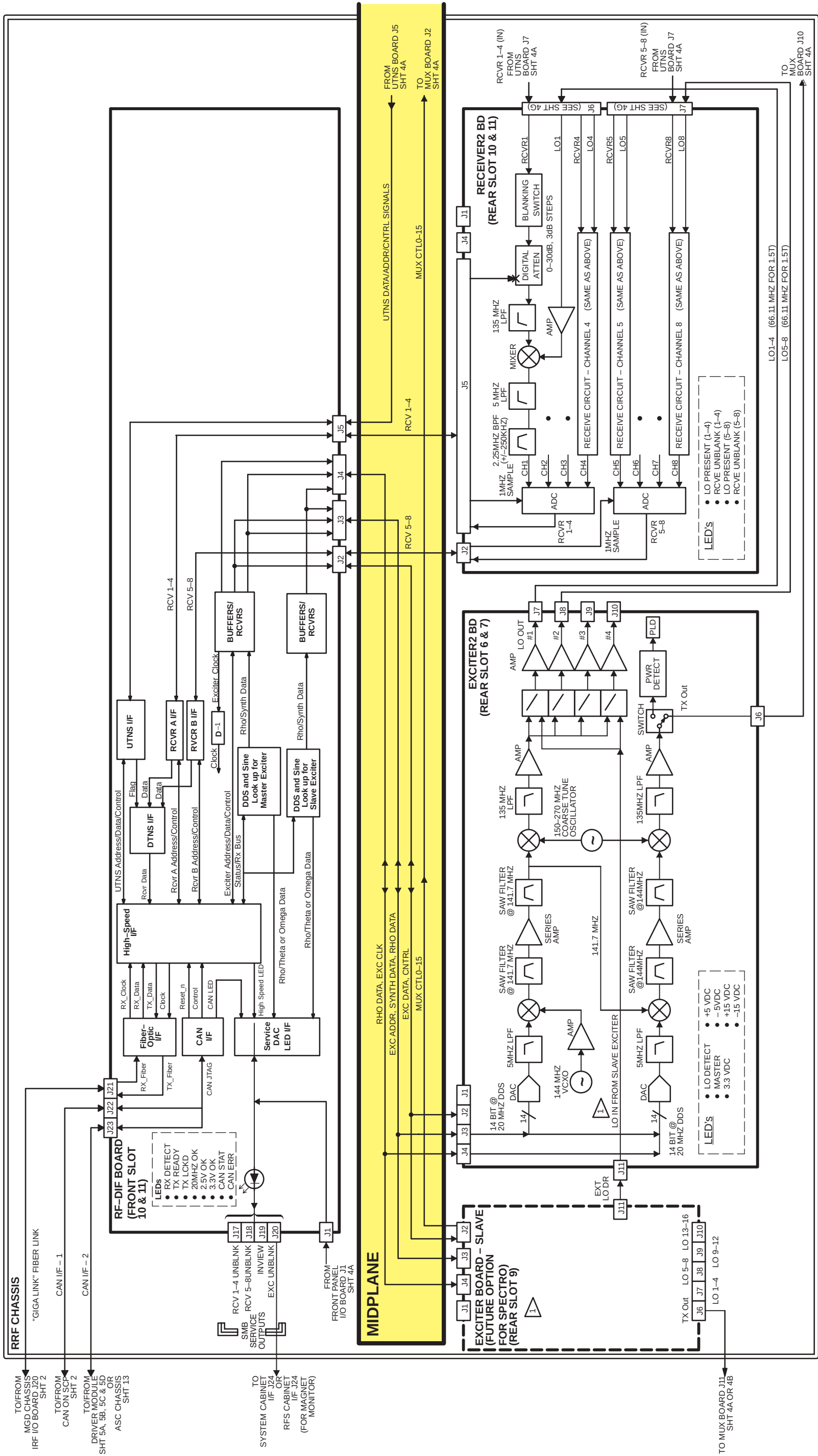
HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(Forward Production & Upgrades)
VOLUME RECON ENGINE
SHEET 1C OF 16



VRE WITH 4 ICN CONFIGURATION WITHOUT INFINIBAND SWITCH



**Signa HD, HDx & HDxt 1.5T (12.x, 14.x & 15.x)
Block Diagrams & Supplemental Schematics**



EXCITE HD (12.x), HDX (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM

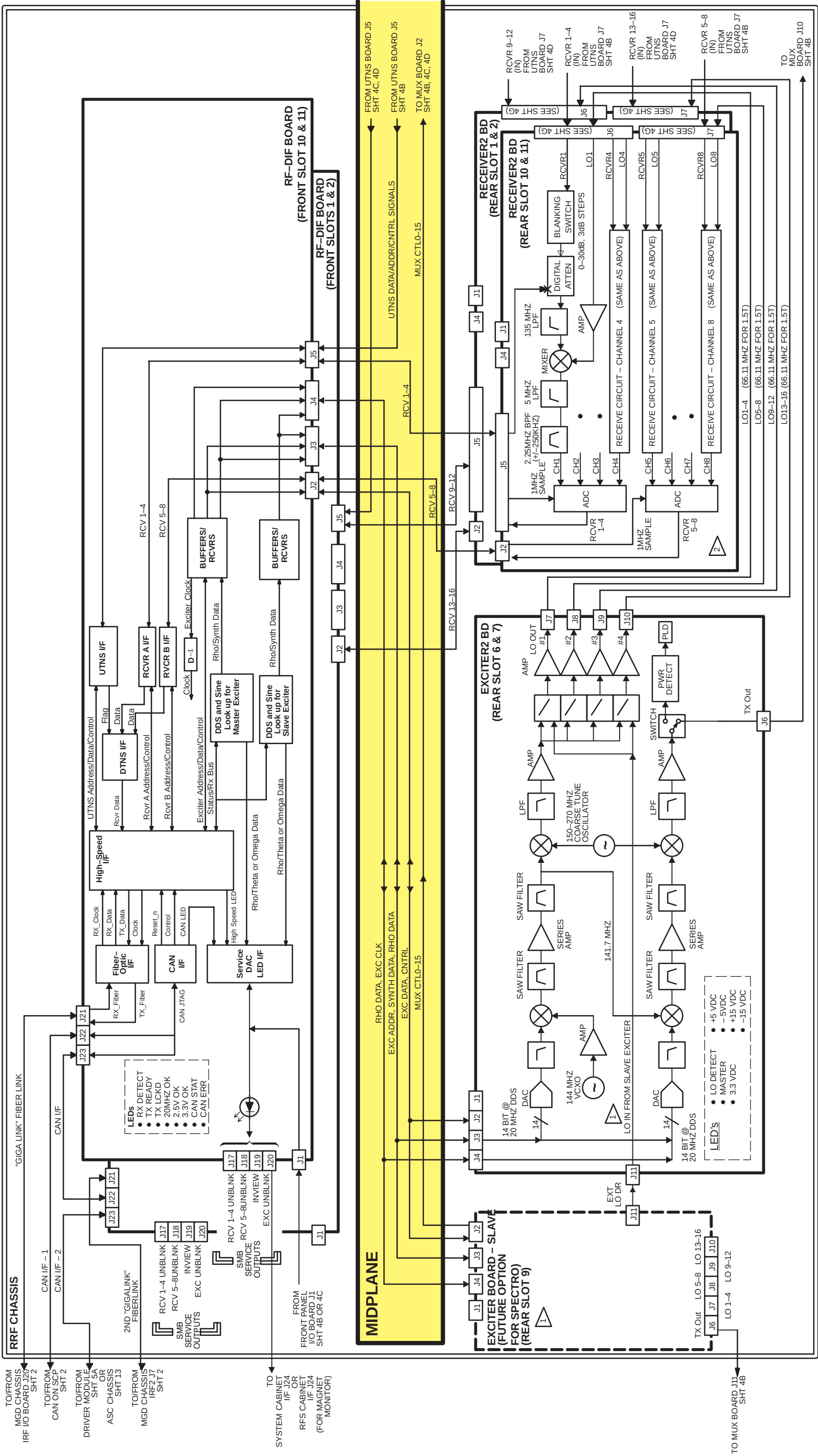
(4 and 8 Channel Systems)

RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 2A

SHEET 3A OF 16

NOTE:

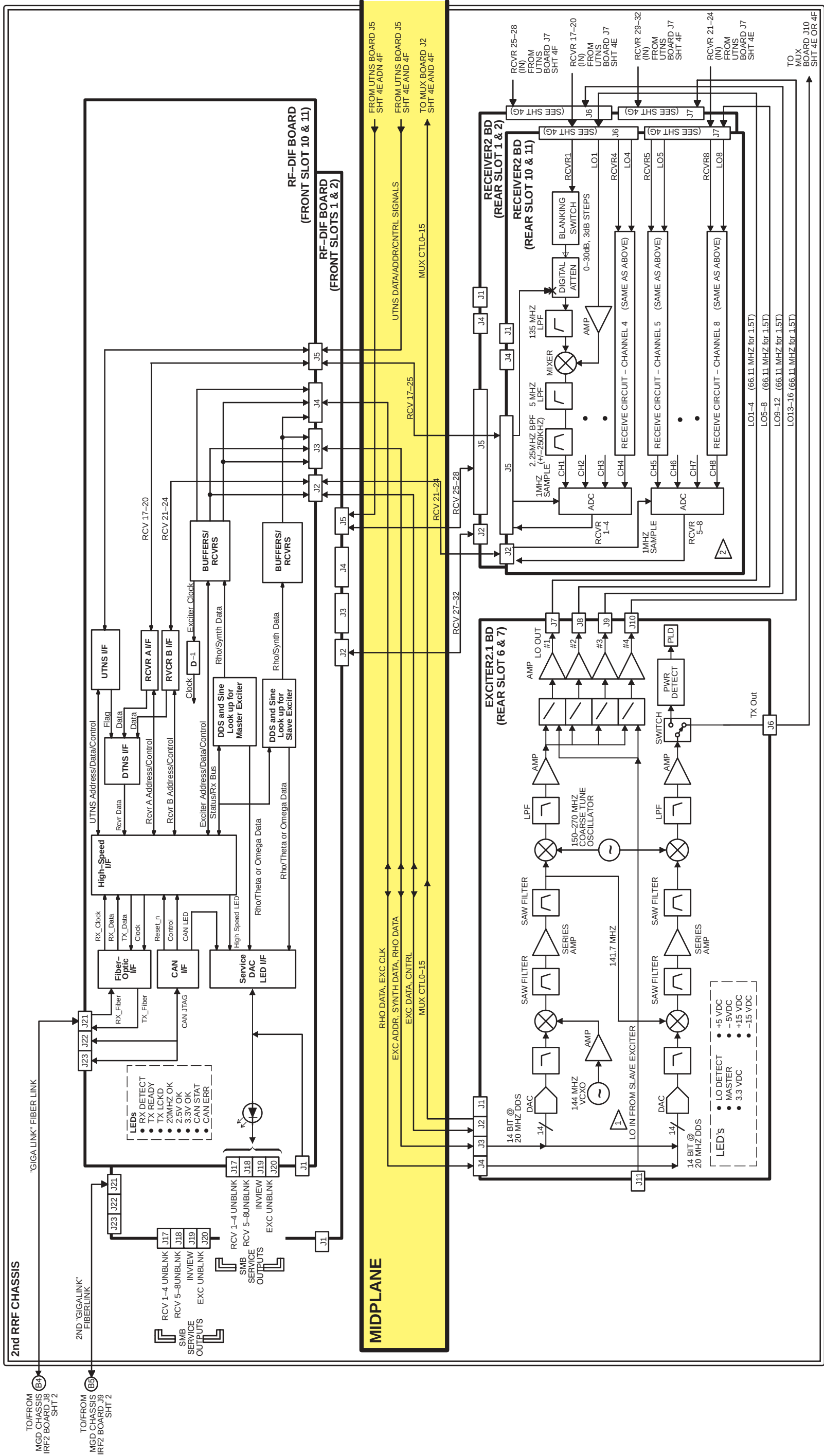
1 Slave Exciter is a future option for Spectro.



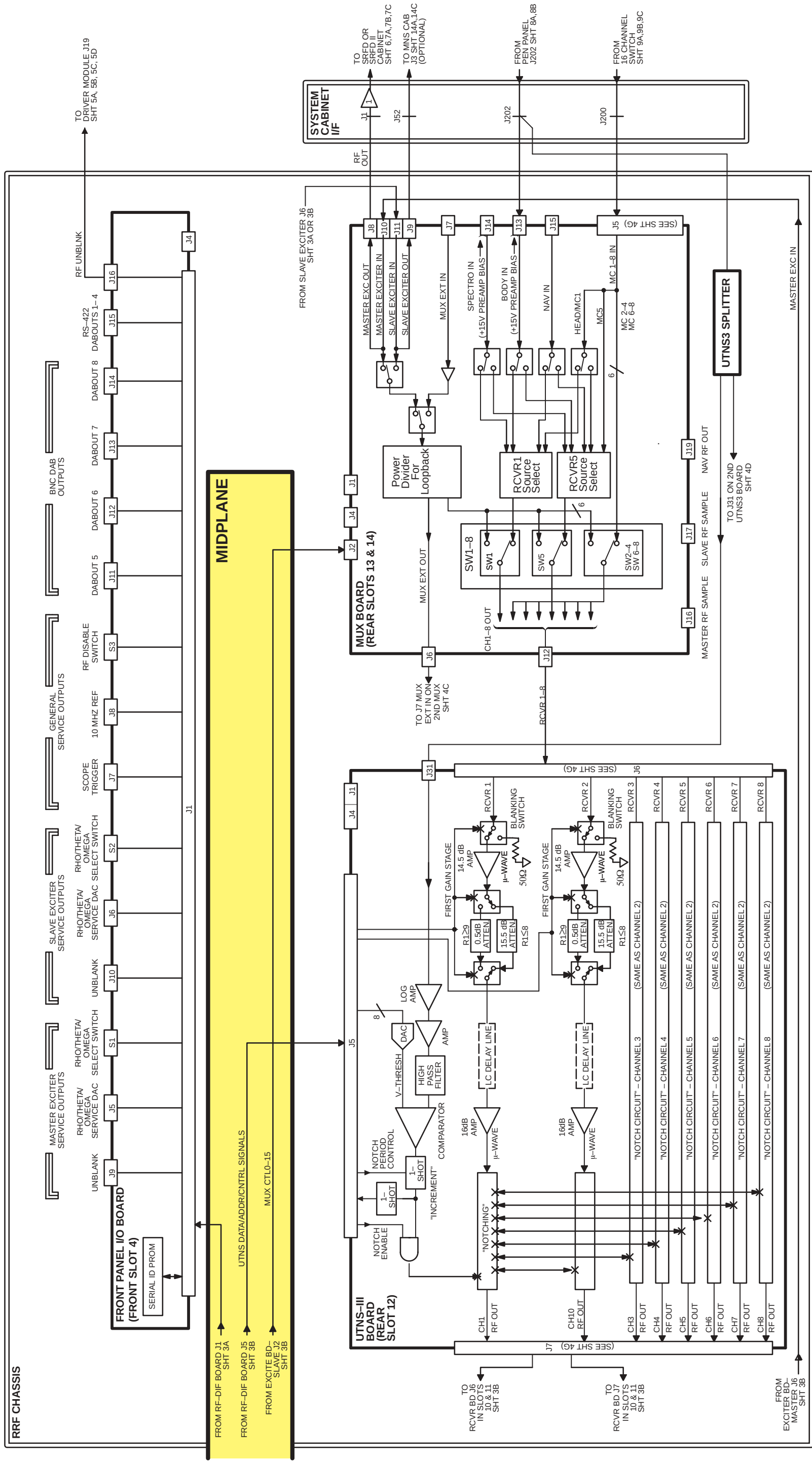
EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
 (16 Channel – Forward Production & Upgrades)

Signa HD, HDx & HDxt 1.5T (12.x, 14.x & 15.x)
Block Diagrams & Supplemental Schematics

DIRECTION 5133310



HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(32 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 4C
SHEET 3C OF 16



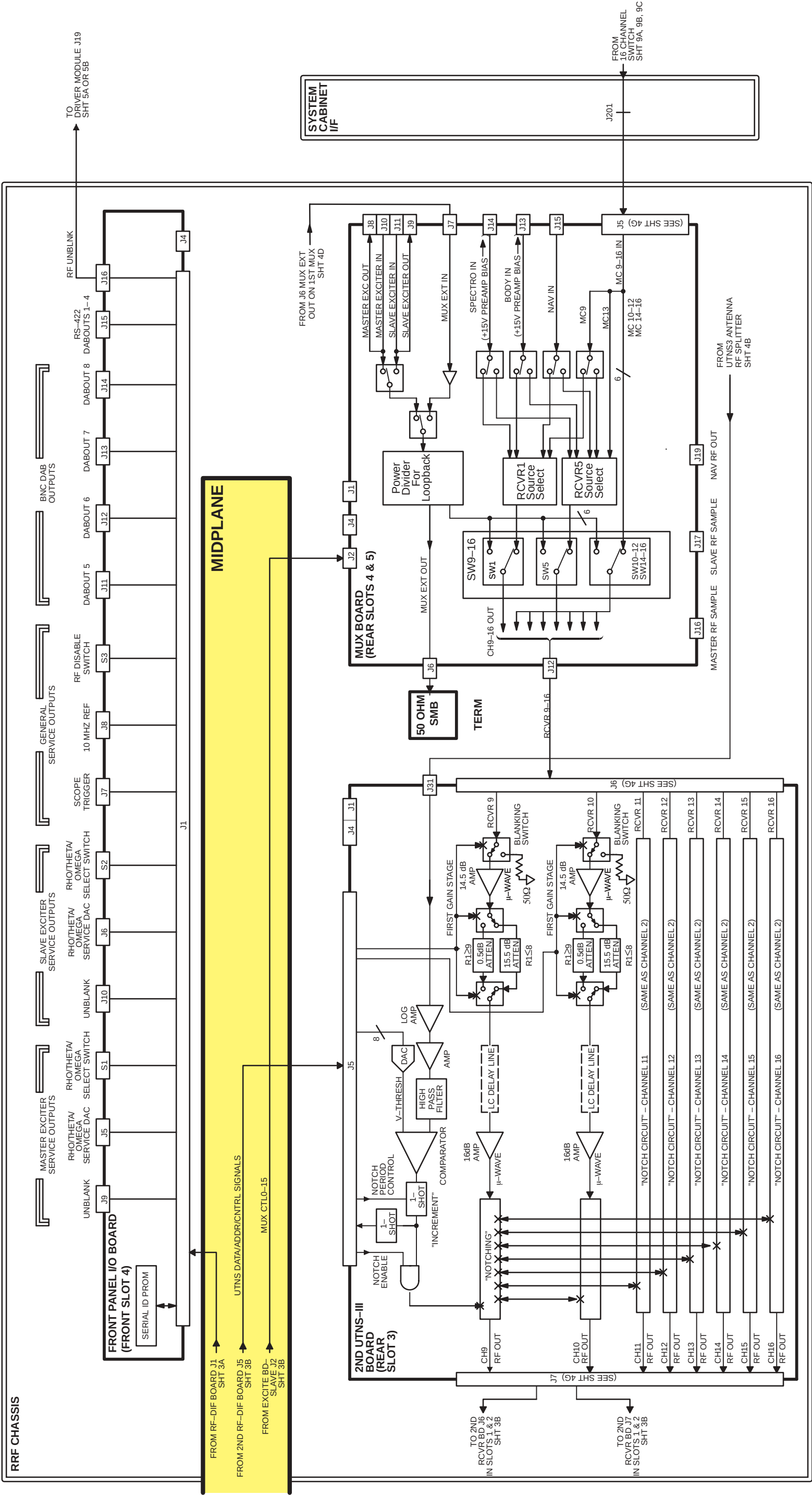
EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM

(Channels 1-8 of 16 Channel – Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2a OF 2B (16 Channel)

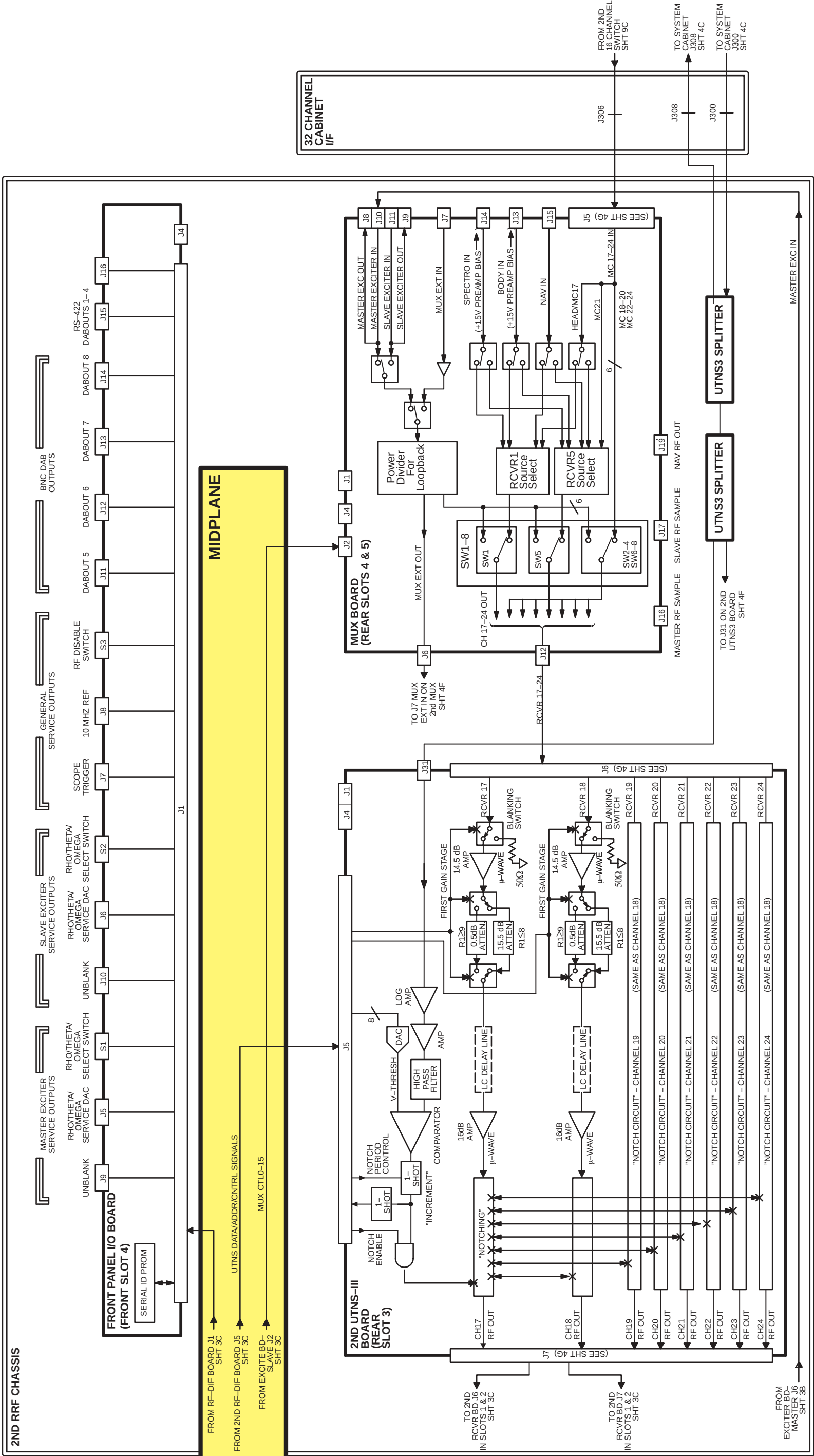
SHEET 4B OF 16

NOTE:

△ J1 not present in RF System Cabinet.

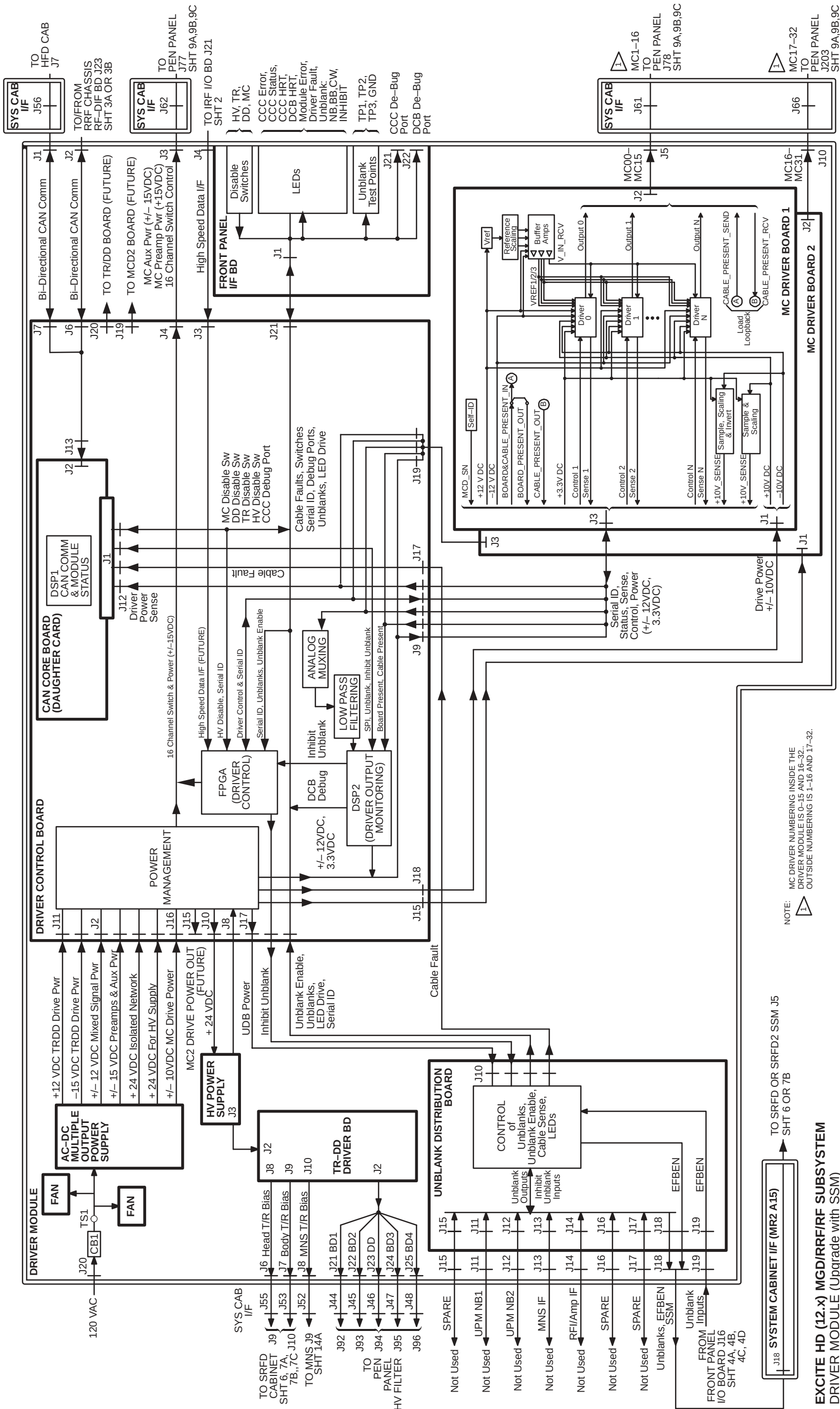


EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(Channels 9-16 of 16 or 32 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2b OF 2B (16 Channel)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2b OF 4C (32 Channel)



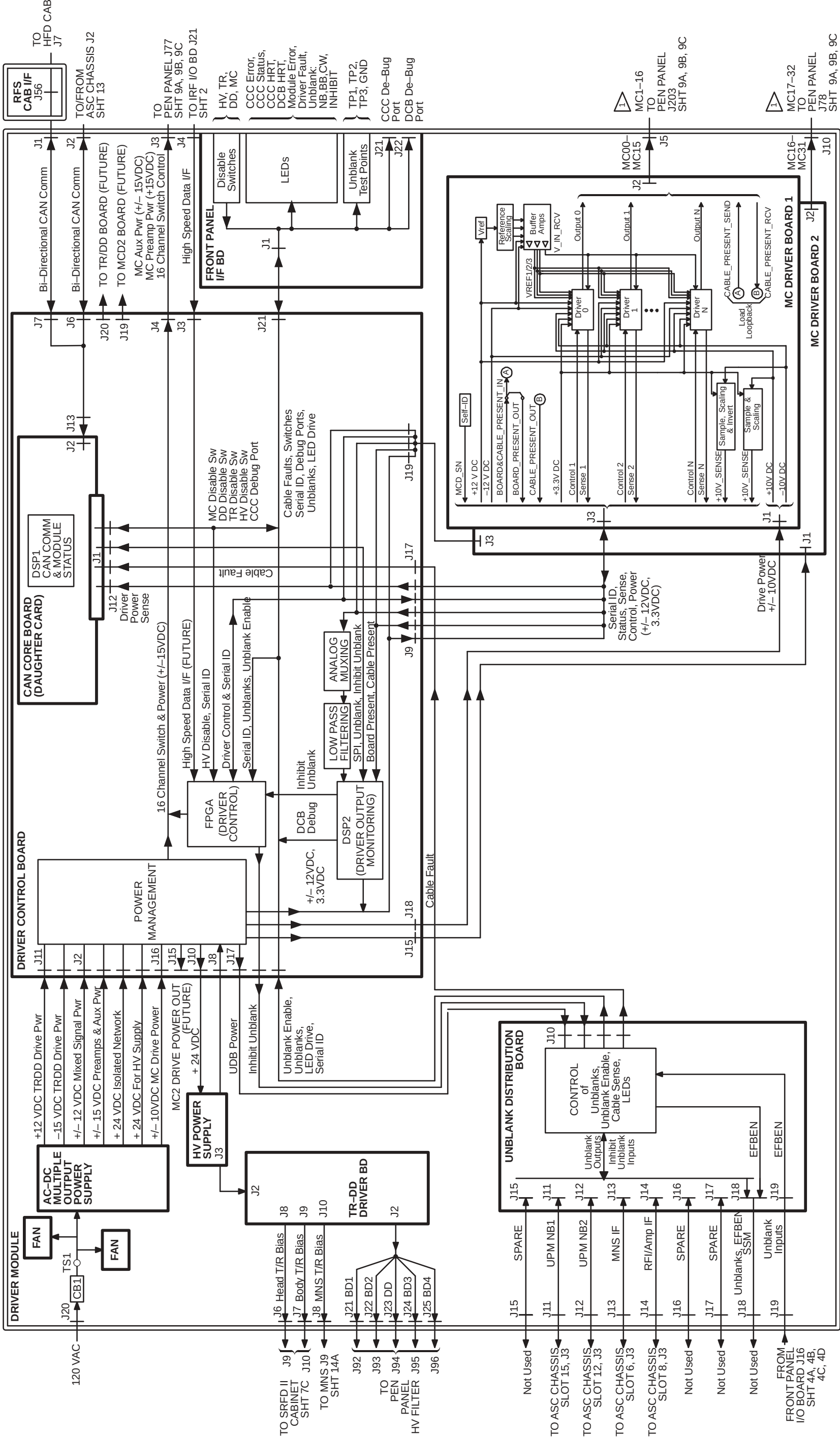
HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
 (Channels 17-24 of 32 Channel - Forward Production & Upgrades)
 2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2c OF 4C (32 Channel)
 SHEET 4E OF 16





EXCITE HD (12.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Upgrade with SSM)

SHEET 5A OF 16



HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
 DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)
 SHEET 5D OF 16

NOTE:  MC DRIVER NUMBERING INSIDE THE DRIVER MODULE IS 0-15 AND 16-32... OUTSIDE NUMBERING IS 1-16 AND 17-32.



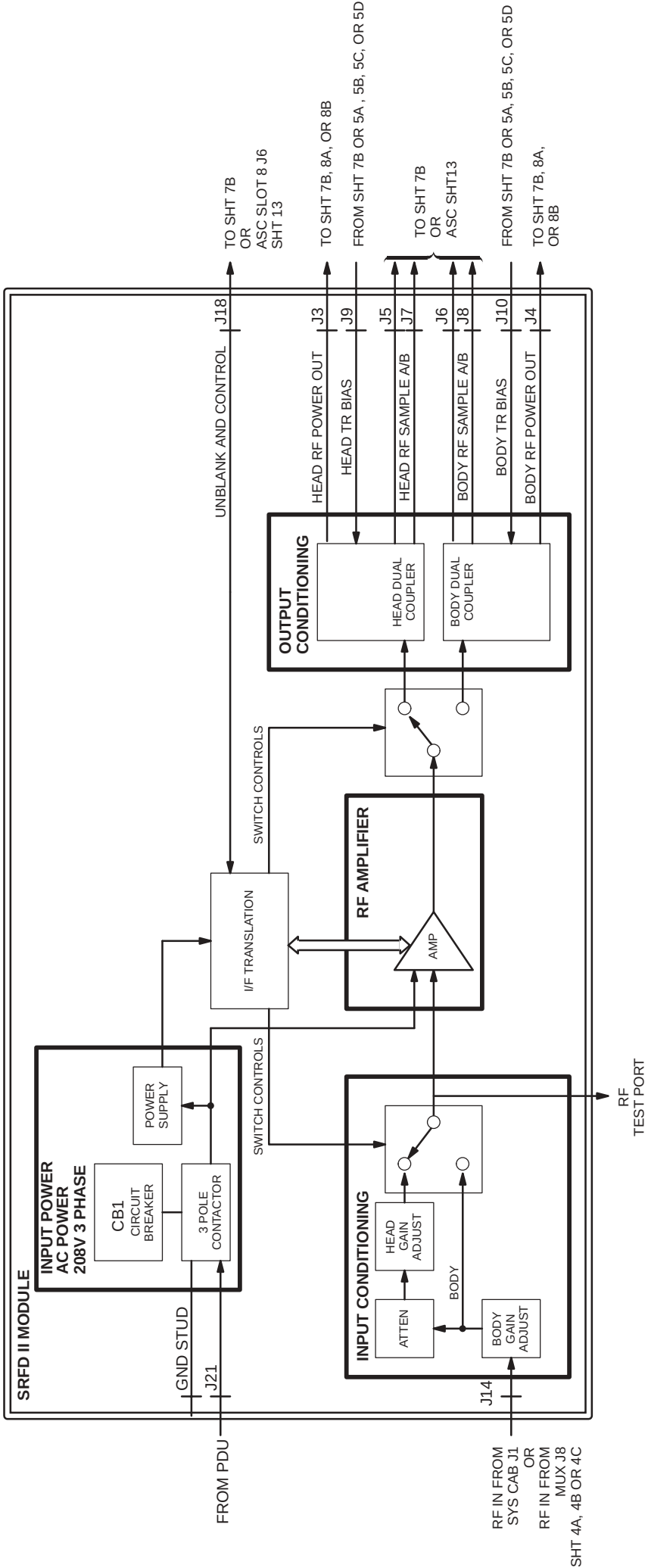
NOTE:
1) GRADIENT
2) PATIENT HANDLING

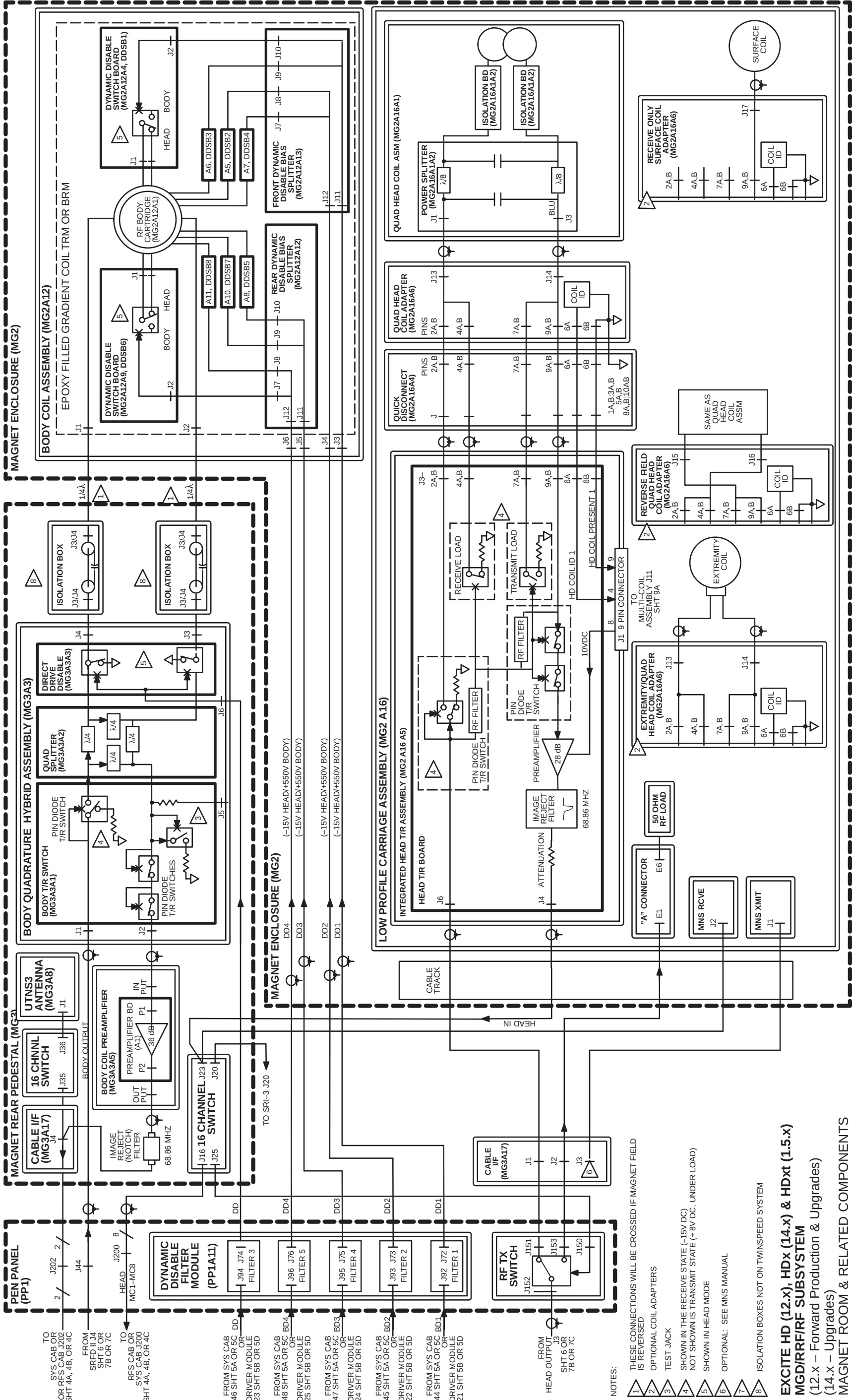


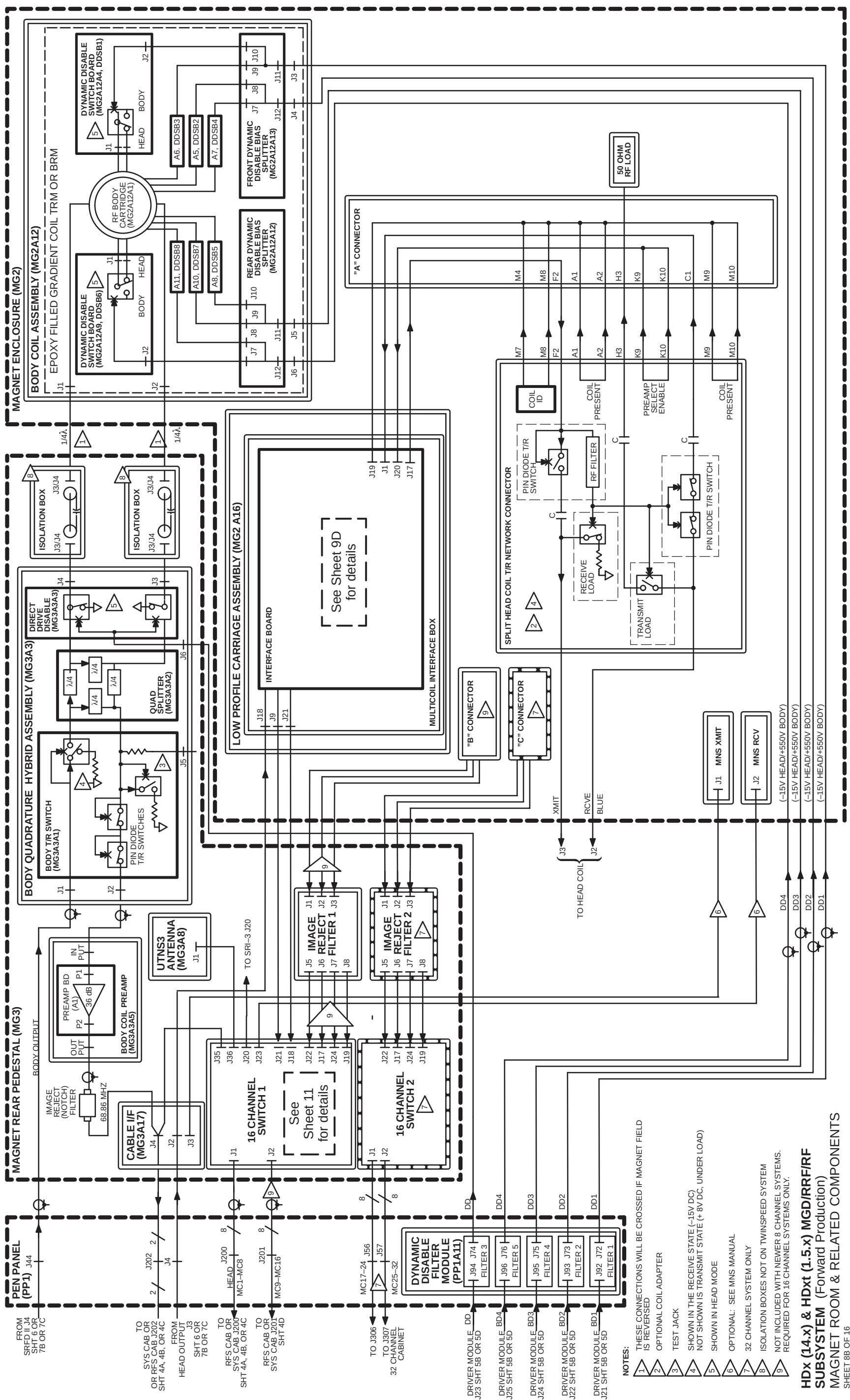
(Upgrades Only)

SHEET 7B OF 16

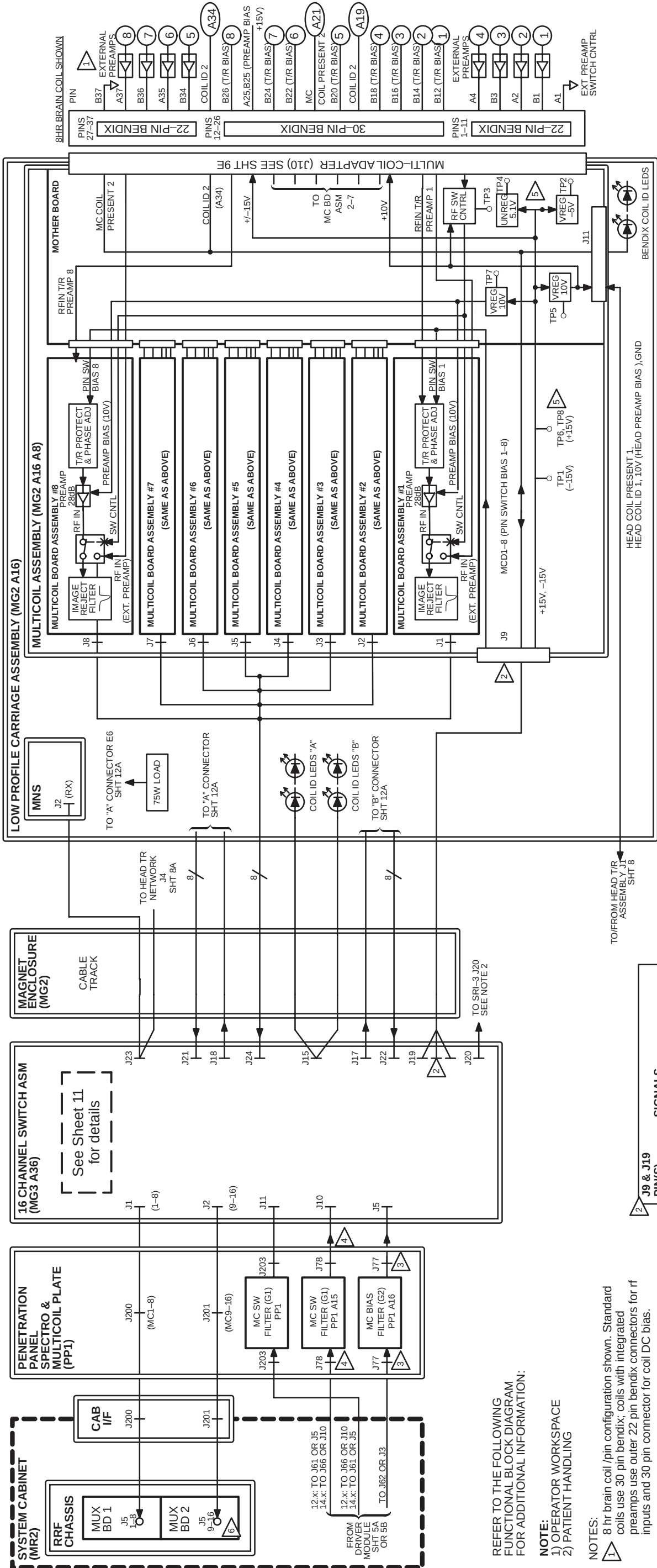
NOTE:
1) GRADIENT
2) PATIENT HANDLING







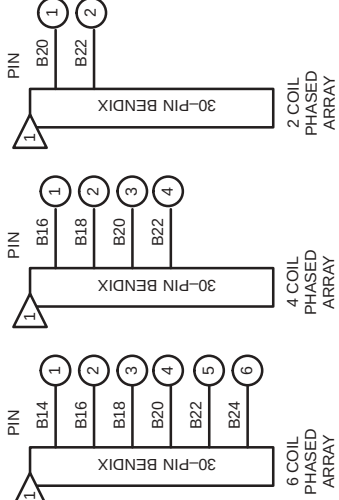
HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF
SUBSYSTEM (Forward Production)
MAGNET ROOM & RELATED COMPONENTS
SHEET 8B OF 16



REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING

NOTES:
1. 8 hr brain coil /pin configuration shown. Standard coils use 30 pin bendix; coils with integrated preamps use outer 22 pin bendix connectors for rf inputs and 30 pin connector for coil DC bias.



**EXCITE HD (12.x) or HDx (14.x Upgrade)
MGD/RRF/RF SUBSYSTEM
MULTI-COIL MODULES**
SHEET 9A OF 16

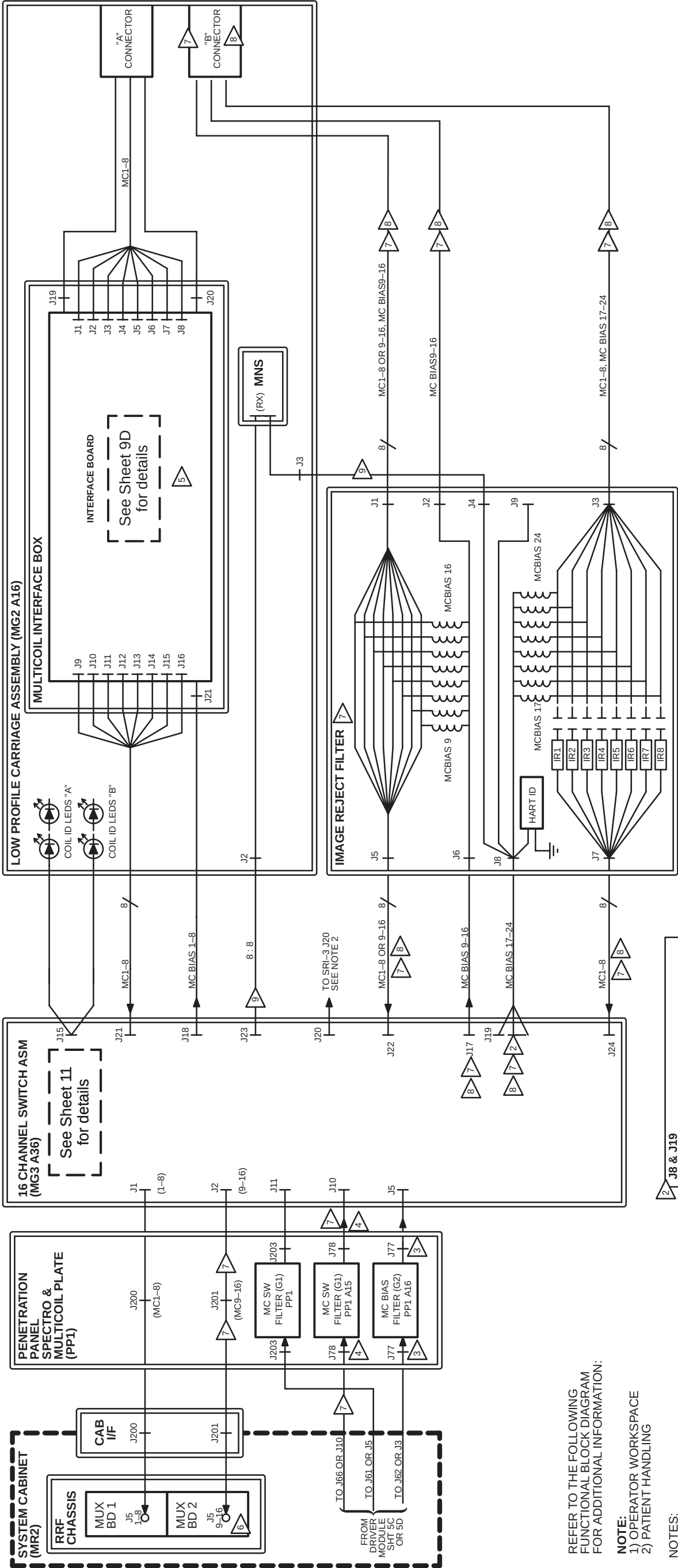
J9 & J19 PIN(S)	SIGNALS
1,14	-15V
2,15	+15V
3, 4, 23-25	... GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2
NOTE 1: PIN 13 DEFAULT LOW. LOW: PREAMP POWER ON. HIGH: PREAMP POWER OFF.	

J62 OR J3 & J77 PIN	SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)
NOTE 1: ANALOG POWER FOR PREAMPS (MC & HEAD), CROSSPOINT & MC COILS.	

J10 & J78 PIN(S)	SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND
NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V). NOTE 2: THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK. NOTE 3: CROSSPOINT SWITCH ASSEMBLY.	

LOCATED BEHIND MULTICOIL BD ASSY	TEST POINT	FUNCTION
7	 TP1	 -15V
6	 TP2	 -5V
6	 TP3	.. RF SW CNTRL SIG
2, 3	 TP4	 5.1V
1, 2	 TP5	 +10V B
1, 2	 TP6	 +15V
7, 8	 TP7	 +10V A
7, 8	 TP8	 +15V

Only 16 Channel Systems have a
second MUX board. J201 not used
in 8 and 4 Channel systems.



REFER TO THE FOLLOWING
 FUNCTIONAL BLOCK DIAGRAM
 FOR ADDITIONAL INFORMATION:

NOTE:
 1) OPERATOR WORKSPACE
 2) PATIENT HANDLING

NOTES:



J8 & J19 PIN(S)	SIGNALS
1,14	-15V
2,15	+15V
3, 4, 23-25 ..	GND
5	MCD8
6	MCD4
7	MCD7
8	MCD3
9	MCD6
10	MCD2
11	MCD5
12	MCD1
13	CTL_PREAMP (NOTE 1)
16	RED_LED
17	YEL_LED
18	GRN_LED
19	COIL_PRESENT #1
20	COIL_PRESENT #2
21	COIL ID #1
22	COIL ID #2
NOTE 1:	PIN 13 DEFAULT LOW. LOW: PREAMP POWER ON. HIGH: PREAMP POWER OFF.

J62 OR J3 & J77 PIN	SIGNAL NAME
1	XPT_DATA_POS
2	XPT_STROBE_POS
3	XPT_OE_N_POS
4	AGND
5	AGND
6	+15V (NOTE 1)
7	+15V (NOTE 1)
8	+15V (NOTE 1)
9	XPT_DATA_NEG
10	XPT_STROBE_NEG
11	XPT_OE_N_NEG
12	AGND
13	AGND
14	-15V (MC COILS)
15	-15V (MC COILS)
NOTE 1:	ANALOG POWER FOR PREAMPS (MC & HEAD), CROSSPOINT & MC COILS.

J10 & J66 PIN(S)	SIGNAL
1-8	MCD1-8 (NOTE 1)
9	MC CABLE PRESENT_O IN (NOTE 2)
10,11	NO CONNECT
12-19	MCD9-16 (NOTE 3)
20-27	GND
29	MC CABLE PRESENT_1 IN (NOTE 2)
30-37	GND
NOTE 1:	MULTICOIL BIAS (+3V,+5V,+7V,-5V)
NOTE 2:	THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3:	NOT USED/CONNECTED AT J10 INSIDE CROSSPOINT SWITCH ASSEMBLY.

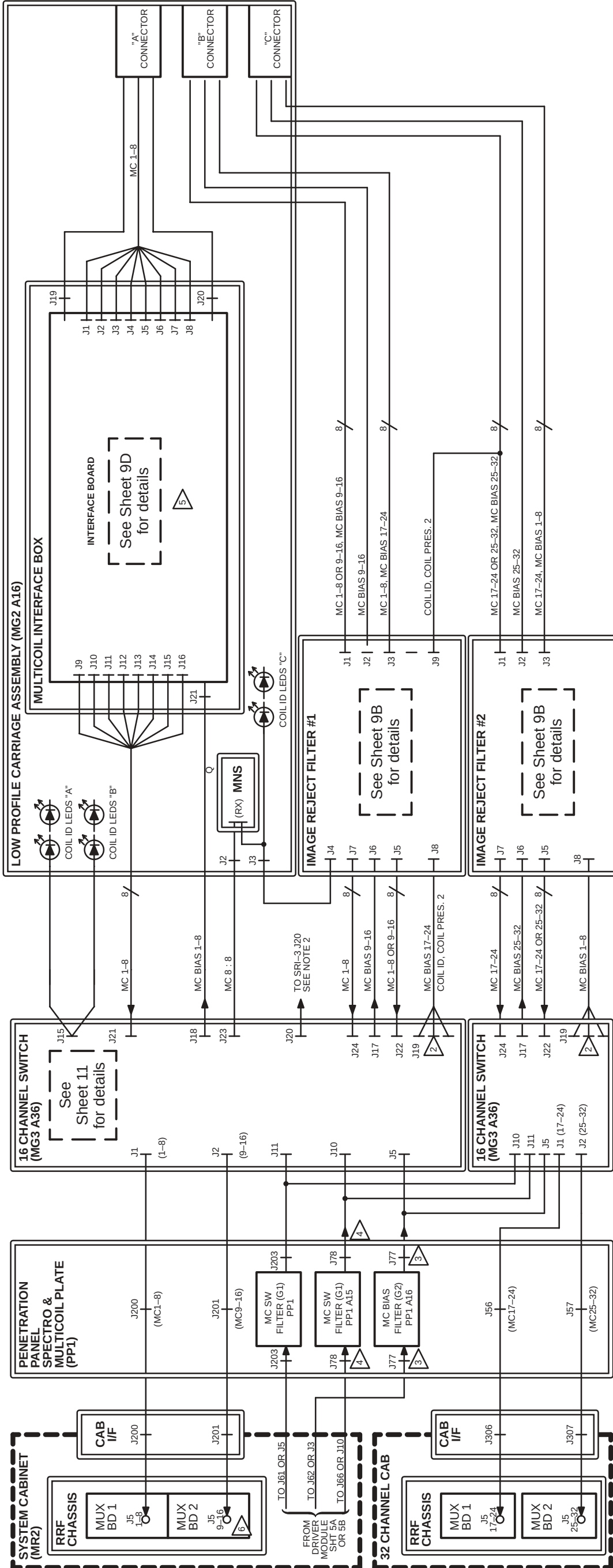
TESTPOINT .	FUNCTION
TP1	SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	PREAMP SELECT 1
TP3	PREAMP SELECT 2
TP4	8.5V_A
TP5	+10V_B_CNTRL
TP6	+15V
TP7	+5.1V
TP8	+15V
TP9	+15V
TP10	8.5V_B
TP11	8.5V_C
TP12	-15V
TP13	-8.5V

Only 16 Channel Systems have a second MUX board.

Required on 16 or 32 Channel systems.

B Connector and MC Bias lines are not included in newer
 8 Channel systems.

Optional: See MNS manual



REFER TO THE FOLLOWING
 FUNCTIONAL BLOCK DIAGRAM
 FOR ADDITIONAL INFORMATION:

- NOTE:**
 1) OPERATOR WORKSPACE
 2) PATIENT HANDLING

NOTES:

J8 & J19 PIN(S)	SIGNALS
1,14	 -15V
2,15	 +15V
3, 4, 23-25	... GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL_PRESENT #1
20	 COIL_PRESENT #2
21	 COIL ID #1
22	 COIL ID #2
NOTE 1: PIN J13 DEFAULT LOW. LOW: PREAMP POWER ON. HIGH: PREAMP POWER OFF.	

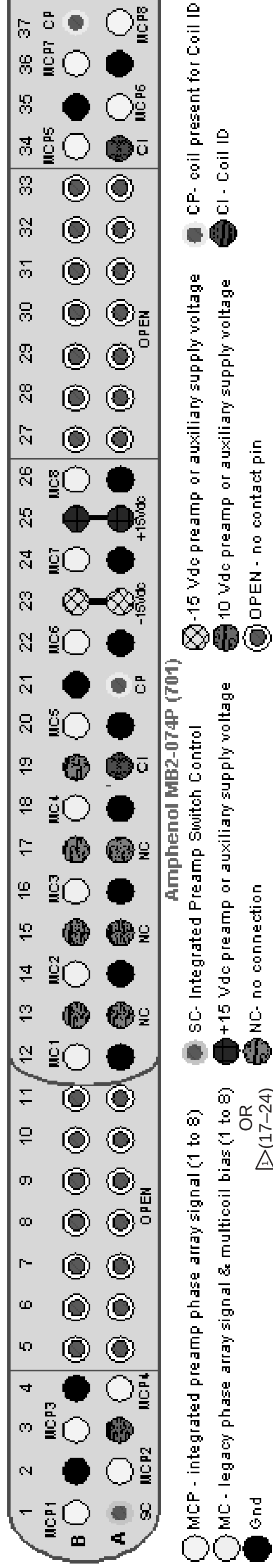
J62 OR J3 & J77 PIN	SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)
NOTE 1: ANALOG POWER FOR PREAMPS (MC & HEAD), CROSSPOINT & MC COILS.	

J10 & J66 PIN(S)	SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND
NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V). NOTE 2: THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK. NOTE 3: NOT USED/CONNECTED AT J10 INSIDE CROSSPOINT SWITCH ASSEMBLY.	

TESTPOINT	FUNCTION
TP1	 SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	 PREAMP SELECT 1
TP3	 PREAMP SELECT 2
TP4	 8.5V_A
TP5	 +10V_B CNTRL
TP6	 +15V
TP7	 +5.1V
TP8	 +15V
TP9	 +15V
TP10	 8.5V_B
TP11	 8.5V_C
TP12	 -15V
TP13	 -8.5V

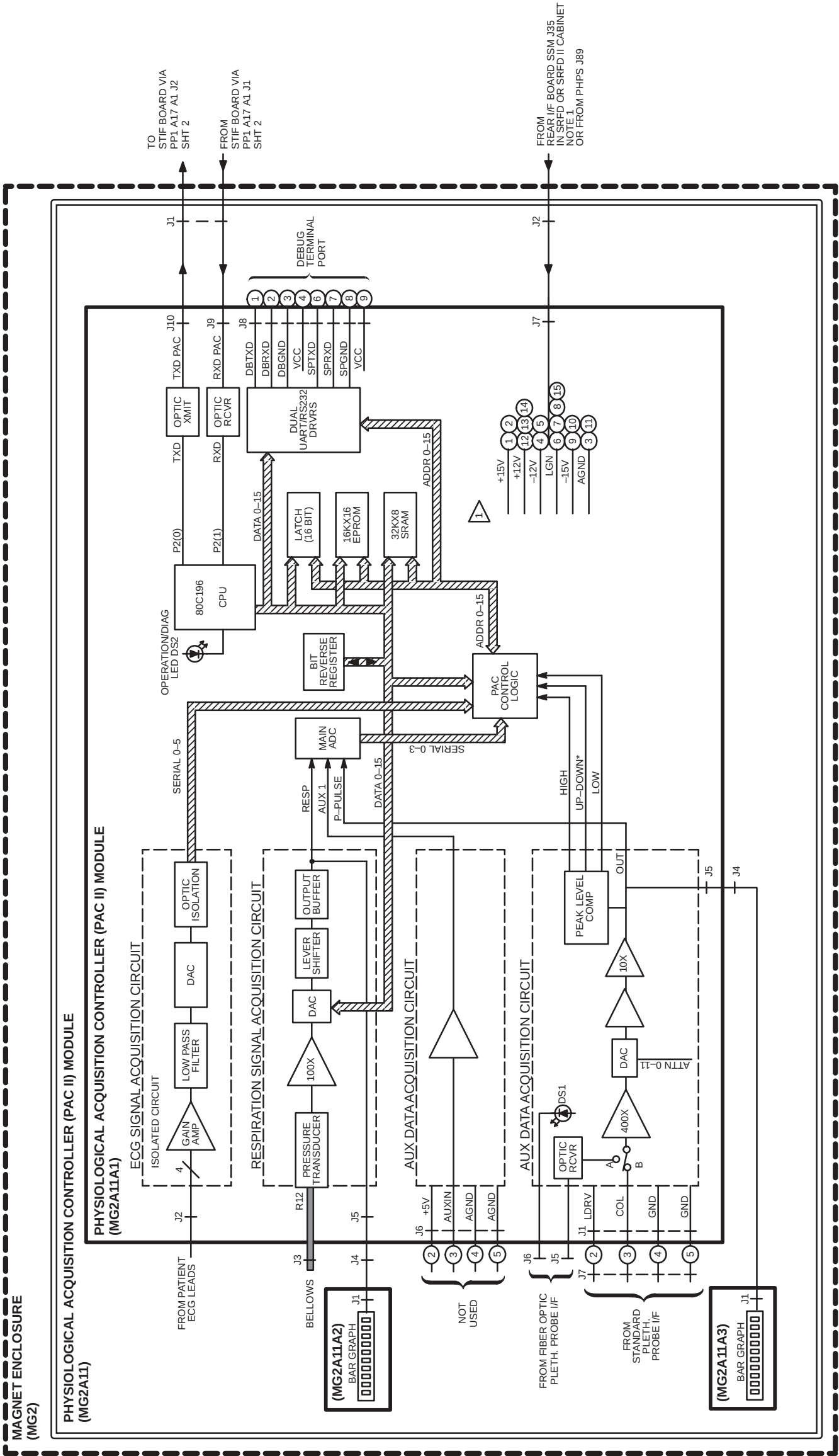
HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF
 SUBSYSTEM
 32 CHANNEL MULTI-COIL
 SHEET 9C OF 16

Only 16 Channel Systems have a
 second MUX board. J201 not used
 in 8 and 4 Channel systems.



NOTES:

MC BIAS is 17–24 on 14.x systems.



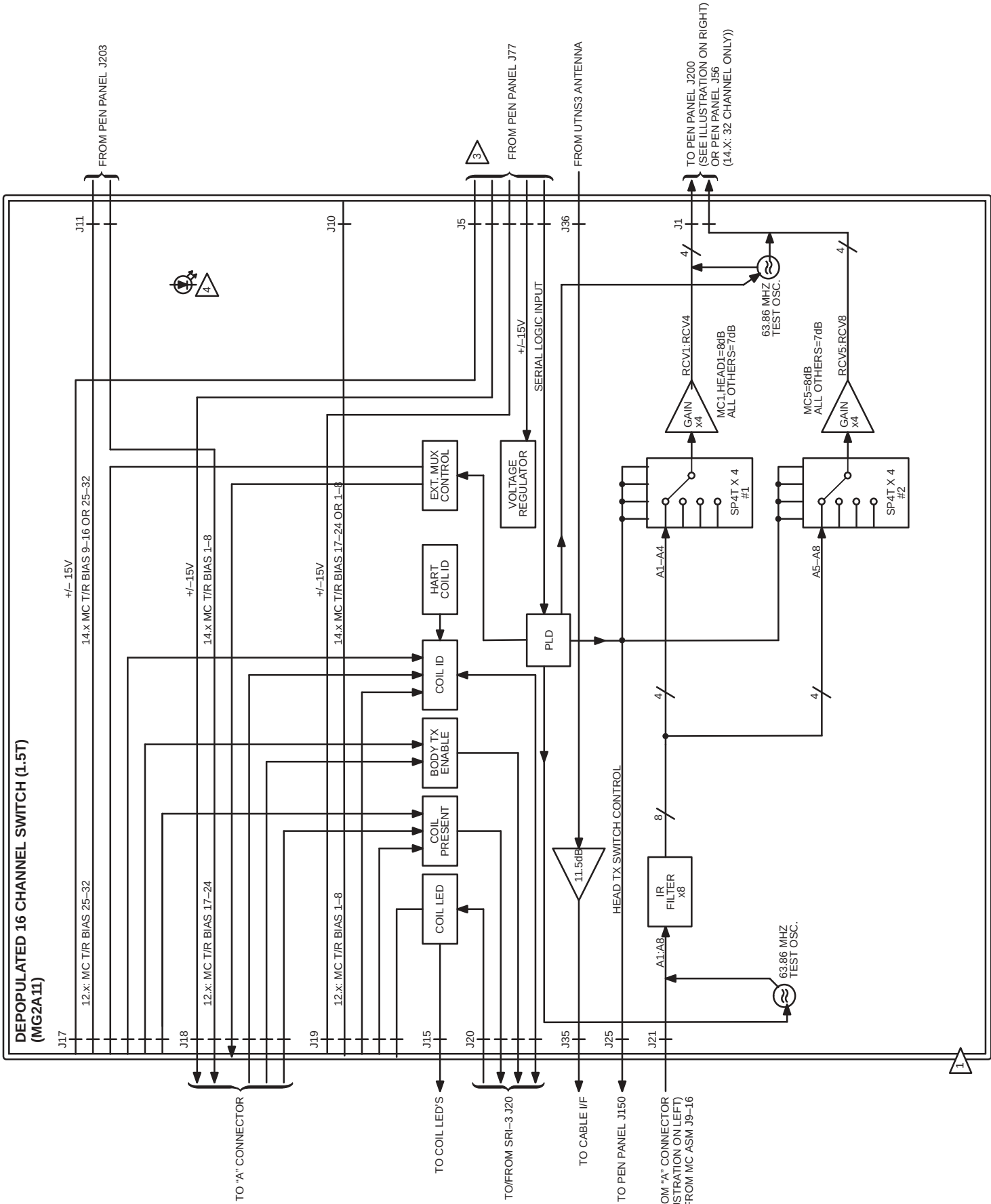
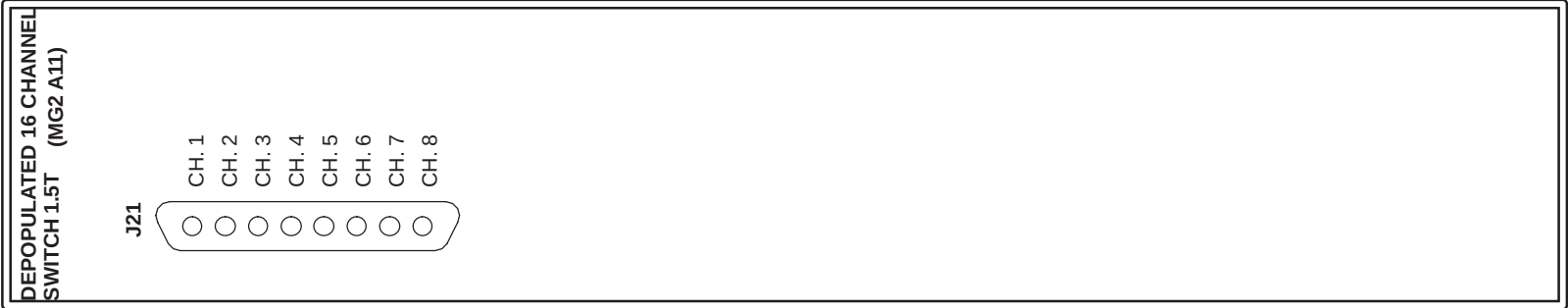
NOTES:

1 MULTIPLE WIRES/PINS WITH SAME SIGNAL ARE NOT SHOWN

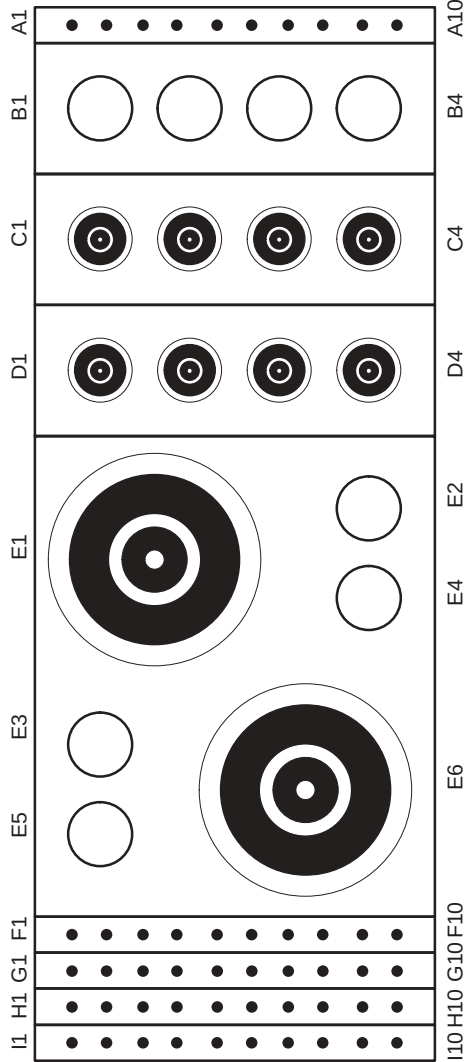
REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:

1) PATIENT HANDLING



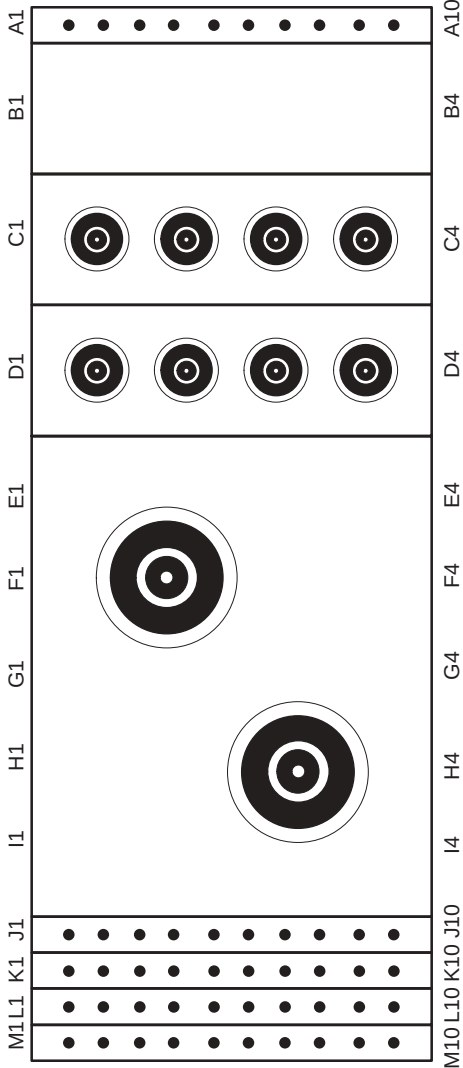
View from CONNECTOR SIDE of Connector Assembly



Contact Function Allocation		
Pin	Twisted with	Description Comments
A1	A2	coil present
A2	A1	coil present return
A3	A4	multiplexerSelect0
A4	A3	multiplexerSelect0 return
A5	A6	multiplexerSelect1
A6	A5	multiplexerSelect1 return
A7	A8	multiplexerSelect2
A8	A7	multiplexerSelect2 return
A9	A10	multiplexerSelect3
A10	A9	multiplexerSelect3 return
B1	n/a	Key blocked in receptacle E
B2	n/a	Key blocked in receptacle F
B3	n/a	Key blocked in receptacle G
B4	n/a	Key blocked in receptacle H
C1	n/a	MC–1A,B or MC–9B
C2	n/a	MC–2A,B or MC–10B
C3	n/a	MC–3A,B or MC–11B
C4	n/a	MC–4A,B or MC–12B
D1	n/a	MC–5A,B or MC–13B
D2	n/a	MC–6A,B or MC–14B
D3	n/a	MC–7A,B or MC–15B
D4	n/a	MC–8A,B or MC–16B

Contact Function Allocation		
Pin	Twisted with	Description Comments
E1	n/a	Transmit signal up to 2kw (1.5T) or 4kw (3.0T) PEP. “A” Connector only. Not present in “B” Connector
E2	n/a	Key blocked in receptacle C
E3	n/a	Key blocked in receptacle A Blocked for “A”
E4	n/a	Key blocked in receptacle B Blocked for “B”
E5	n/a	Key blocked in receptacle D
E6	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS. “A” Connector only. Not present in “B” Connector
F1	G1	MC bias 1 MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2 MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3 MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4 MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5 MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6 MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7 MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8 MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable
F10	F9	body transmit enable return
G1	F1	MC bias return 1 MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2 MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3 MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4 MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5 MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6 MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7 MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8 MC T/R Bias Line 24 for “A”, 32 for “B”
G9	n/a	Reserved
G10	n/a	Reserved
H1	n/a	Reserved
H2	n/a	Reserved
H3	n/a	Reserved
H4	n/a	Reserved
H5	n/a	Reserved
H6	n/a	Reserved
H7	n/a	Reserved
H8	n/a	Reserved
H9	n/a	Reserved
H10	n/a	Reserved
I1	n/a	Reserved
I2	I5	+15Vdc (auxiliary & preamp power)
I3	I6	+10Vdc (auxiliary & preamp power)
I4	n/a	–15Vdc (auxiliary power)
I5	I2	Return for aux power
I6	I3	Return for aux power
I7	I8	coil ID (Dallas OneWire protocol) All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol) All coils MUST have coil ID.
I9	I10	coil present’
I10	I9	coil present return’

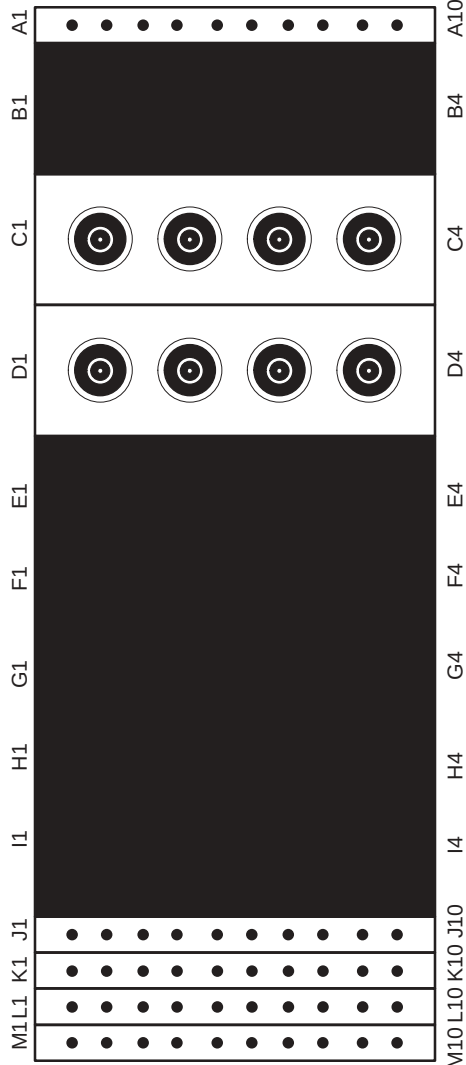
View from CONNECTOR SIDE of Connector Assembly



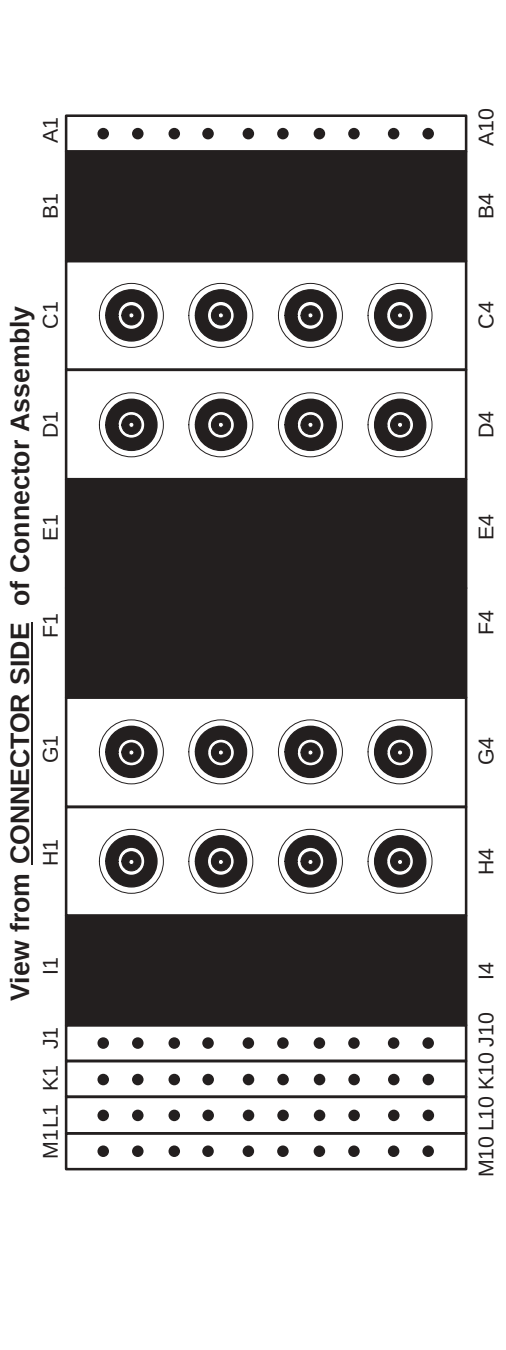
Contact Function Allocation		
Pin	Twisted with	Comments
A1	A2	coil present
A2	A1	coil present return
A3	A4	multiplexerSelect0
A4	A3	multiplexerSelect0 return
A5	A6	multiplexerSelect1
A6	A5	multiplexerSelect1 return
A7	A8	multiplexerSelect2
A8	A7	multiplexerSelect2 return
A9	A10	multiplexerSelect3
A10	A9	multiplexerSelect3 return
B1	n/a	Key blocked in receptacle E
B2	n/a	Key blocked in receptacle F
B3	n/a	Key blocked in receptacle G
B4	n/a	Key blocked in receptacle H
C1	n/a	MC--A1
C2	n/a	MC--A2
C3	n/a	MC--A3
C4	n/a	MC--A4
D1	n/a	MC--A5
D2	n/a	MC--A6
D3	n/a	MC--A7
D4	n/a	MC--A8
		Preamp out 1, no bias, DC blocked
		Preamp out 2, no bias, DC blocked
		Preamp out 3, no bias, DC blocked
		Preamp out 4, no bias, DC blocked
		Preamp out 5, no bias, DC blocked
		Preamp out 6, no bias, DC blocked
		Preamp out 7, no bias, DC blocked
		Preamp out 8, no bias, DC blocked

Contact Function Allocation			
Pin	Twisted with	Description	Comments
F2	n/a	Transmit Signal up to 8kw PEP	
H3	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS	"A" connector only
J1	K1	MC bias 1	
J2	K2	MC bias 2	
J3	K3	MC bias 3	
J4	K4	MC bias 4	
J5	K5	MC bias 5	
J6	K6	MC bias 6	
J7	K7	MC bias 7	
J8	K8	MC bias 8	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1	
K2	J2	MC bias return 2	
K3	J3	MC bias return 3	
K4	J4	MC bias return 4	
K5	J5	MC bias return 5	
K6	J6	MC bias return 6	
K7	J7	MC bias return 7	
K8	J8	MC bias return 8	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	−15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

View from CONNECTOR SIDE of Connector Assembly

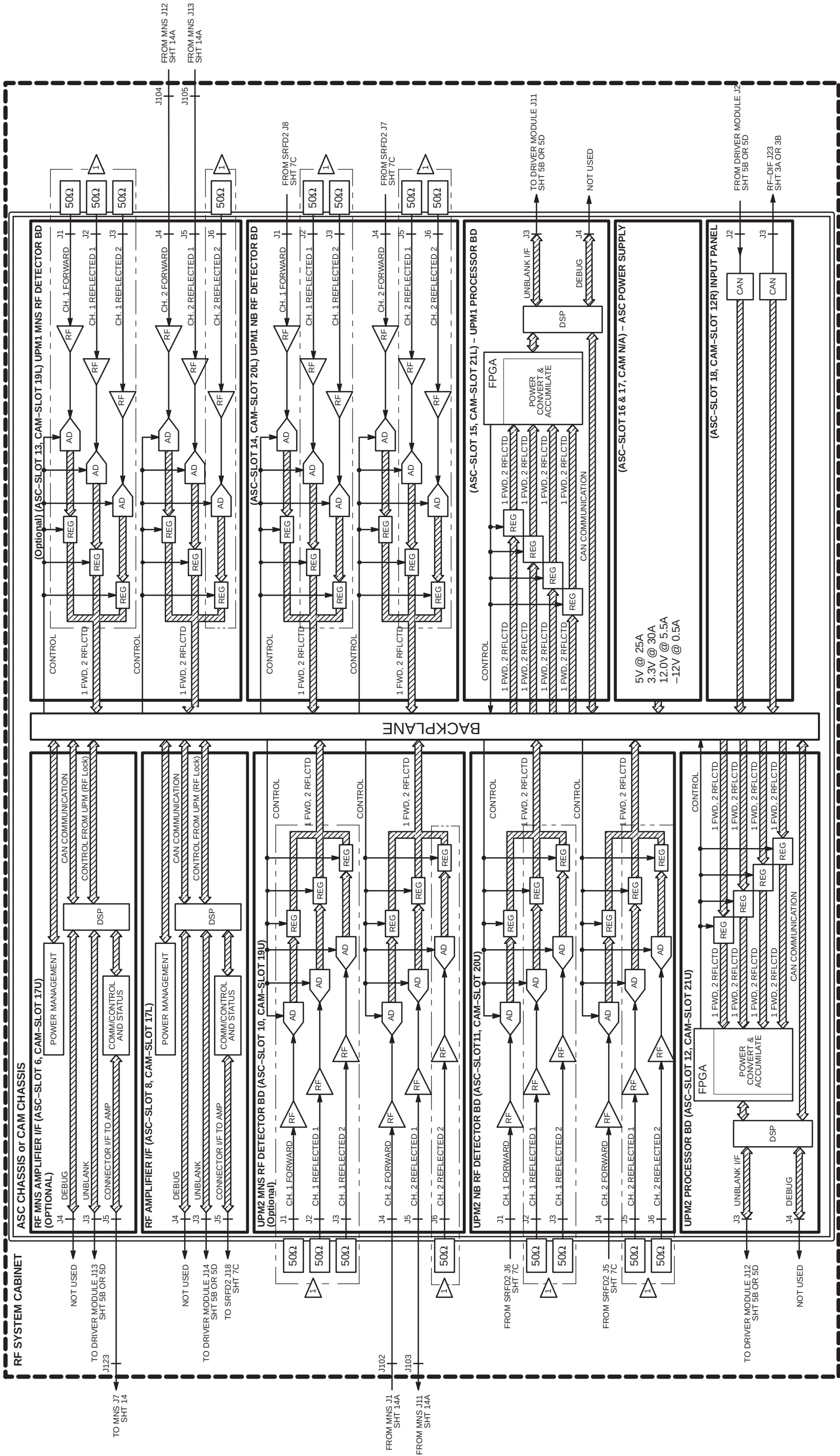


Contact Function Allocation		
Pin	Twisted with	Comments
A1	A2	coil present
A2	A1	coil present return
A3	A4	multiplexerSelect0
A4	A3	multiplexerSelect0 return
A5	A6	multiplexerSelect1
A6	A5	multiplexerSelect1 return
A7	A8	multiplexerSelect2
A8	A7	multiplexerSelect2 return
A9	A10	multiplexerSelect3
A10	A9	multiplexerSelect3 return
B1	n/a	
B2	n/a	
B3	n/a	
B4	n/a	
C1	n/a	MC-B1
C2	n/a	MC-B2
C3	n/a	MC-B3
C4	n/a	MC-B4
D1	n/a	MC-B5
D2	n/a	MC-B6
D3	n/a	MC-B7
D4	n/a	MC-B8



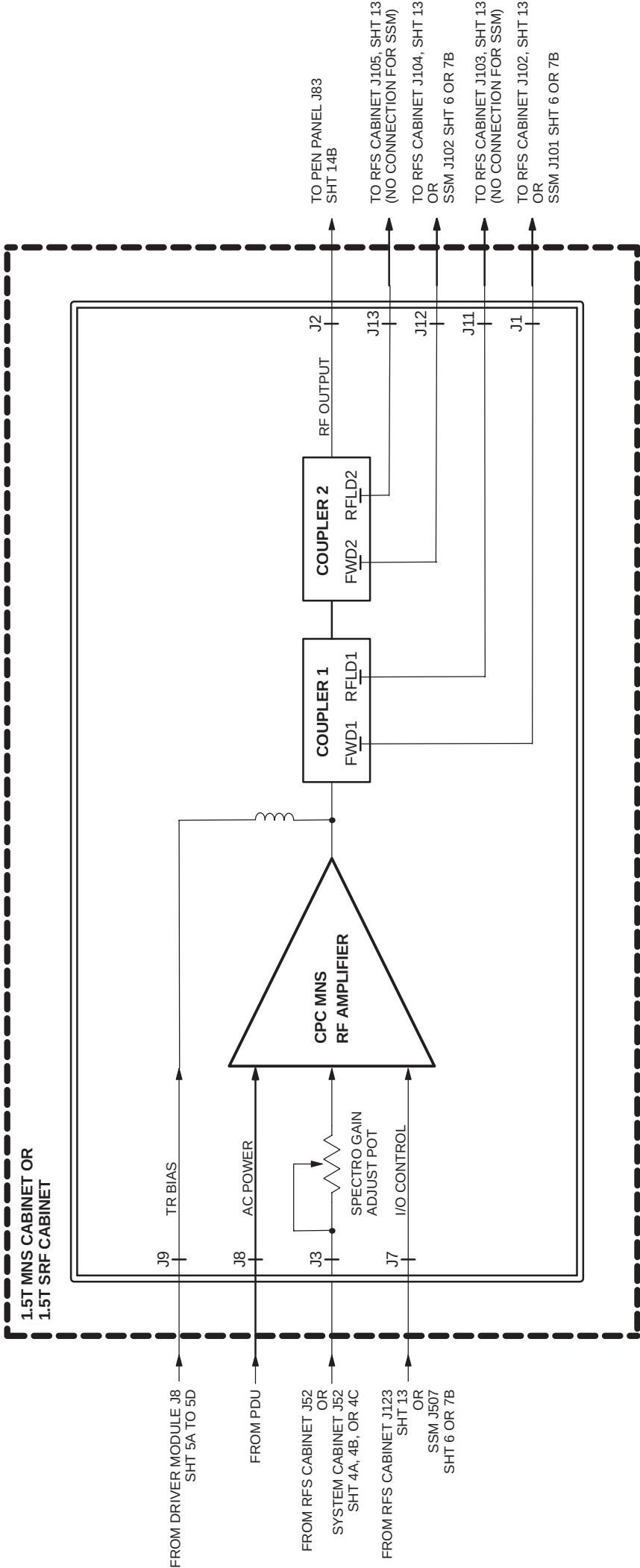
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
C1	n/a	MC-B1 or MC-C1, MC bias 1B (9), 1C (25)	Preamp out 1, MC bias 1B (9), 1C (25)
C2	n/a	MC-B2 or MC-C2, MC bias 2B (10), 2C (26)	Preamp out 2, MC bias 2B (10), 2C (26)
C3	n/a	MC-B3 or MC-C3, MC bias 3B (11), 3C (27)	Preamp out 3, MC bias 3B (11), 3C (27)
C4	n/a	MC-B4 or MC-C4, MC bias 4B (12), 4C (28)	Preamp out 4, MC bias 4B (12), 4C (28)
D1	n/a	MC-B5 or MC-C5, MC bias 5B (13), 5C (29)	Preamp out 5, MC bias 5B (13), 5C (29)
D2	n/a	MC-B6 or MC-C6, MC bias 6B (14), 6C (30)	Preamp out 6, MC bias 6B (14), 6C (30)
D3	n/a	MC-B7 or MC-C7, MC bias 7B (15), 7C (31)	Preamp out 7, MC bias 7B (15), 7C (31)
D4	n/a	MC-B8 or MC-C8, MC bias 8B (16), 8C (32)	Preamp out 8, MC bias 8B (16), 8C (32)
G1	n/a	MC-B9 or MC-C9, MC bias 9B (25), 9C (1)	Preamp out 1, MC bias 9B (25), 9C (1)
G2	n/a	MC-B10 or MC-102, MC bias 10B (26), 10C (2)	Preamp out 2, MC bias 10B (26), 10C (2)
G3	n/a	MC-B11 or MC-C11, MC bias 11B (27), 11C (3)	Preamp out 3, MC bias 11B (27), 11C (3)
G4	n/a	MC-B12 or MC-C12, MC bias 12B (28), 12C (4)	Preamp out 4, MC bias 12B (28), 12C (4)
H1	n/a	MC-B13 or MC-C13, MC bias 13B (29), 13C (5)	Preamp out 5, MC bias 13B (29), 13C (5)
H2	n/a	MC-B14 or MC-C14, MC bias 14B (30), 14C (6)	Preamp out 6, MC bias 14B (30), 14C (6)
H3	n/a	MC-B15 or MC-C15, MC bias 15B (31), 15C (7)	Preamp out 7, MC bias 15B (31), 15C (7)
H4	n/a	MC-B16 or MC-C16, MC bias 16B (32), 16C (8)	Preamp out 8, MC bias 16B (32), 16C (8)

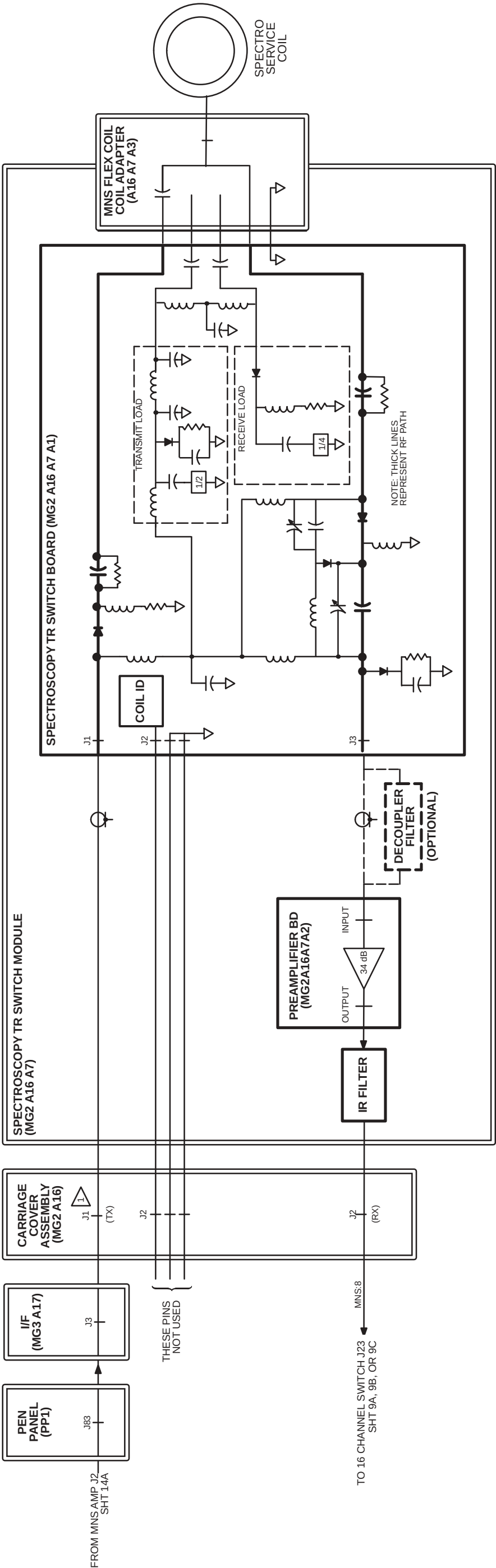
Contact Function Allocation			
Pin	Twisted with	Description	Comments
J1	K1	MC bias 1 (9)-B / MC bias 1 (25)-C	
J2	K2	MC bias 2 (10)-B / MC bias 2 (26)-C	
J3	K3	MC bias 3 (11)-B / MC bias 3 (27)-C	
J4	K4	MC bias 4 (12)-B / MC bias 4 (28)-C	
J5	K5	MC bias 5 (13)-B / MC bias 5 (29)-C	
J6	K6	MC bias 6 (14)-B / MC bias 6 (30)-C	
J7	K7	MC bias 7 (15)-B / MC bias 7 (31)-C	
J8	K8	MC bias 8 (16)-B / MC bias 8 (32)-C	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1 (25)-B / MC bias return 1 (1)-C	
K2	J2	MC bias return 2 (26)-B / MC bias return 2 (2)-C	
K3	J3	MC bias return 3 (27)-B / MC bias return 3 (3)-C	
K4	J4	MC bias return 4 (28)-B / MC bias return 4 (4)-C	
K5	J5	MC bias return 5 (29)-B / MC bias return 5 (5)-C	
K6	J6	MC bias return 6 (30)-B / MC bias return 6 (6)-C	
K7	J7	MC bias return 7 (31)-B / MC bias return 7 (7)-C	
K8	J8	MC bias return 8 (32)-B / MC bias return 8 (8)-C	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	-15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	



EXCITE HD (12.x), HDx & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
ASC CHASSIS
SHEET 13 OF 16

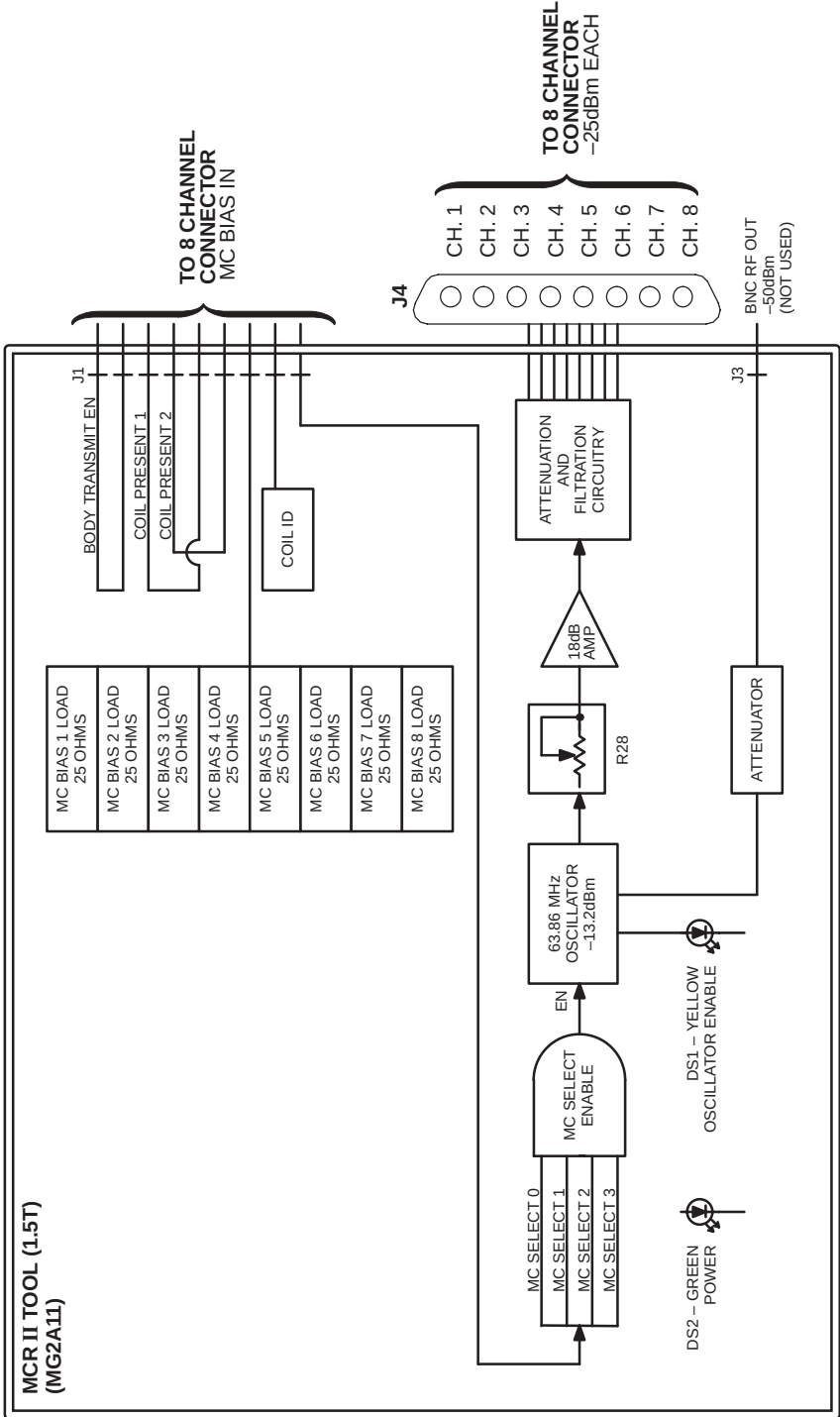
NOTES:
1 Sections enclosed by dotted lines are not present with new Detector Board, part # 5250154.





NOTES:

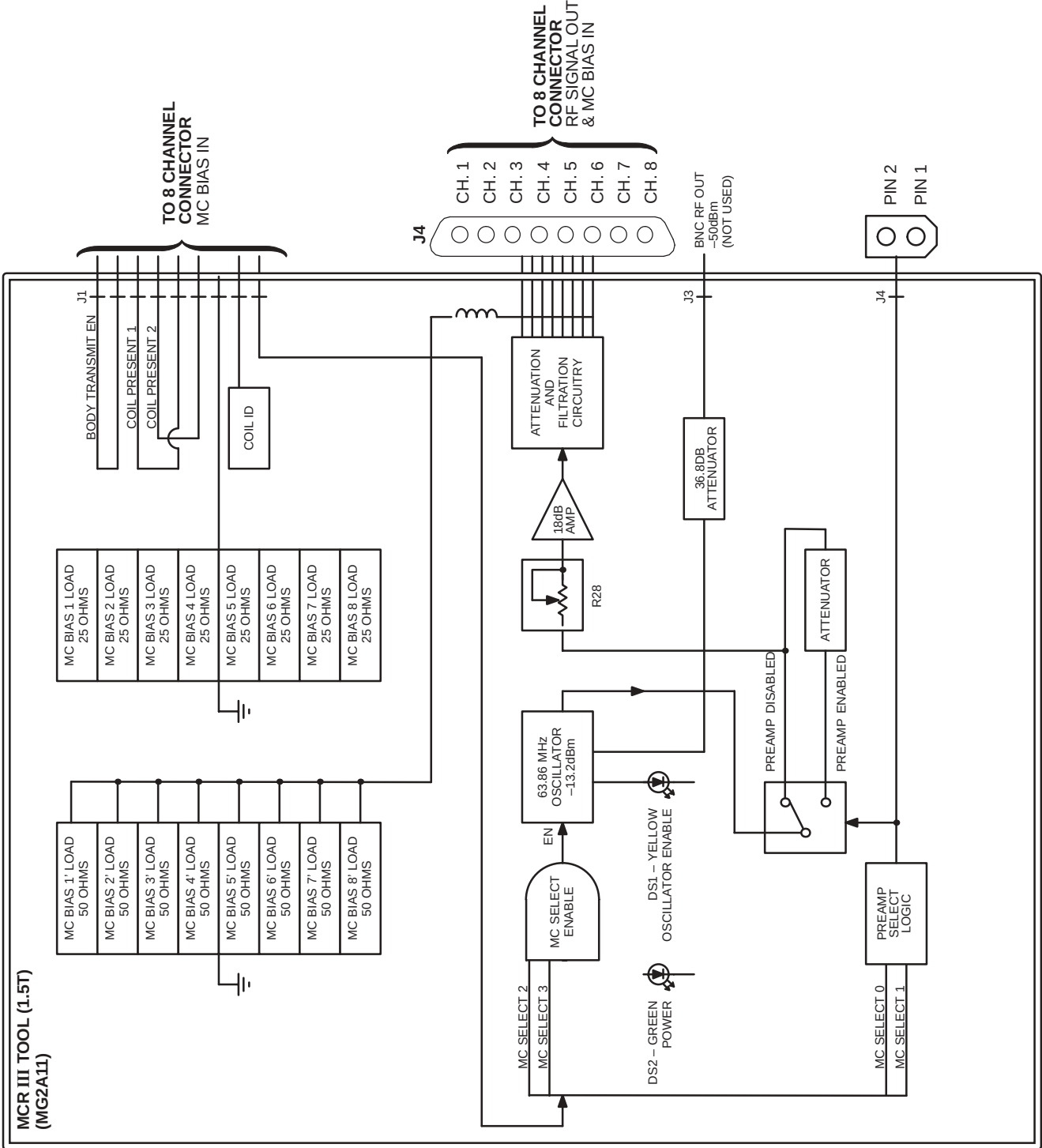
1 SHV Thick Coax Pasternack PE-4194 connector.



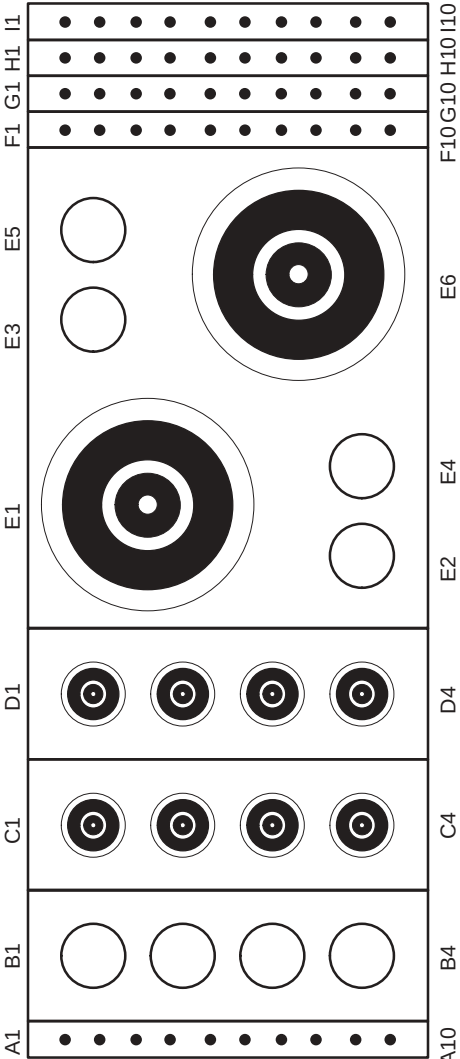
J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	-15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return

J4	
PINS	SIGNAL
1	Preampl Enable
2	Preampl Enable RTN

J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	-15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return



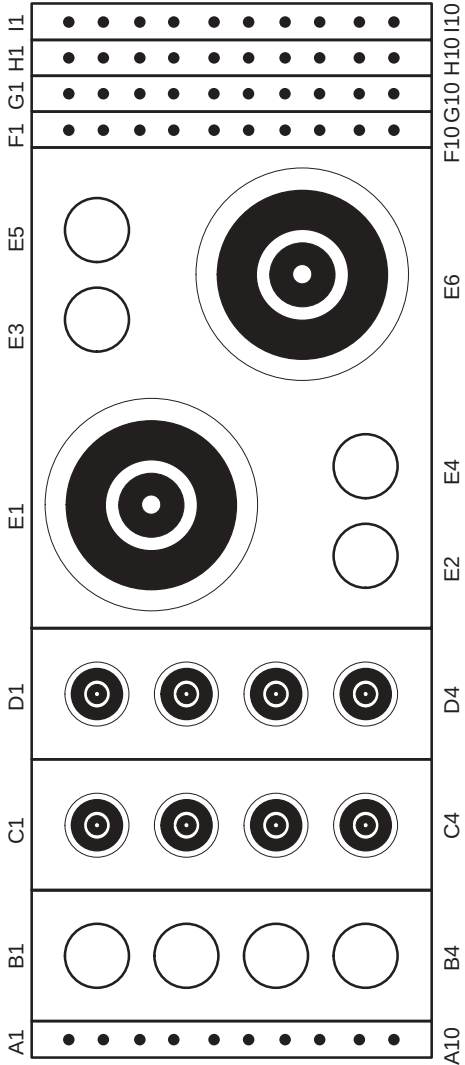
View from CONNECTOR SIDE of 8 Channel Connector Assembly



Contact Function Allocation				
Pin	Twisted with	Description	Comments	
E1	n/a	Open	Not used	
E2	n/a	Open	Not used	
E3	n/a	Open	Not used	
E4	n/a	Open	Not used	
E5	n/a	Open	Not used	
E6	n/a	Open	Not used	
F1	G1	MC bias 1	MC T/R Bias Line 17 for "A", 25 for "B"	
F2	G2	MC bias 2	MC T/R Bias Line 18 for "A", 26 for "B"	
F3	G3	MC bias 3	MC T/R Bias Line 19 for "A", 27 for "B"	
F4	G4	MC bias 4	MC T/R Bias Line 20 for "A", 28 for "B"	
F5	G5	MC bias 5	MC T/R Bias Line 21 for "A", 29 for "B"	
F6	G6	MC bias 6	MC T/R Bias Line 22 for "A", 30 for "B"	
F7	G7	MC bias 7	MC T/R Bias Line 23 for "A", 31 for "B"	
F8	G8	MC bias 8	MC T/R Bias Line 24 for "A", 32 for "B"	
F9	F10	body transmit enable		
F10	F9	body transmit enable return		
G1	F1	MC bias return 1	MC T/R Bias Line 17 for "A", 25 for "B"	
G2	F2	MC bias return 2	MC T/R Bias Line 18 for "A", 26 for "B"	
G3	F3	MC bias return 3	MC T/R Bias Line 19 for "A", 27 for "B"	
G4	F4	MC bias return 4	MC T/R Bias Line 20 for "A", 28 for "B"	
G5	F5	MC bias return 5	MC T/R Bias Line 21 for "A", 29 for "B"	
G6	F6	MC bias return 6	MC T/R Bias Line 22 for "A", 30 for "B"	
G7	F7	MC bias return 7	MC T/R Bias Line 23 for "A", 31 for "B"	
G8	F8	MC bias return 8	MC T/R Bias Line 24 for "A", 32 for "B"	
G9	n/a	Reserved		
G10	n/a	Reserved		
H1	n/a	Reserved		
H2	n/a	Reserved		
H3	n/a	Reserved		
H4	n/a	Reserved		
H5	n/a	Reserved		
H6	n/a	Reserved		
H7	n/a	Reserved		
H8	n/a	Reserved		
H9	n/a	Reserved		
H10	n/a	Reserved		
I1	n/a	Reserved		
I2	I5	+15Vdc (auxiliary & preamp power)		
I3	I6	+10Vdc (auxiliary & preamp power)		
I4	n/a	−15Vdc (auxiliary power)		
I5	I2	Return for aux power		
I6	I3	Return for aux power		
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.	
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.	
I9	I10	coil present'		
I10	I9	coil present return'		

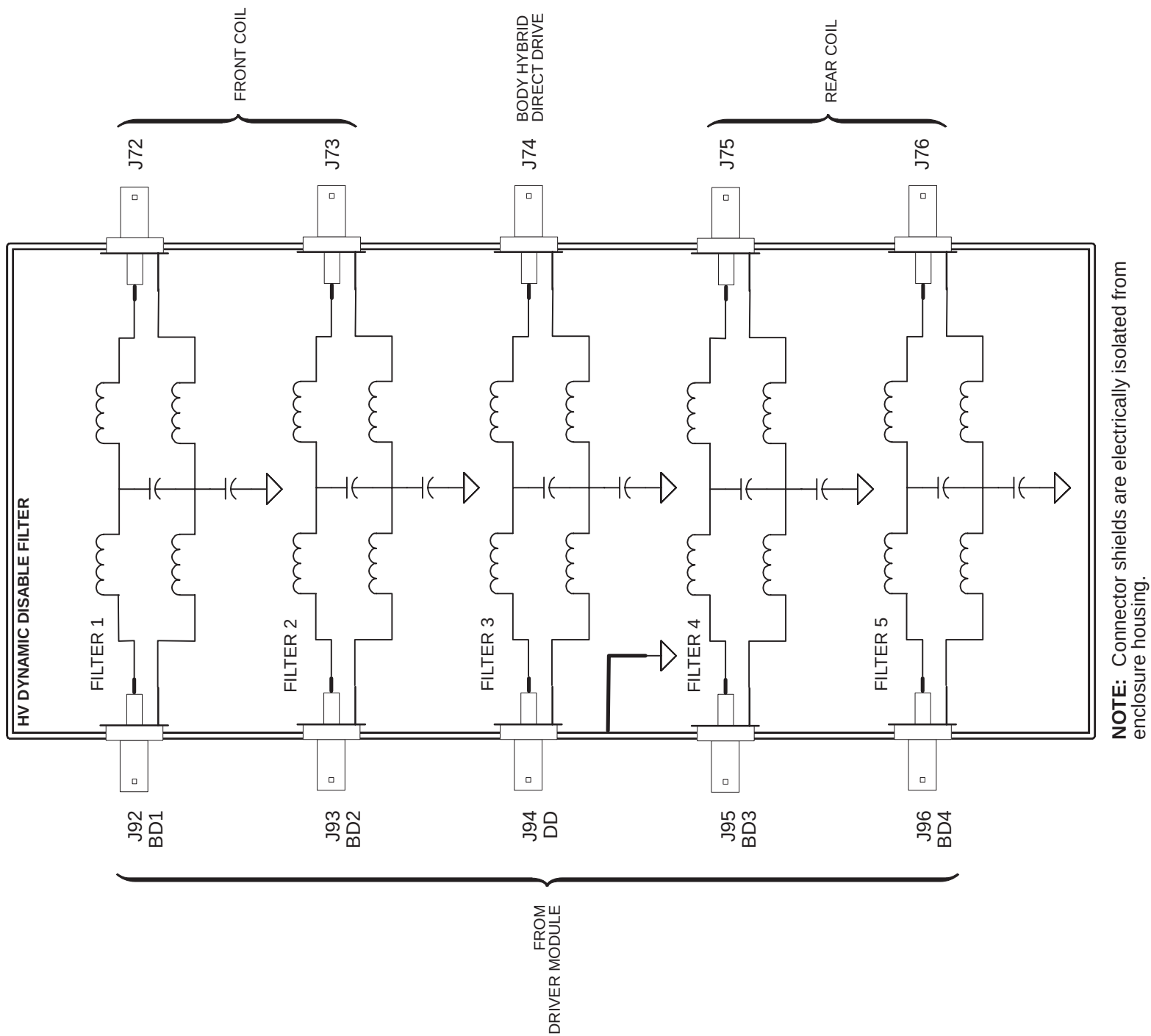
Contact Function Allocation		
Pin	Twisted with	Comments
A1	A2	coil present
A2	A1	coil present return
A3	A4	multiplexerSelect0
A4	A3	multiplexerSelect0 return
A5	A6	multiplexerSelect1
A6	A5	multiplexerSelect1 return
A7	A8	multiplexerSelect2
A8	A7	multiplexerSelect2 return
A9	A10	multiplexerSelect3
A10	A9	multiplexerSelect3 return
B1	n/a	Not used
B2	n/a	Not used
B3	n/a	Not used
B4	n/a	Not used
C1	n/a	MC−1A,B or MC−9B
C2	n/a	MC−2A,B or MC−10B
C3	n/a	MC−3A,B or MC−11B
C4	n/a	MC−4A,B or MC−12B
D1	n/a	MC−5A,B or MC−13B
D2	n/a	MC−6A,B or MC−14B
D3	n/a	MC−7A,B or MC−15B
D4	n/a	MC−8A,B or MC−16B

View from CONNECTOR SIDE of 8 Channel Connector Assembly



Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		Not used
B2	n/a		Not used
B3	n/a		Not used
B4	n/a		Not used
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Open	Not used
E2	n/a	Open	Not used
E3	n/a	Open	Not used
E4	n/a	Open	Not used
E5	n/a	Open	Not used
E6	n/a	Open	Not used
F1	G1	MC bias 1	MC T/R Bias Line 17 for "A", 25 for "B"
F2	G2	MC bias 2	MC T/R Bias Line 18 for "A", 26 for "B"
F3	G3	MC bias 3	MC T/R Bias Line 19 for "A", 27 for "B"
F4	G4	MC bias 4	MC T/R Bias Line 20 for "A", 28 for "B"
F5	G5	MC bias 5	MC T/R Bias Line 21 for "A", 29 for "B"
F6	G6	MC bias 6	MC T/R Bias Line 22 for "A", 30 for "B"
F7	G7	MC bias 7	MC T/R Bias Line 23 for "A", 31 for "B"
F8	G8	MC bias 8	MC T/R Bias Line 24 for "A", 32 for "B"
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for "A", 25 for "B"
G2	F2	MC bias return 2	MC T/R Bias Line 18 for "A", 26 for "B"
G3	F3	MC bias return 3	MC T/R Bias Line 19 for "A", 27 for "B"
G4	F4	MC bias return 4	MC T/R Bias Line 20 for "A", 28 for "B"
G5	F5	MC bias return 5	MC T/R Bias Line 21 for "A", 29 for "B"
G6	F6	MC bias return 6	MC T/R Bias Line 22 for "A", 30 for "B"
G7	F7	MC bias return 7	MC T/R Bias Line 23 for "A", 31 for "B"
G8	F8	MC bias return 8	MC T/R Bias Line 24 for "A", 32 for "B"
G9	G10	Preamp Enable	
G10	G9	Preamp Enable Return	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	–15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	

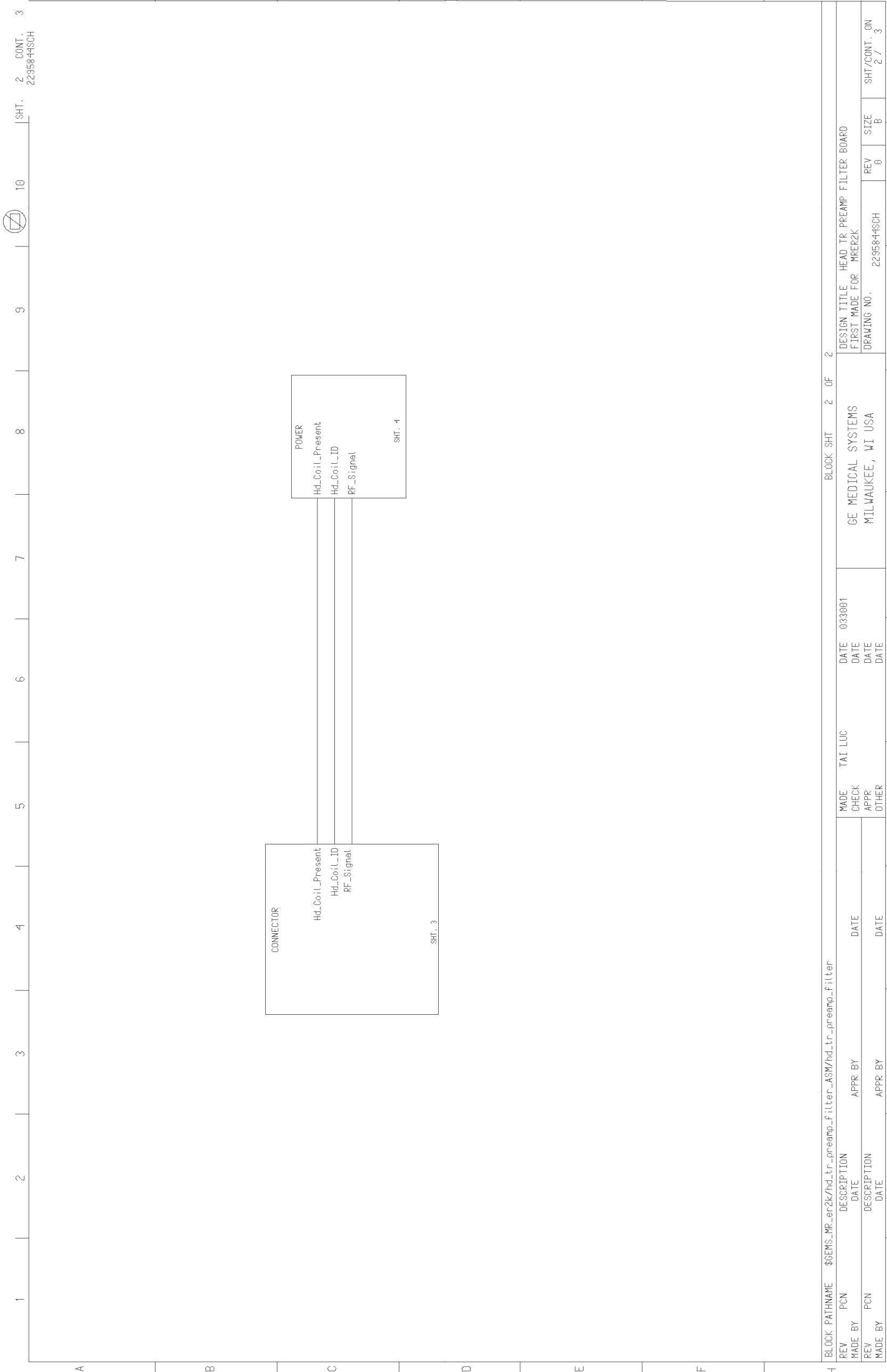


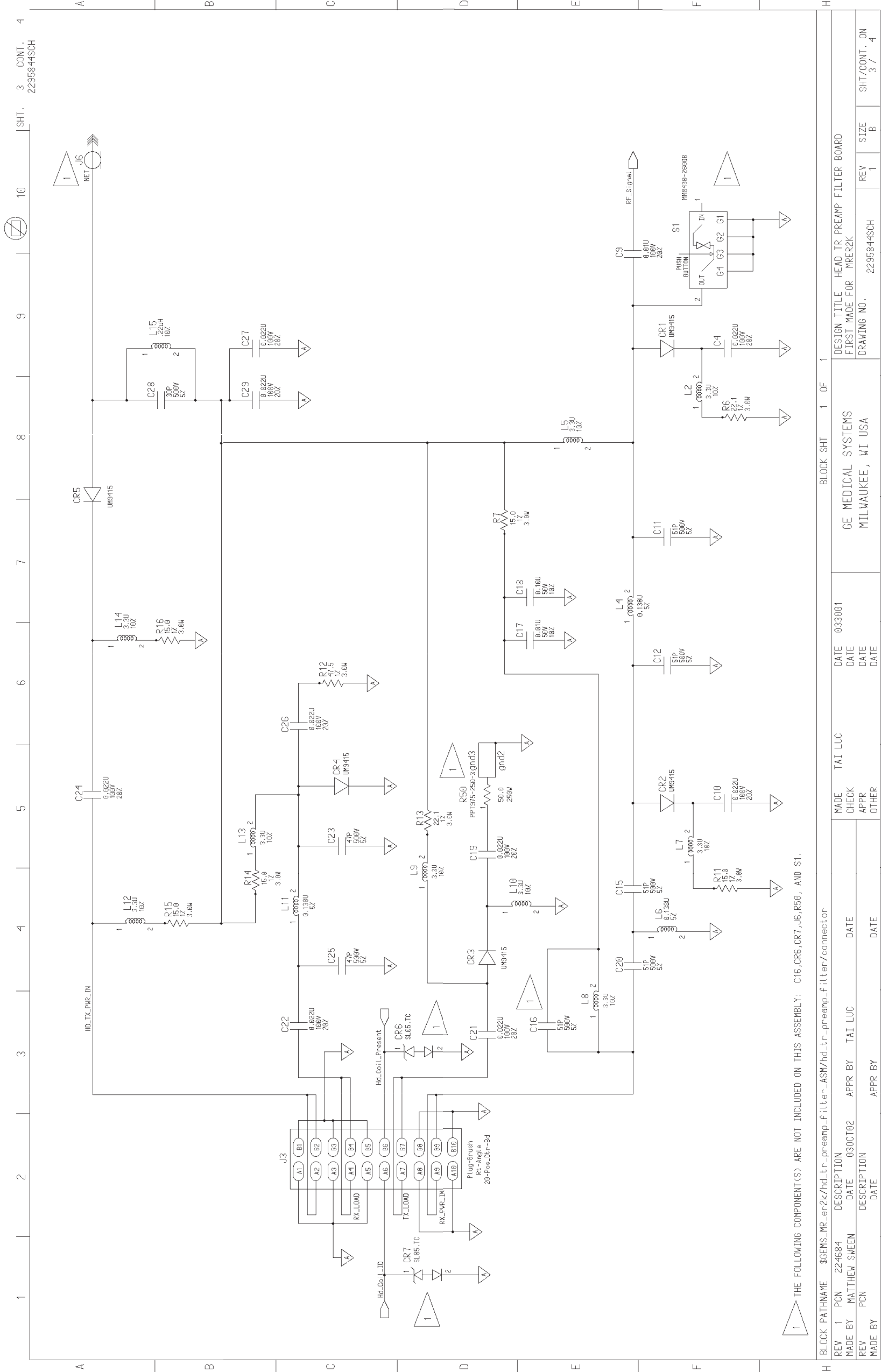
GE Healthcare

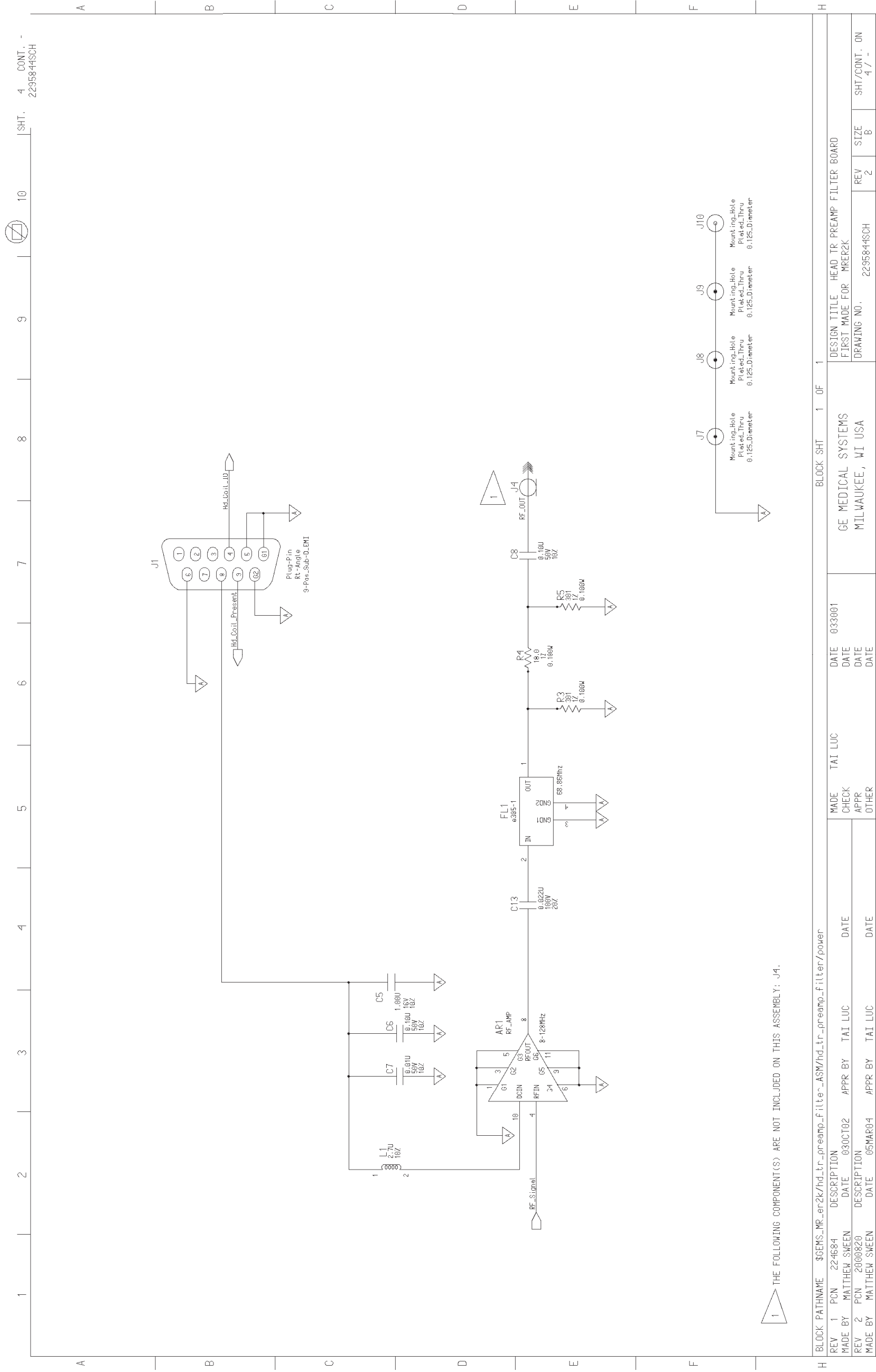
Signal HD, HDx & HDxt 1.5T (12.x ,14.x & 15.x)
Block Diagrams & Supplemental Schematics

REV 10

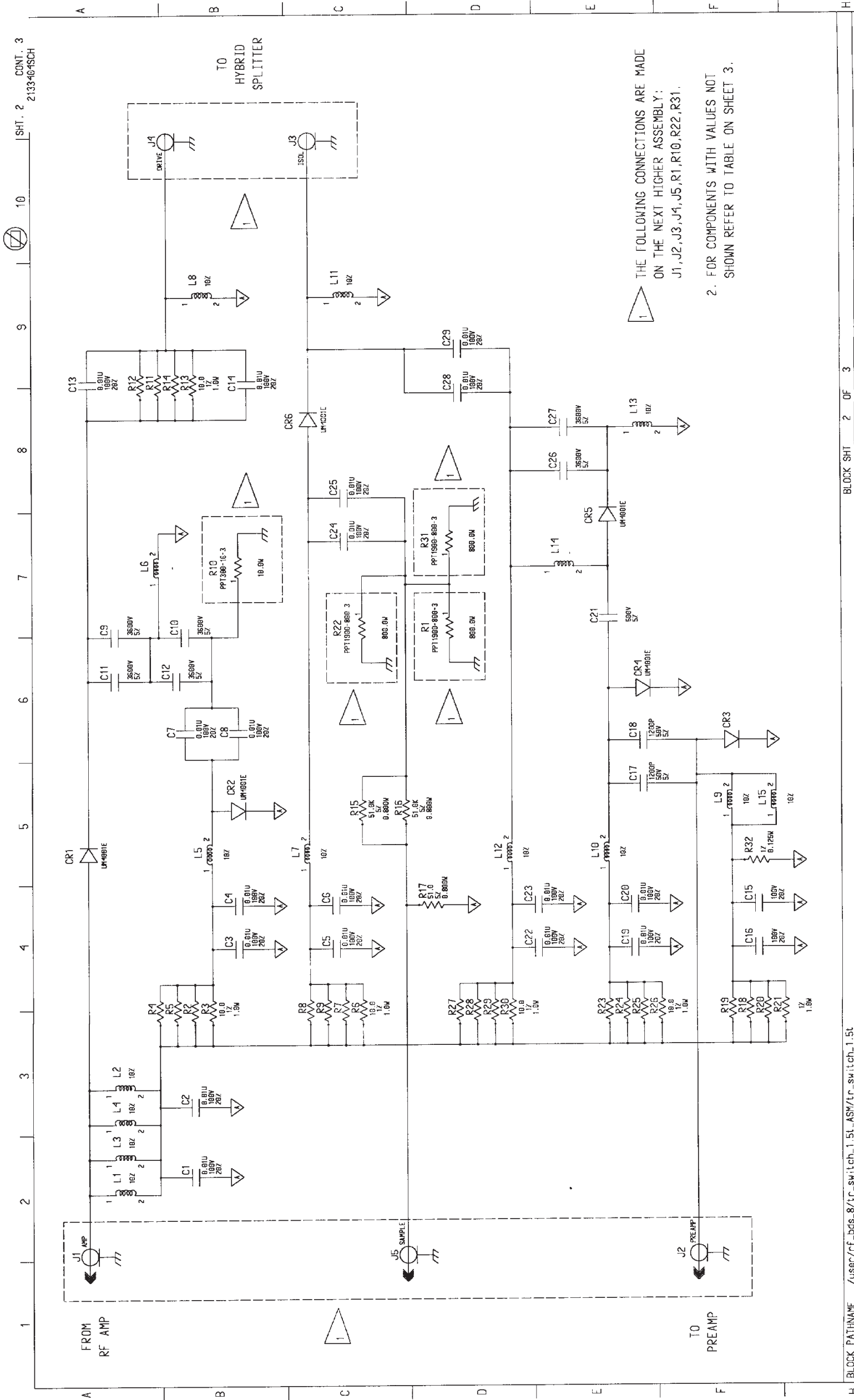
DIRECTION 5133310







EXCITE HD (12.x) and HDx (14.x) MGD/RRF/RF SUBSYSTEM



THE FOLLOWING CONNECTIONS ARE MADE
 ON THE NEXT HIGHER ASSEMBLY:
 J1, J2, J3, J4, J5, R1, R10, R22, R31.

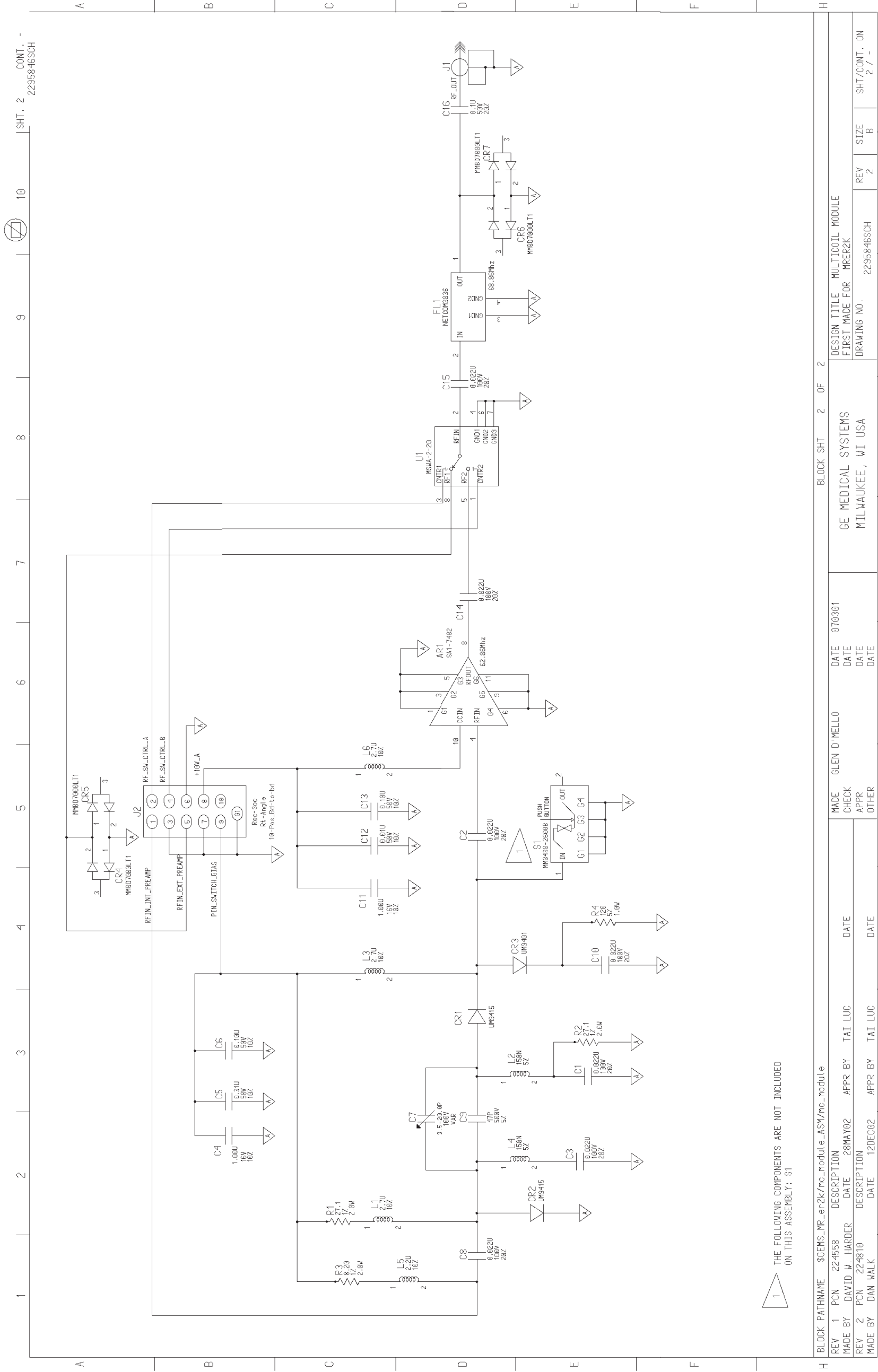
2. FOR COMPONENTS WITH VALUES NOT
 SHOWN REFER TO TABLE ON SHEET 3.

BLOCK PATHNAME		BLOCK SHT		2 OF 3		DESIGN TITLE		T/R BOARD	
REV	2	PCN	197671	DESCRIPTION	7/NOV/97	DATE	10NOV97	MADE BY	MARK WITTE
REV	PCN	DESCRIPTION	DATE	DATE	DATE	DATE	DATE	APPR	DAVE HARDER
MADE BY	DATE	DATE	DATE	DATE	DATE	DATE	DATE	OTHER	ROBERT JACOBS

EXCITE HD (12.x) and HDx (14.x) MGD/RRF/RF SUBSYSTEM

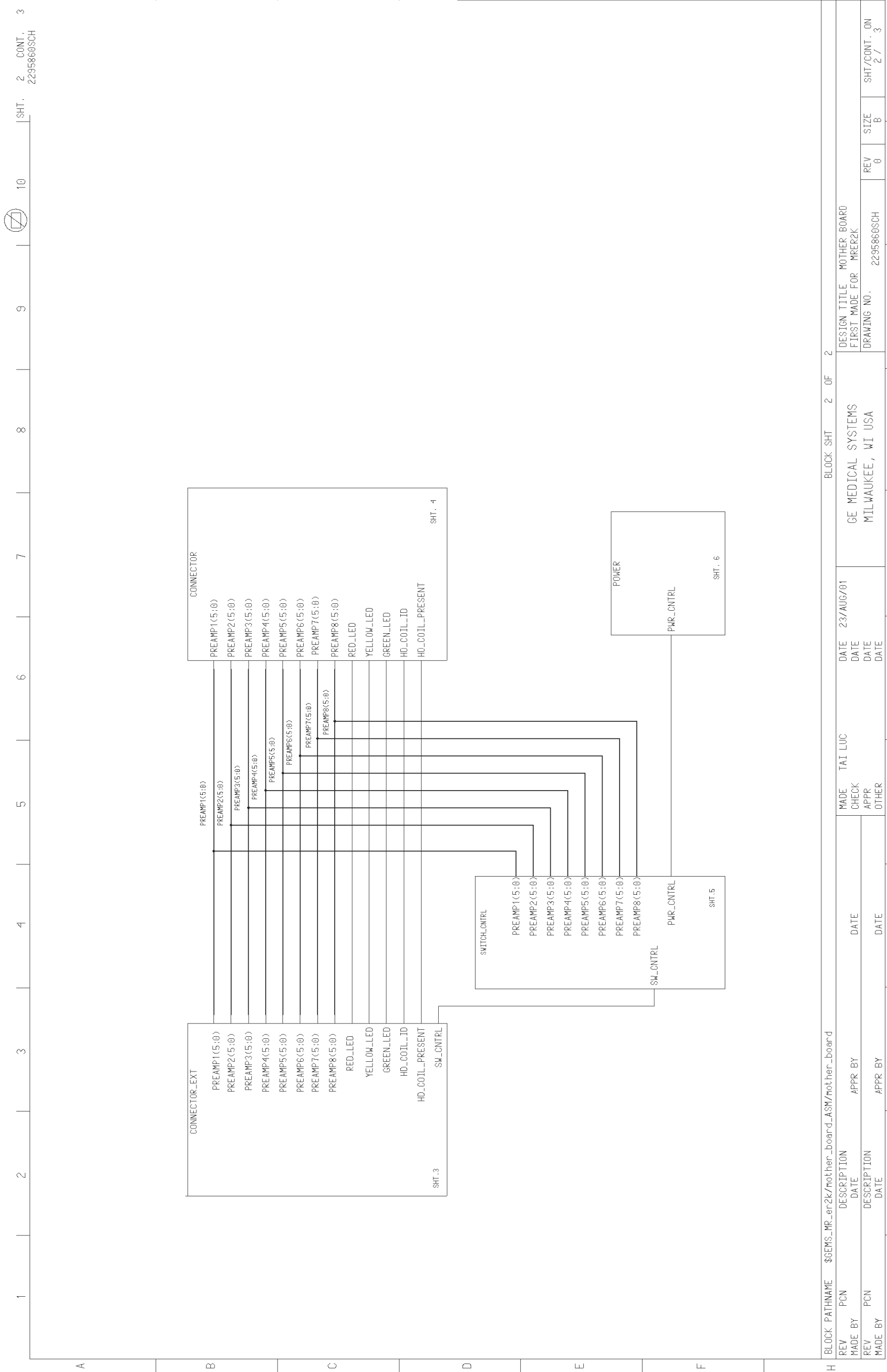
Signa HD, HDx & HDxt 1.5T (12.x ,14.x & 15.x)
Block Diagrams & Supplemental Schematics

DIRECTION 5133310



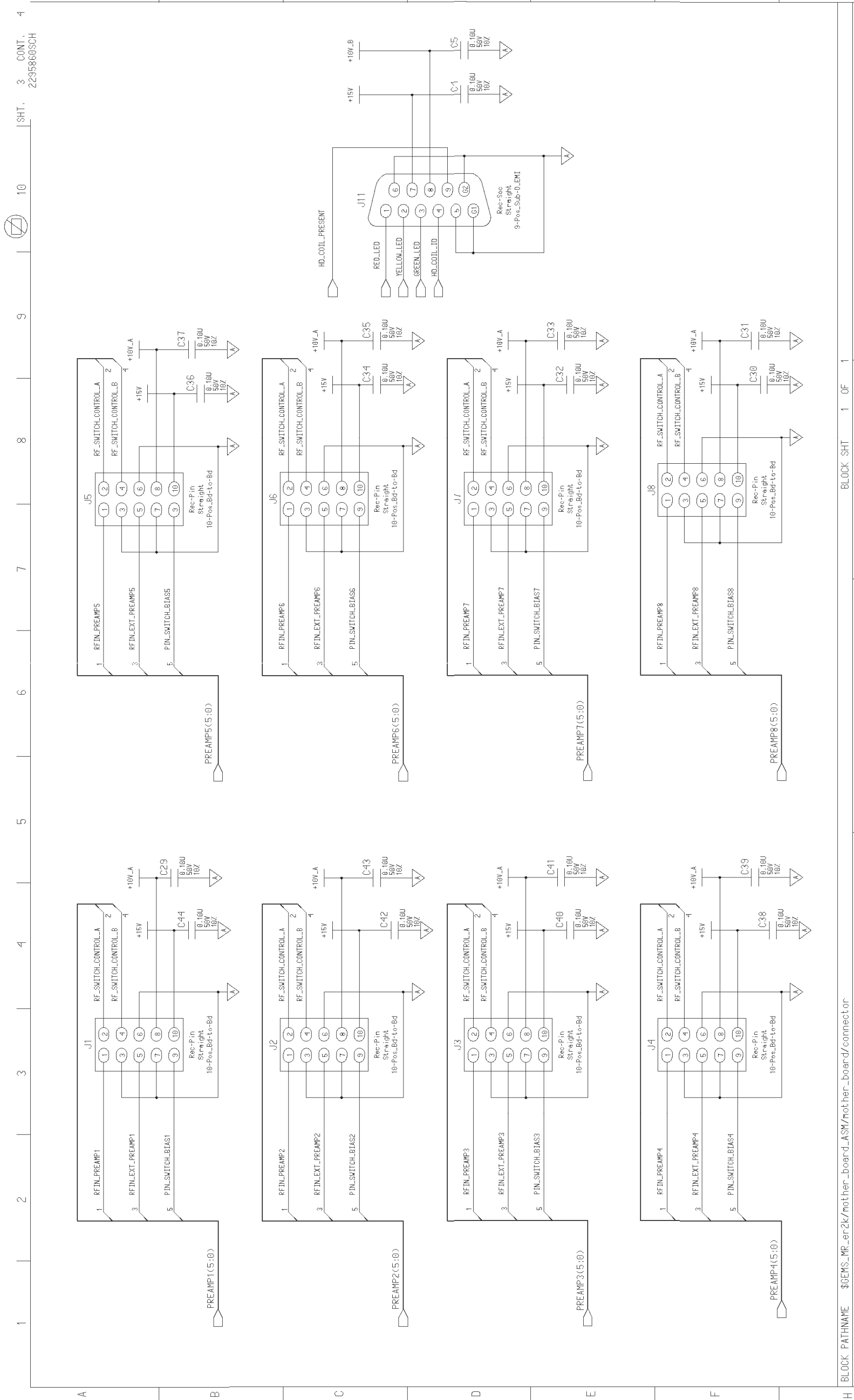
BLOCK PATHNAME \$6EMS_MR_er2k/mc_modul e_ASM/mc_modul e				BLOCK SHT 2 OF 2		DESIGN TITLE MULTICOIL MODULE		
REV 1	PCN 22-4558	DESCRIPTION	DATE 28MAY02	APPR BY TAI LUC	DATE	FIRST MADE FOR MPR2K		
MADE BY DAVID W. HARDER	DATE	DATE	DATE	DATE	DATE	DRAWING NO. 2295846SCH		
REV 2	PCN 22-4810	DESCRIPTION	DATE 12DEC02	APPR BY TAI LUC	DATE	REV 2	SIZE B	SHT/CONT. ON 2 / -
MADE BY DAN WALK	DATE	DATE	DATE	DATE	DATE			

EXCITE HD (12.x) and HDx (14.x) MGD/RRF/RF SUBSYSTEM



Signa HD, HDx & HDxt 1.5T (12.x ,14.x & 15.x)
Block Diagrams & Supplemental Schematics

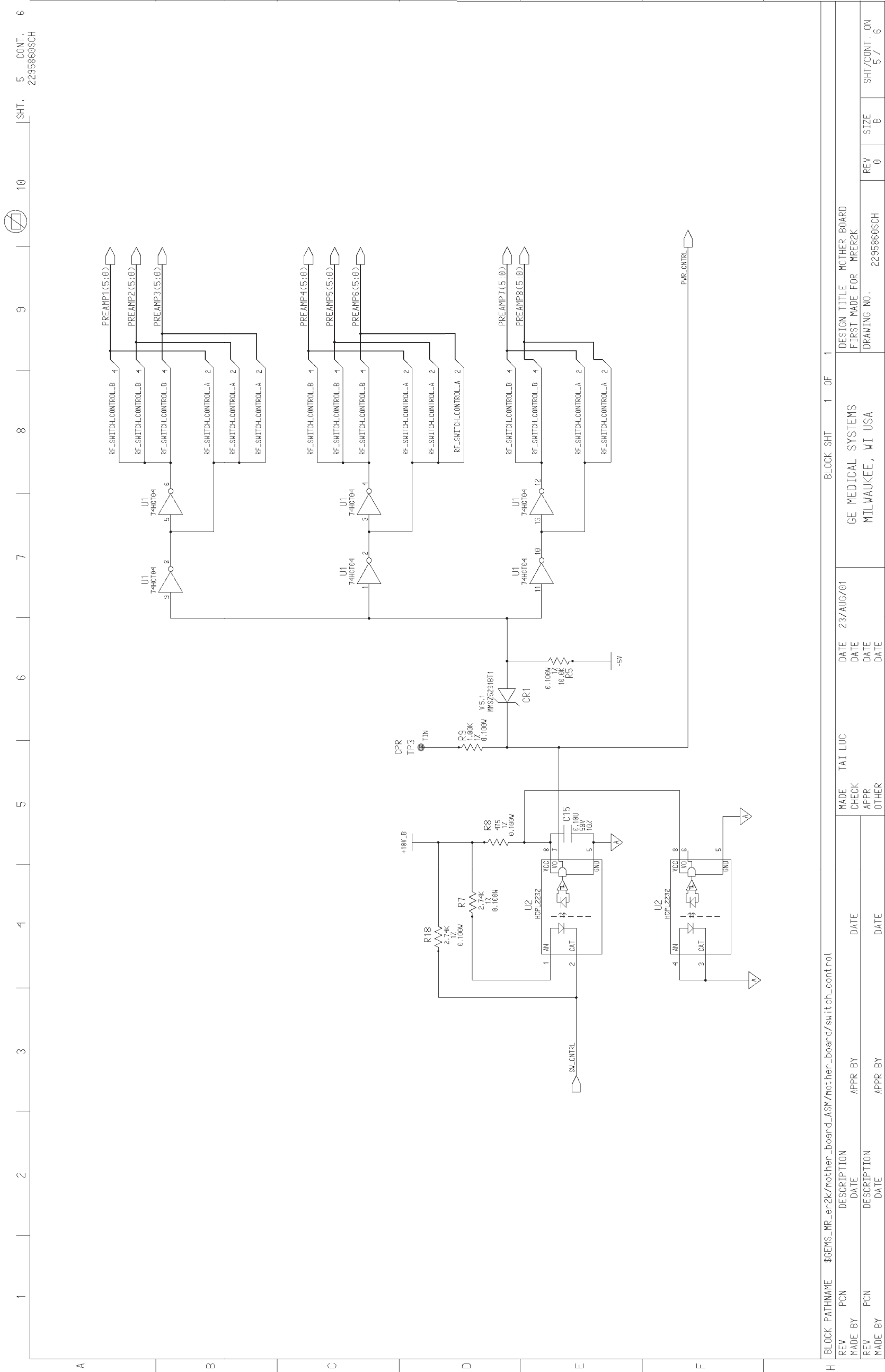
DIRECTION 5133310



EXCITE HD (12.x) and HDx (14.x) MGD/RRF/RF SUBSYSTEM

Signa HD, HDx & HDxt 1.5T (12.x ,14.x & 15.x)
Block Diagrams & Supplemental Schematics

DIRECTION 5133310



COOLING SYSTEM

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Water Chillers Subsystem Group Interconnect Diagrams

ILLUSTRATION	PAGE 1-1
1 OUTDOOR TSCC & RCP	
2 INDOOR WATER COOLED TSCC	
3 INDOOR AIR COOLED TSCC	
4 OUTDOOR TGWC & RCP	
5 INDOOR WATER COOLED TGWC	
6 INDOOR AIR COOLED TGWC	
7 DUAL INDOOR MRCC	
8 INDOOR MRCC	
9 DUAL OUTDOOR MRCC	
10 OUTDOOR MRCC	
11 INDOOR GWHX	

Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)

